

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of:

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

**RESPONDENTS OURARING INC. AND OURA HEALTH OY'S MOTION FOR
SUMMARY DETERMINATION OF NON-INFRINGEMENT BY OURA'S
ALTERNATIVE PRODUCTS**

GROUND RULE 5.1 CERTIFICATION

Pursuant to Ground Rule 5.1, Oura states that it made reasonable, good-faith efforts to resolve the matter with Samsung prior to filing the Motion. Samsung has indicated that it opposes this motion and agrees that the parties are at an impasse.

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Respondents Ouraring Inc. and Oura Health Oy (collectively, “Oura”) respectfully move for summary determination of non-infringement of U.S. Patent No. 10,978,789 (the “’789 Patent”) as to Oura Ring Gen. 5 [REDACTED] of which are products-at-issue in this Investigation and neither of which Samsung accuses of infringing any of the Asserted Patents.¹

I. INTRODUCTION

Samsung bears the burden of proof on infringement. Yet, despite Oura identifying that it would seek adjudication of the [REDACTED], and despite Oura providing extensive discovery on [REDACTED] in both this Investigation and the co-pending 337-TA-1468 Investigation for which there is a stipulation for the cross-use of discovery,² Samsung accuses only one ring: Oura Ring Gen. 4. Samsung never accuses [REDACTED].

Nor has Samsung properly offered any expert testimony or even infringement contentions against [REDACTED]. In contrast, Oura’s technical expert has provided unrebutted expert testimony on non-infringement as to [REDACTED]. Samsung’s failure to proffer any allegations or evidence of infringement as to [REDACTED] entitles Oura to summary determination of non-infringement. *See Certain Crafting Machines and Components Thereof*, Inv. No. 337-TA-1426, Init. Det. at 40 (Jan. 21, 2026) (Cheney, Chief ALJ) (EDIS Doc. ID 869643)

¹ In view of Ground Rule 5.6.2.1’s limits to summary determination motions, Oura is moving with respect to the ’789 Patent only. Oura intends to seek adjudication of the Oura Ring Gen. 5 [REDACTED] in this Investigation at a later time with respect to the remaining asserted patents. *See Certain Photovoltaic Trunk Bus Cable Assemblies and Components Thereof*, Inv. No. 337-TA-1438, Init. Det. at 89-113 (Feb. 6, 2026) (adjudicating alternative designs after the evidentiary hearing).

² The Parties agreed to a Joint Discovery Stipulation that permits cross-use of discovery from 1468 Inv. in the present 1478 Investigation. *See* Joint Discovery Stipulation, EDIS No. 878423, at ¶ 29 (“The Parties agree that all discovery taken among the Parties in connection with . . . 337-TA-1468 . . . including produced documents, discovery responses, and 30(b)(6) and 30(b)(1) depositions of Party employees . . . may be used in the present Investigation”).

[REDACTED]

[REDACTED]

("[Complainant] has not provided any allegations or evidence that the redesigned products, which are ripe for adjudication, infringe the asserted patents. [] Consequently, [Complainant] cannot satisfy its burden of proof on infringement, and [Respondent] is entitled to summary determination of non-infringement.").

II. BACKGROUND

A. *Samsung Complaint*

On December 12, 2025, Samsung filed a complaint seeking 337 Investigation against Oura by alleging that Oura infringes four asserted patents: U.S. Patents Nos. 10,642,359 ("359 patent"), 10,945,677 ("677 patent"), 10,231,675 ("675 patent"), and the '789 patent (collectively, "Asserted Patents"). *See* Original Complaint, EDIS Doc. ID 866474, ¶ 1; Ex. 1 ('789 Patent). Samsung identified the Accused Products as "certain wearable devices, such as smart rings." *Id.* ¶ 3. But the infringement charts accompanying the complaint alleged infringement of only Oura Ring Gen. 4. *Id.*, Exs. 13-16. On December 31, 2025, Samsung filed a first supplement to the Complaint. EDIS No. 867722.

On January 14, 2026, the Commission instituted the Investigation defining the scope of the investigation as "wearable devices, specifically wearable biosignal interfaces and an operation method thereof as well as wearable electronic devices for detecting biometric information." Notice of Institution of Investigation (Jan. 14, 2026) (EDIS Doc. ID 868875).

On January 30, 2026, after setting a 16-month target date (Order No. 4), the ALJ entered a procedural schedule for this Investigation. *See* Order No. 6. Relevant to this motion, the ALJ set two deadlines for responses to contention burden interrogatories: March 16, 2026 for the initial burden contentions and May 7, 2026 for final supplemental of burden contentions. *Id.* On May 5, 2026, at the request of Samsung seeking "to amend the procedural schedule to . . . complete fact

and expert discovery,” the ALJ adjusted certain deadlines, including deadline for contention burden interrogatories to May 13, 2026 and close of fact discovery to May 27. Order No. 8.

B. Discovery in the Parallel 1468 Investigation and Cross-Use Agreement

Before the 1478 Investigation was even instituted, Oura and Samsung were in the midst of another investigation that also involved smart ring products. *Certain Smart Wearable Devices, Systems, and Components Thereof*, Inv. No. 337-TA-1468 (November 18, 2025). In the 1468 complaint, Oura identified Oura Ring Generation (“Gen.”) 3, Oura Ring Gen. 4, and [REDACTED] as Oura’s domestic industry products. Ex. 2 (Oura 1468 Complaint), ¶ 5 (“Each of Oura Ring Gen. 3, Gen. 4, [REDACTED] are Oura’s Domestic Industry Products.”) (EDIS Doc. ID 864274). The same counsel represents each party in both Investigations.

Given the significant overlap in the scope of the two investigations, and at the recommendation of the ALJ during the January 29, 2026 preliminary conference for 1478 Investigation, the parties agreed to a discovery cross-use agreement between the 1468 and 1478 Investigations that obviated the need for re-production of the same materials in this Investigation. Ex. 3 (February 12, 2026 Correspondence between Oura and Samsung counsel); *See* Joint Discovery Stipulation, EDIS Doc. ID 878423, at ¶ 29.

The cross-use agreement is significant given the early discovery that Oura provided to Samsung in the 1468 Investigation related to [REDACTED] that therefore did not need to be re-produced in this Investigation. Key dates and productions related to [REDACTED] are the following:

- **January 26, 2026:** Oura produced 99 technical documents comprising photographs of [REDACTED] (OURA1468_00023843 - 847), [REDACTED] hardware tech specs (OURA1468_00023857 - 865), [REDACTED] testing and teardown photographs (OURA1468_00028664 - 760),

and a [REDACTED] design overview (OURA1468_00056564 - 575). *See* Ex. 4 (Jan. 26, 2026 1468 Inv. Production Letter).

- **February 9, 2026:** Oura again identified the [REDACTED] products as domestic industry products in its initial burden domestic industry contentions that Oura would rely on in Investigation No. 337-TA-1468. *See* Ex. 5 at 10. Along with the identification, Oura also served charts that included images of the physical samples, teardowns, and design files for [REDACTED]. *See* Ex. 6 ([REDACTED]) and Ex. 7 ([REDACTED]).
- **February 11, 2026:** Oura made the source code for all its domestic industry products available for inspection by Samsung, including firmware and application code related to [REDACTED]. Ex. 8.
- **February 25, 2026:** Oura produced 28 technical documents comprising [REDACTED] (OURA1468_00218121 - 131), [REDACTED] files (OURA1468_00218132 - 137), and [REDACTED] bill of materials, design schematics, and a PWB stackup specification (OURA1468_00219723 - 738). *See* Ex. 9 (Feb. 25, 2026 1468 Inv. Production Letter).
- **March 17, 2026:** Oura produced 85 technical documents comprising [REDACTED] teardown documents (OURA1468_00254628 - 645), [REDACTED] teardown documents (OURA1468_00254646 - 659), [REDACTED] bill of materials (OURA1468_00254743 - 5091), [REDACTED] feature and design documents (OURA1468_00255092 - 338), [REDACTED] schematic design documents and PCB stackup (OURA1468_00255339 - 375), [REDACTED] photographs (OURA1468_00255376 - 394), a [REDACTED] sensor overview (OURA1468_00255395), and [REDACTED] technical features specifications (OURA1468_00255396 - 447). *See* Ex. 10 (Mar. 17, 2026 1468 Inv. Production Letter).
- **March 26, 2026:** Oura produced a product overview of [REDACTED] with a technical features specification (OURA1468_00256348 - 376). *See* Ex. 11 (Mar. 26, 2026 1468 Inv. Production Letter).
- **March 31, 2026:** Oura made available for inspection physical samples of the [REDACTED] (OURA_PHYS_00000006) and [REDACTED] (OURA_PHYS_00000007), including both new and teardown samples. *See* Ex. 12 (Mar. 31, 2026 1468 Inv. Production Email).

In addition to receiving these technical documents, Samsung also deposed a number of Oura engineers in March 2026, including Miguel Christophy, Vice President of Hardware

Engineering at Oura, whose primary focus is the development and production of [REDACTED]
[REDACTED]:

Q. Have you worked on what's been called [REDACTED] in this case, what I understand to be [REDACTED]?

A. Yes. I've been engaged on [REDACTED] for some months now.

Ex. 13 (Christophy Tr. March 20, 2026) at 28:13-20; *see also id.* at 20:8-11, 26:1-27:16 (establishing Mr. Christophy's role within Oura). During his deposition, Mr. Christophy provided in-depth details on [REDACTED], including [REDACTED], hardware structure, functionality, and engineering schematics of both [REDACTED]. *Id.* 61:10-15 (confirming that [REDACTED]); 64:25-65:3 (confirming that the [REDACTED]); 62:24-64:19 (explaining the hardware and sensors of [REDACTED]); 65:10-72:10 ([REDACTED]); 131:4-146:15 (testifying regarding the CAD and hardware of [REDACTED]); 146:16-150:7 (manufacturing process of [REDACTED] Ring); 150:8-15 (testimony on the PCB structure of [REDACTED]); 154:9-155:15 (discussing the plan of record regarding [REDACTED]); 157:15-178:17 ([REDACTED]).

C. Discovery and Party Positions in the 1478 Investigation

On March 16, 2026, despite obtaining substantial discovery about [REDACTED], Samsung served its initial infringement contentions in the 1478 Investigation that accused only Oura Ring Gen. 4 of infringing the Asserted Patents. Ex. 14 at 1 ("Samsung accuses the following wearable devices and corresponding applications of infringement: Oura Ring 4 and its

corresponding software application, including the Oura App”). Samsung made no mention of either [REDACTED]. *Id.*

On April 29, 2026, Oura amended its interrogatory responses to affirmatively identify the [REDACTED] for which it would seek a determination of noninfringement. For example, in response to Samsung’s Interrogatory No. 1 seeking identification of each Accused Product, Oura specifically identified [REDACTED] [REDACTED]. *See* Ex. 15 (April 29 Interrogatory Responses) at 12-13 (Responding to Interrogatory No. 1 to indicate that in addition to Oura Ring Gen. 4, both [REDACTED] should be considered “Accused Products”), 79 (stating the [REDACTED] “do not directly infringe, induce infringement of, or contribute to the infringement of the ’359 Patent, the ’677 Patent, the ’675 Patent, and the ’789 Patent”). The interrogatory response further identified the technical documents previously produced in the 1468 Investigation at OURA1468_00254743 – OURA1468_00255395, OURA1468_00255396 – OURA1468_00255447, and OURA1468_00256348 - OURA1468_00256376 as being particularly relevant. *Id.* at 13. Oura also noted that “source code for the [REDACTED]” for both firmware and Oura Application code continued to be available for inspection on source code laptop since February 11 2026. *Id.* at 12; Ex. 8 (source code production email). Technical documents related to [REDACTED] were also identified in the context of Samsung Interrogatories 2-5. Ex. 15 at 15, 17, 19-21 (further identifying OURA1468_00219723 – 738).

Oura did not rest on its discovery obligations. Despite having produced over 210 technical documents, teardown images, CAD drawings, physical samples, source code, product roadmaps, and deposition testimony related to [REDACTED], Oura continued to flag the [REDACTED] [REDACTED] issue for Samsung by producing additional relevant documents, including:

- **May 5, 2026:** Oura produced [REDACTED] (OURA1478_00004838); 14 datasheets [REDACTED] (OURA1478_00004839 - 5180); 48

[REDACTED] BOMs (OURA1478_00006234 - 281); 7 additional [REDACTED] component specifications and a PCB stack-up diagram (OURA1478_00006283 - 316). *See* Ex. 16 (May 5, 2026 Production Letter).

- **May 29, 2026:** Oura produced extensive [REDACTED] (OURA1478_00013076 - 4258) and extensive [REDACTED] photographs/testing documents, including cross-sectional views of the PCB (OURA1478_00014259 - 793). *See* Ex. 17 (May 29, 2026 Production Letter). Oura also made available for inspection again all the physical samples of the rings, including [REDACTED] along with the new and tested PCB samples. Ex. 18 (May 29, 2026 Production Email).

Furthermore, in its May 13, 2026 supplemental interrogatory responses, Oura identified OURA1478_00004838 – 5180, OURA1478_00006234 – 281, OURA1478_00006283 – 316 as additional particularly relevant documents for interrogatories 1-5. *See* Ex. 19 at 13, 16, 18, 21, 23. And in its final interrogatory responses on May 27, 2026, Oura noted Samsung’s failure to advance any infringement contentions for “any other Oura product” besides the Oura Gen. 4 ring and stated that “it has produced documents from which Complainants may ascertain relevant information responsive to this Interrogatory, including without limitation those cited in Interrogatory Nos. 1-5.” *See* Ex. 20 at 82-84.

Samsung also deposed a number of Oura employees in the 1478 Investigation. These employees testified about [REDACTED]. For example, Mr. Christopher James Vincent, Senior Director of Sourcing at Oura, testified regarding the production details of [REDACTED]:

Q. [] Beyond the Oura Gen 4 Ring, the Gen 4 Ceramic, and the charger, what other products is Oura currently manufacturing?

A. [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Ex. 21 (Vincent Dep. Tr.) at 21:13-20, 57:6-10. Mr. Vincent further testified that “[REDACTED]

[REDACTED],” [REDACTED]

[REDACTED]

[REDACTED]

Oura also designated three technical corporate witnesses to testify on the Accused Products: (1) Mr. Christophy, VP of Hardware Engineering at Oura, related to hardware of all accused products (2) Mr. Michael Larson, Director of Firmware, to testify on firmware of all accused products, and (3) Mr. Simon Ratner, VP of Software Engineering, to testify regarding software or application of all accused products. Ex. 22 (Oura 2026.05.01 30(b)(6) Dep. Designations). Each of Oura’s technical witness was prepared to testify regarding [REDACTED]. See, e.g., Ex. 23 (Christophy Dep. Tr.) at 10:16-11:21 (“yeah, I’m prepared to answer regarding Gen. 1, Oura 4, as we call it, [REDACTED]”). In fact, Samsung’s own counsel specifically stated that the [REDACTED] were within the scope of the questions being asked during the deposition of Mr. Larson:

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Ex. 24 (Larson Dep. Tr.) at 11:22-12:10; *see also id.* at 26:15-27:5, 44:9-45:6, 61:6-9 (Mr. Larson answering questions regarding the [REDACTED]).

[REDACTED]
[REDACTED]
[REDACTED]. Ex. 21 (Vincent Dep. Tr.) at 63:18-64:6. Fact discovery in this investigation then ended on May 29, 2026. *See* Order No. 8 at 2.

Samsung never complained that Oura failed to meet its discovery obligation regarding [REDACTED]. Nor did Samsung ever suggest that Samsung needed any additional information regarding [REDACTED] that it was not afforded to analyze infringement of either ring.

D. Samsung's expert offers no opinions regarding [REDACTED] while Oura's expert provided un rebutted non-infringement opinions on the two rings.

On June 10, 2026, Samsung's technical expert on infringement of the '789 Patent, Dr. Steven Warren, submitted his opening expert report. Ex. 25 (Warren Inf. Rep.). Dr. Warren again only alleged infringement of Oura Ring Gen. 4. *Id.*, ¶ 3 [REDACTED]
[REDACTED]

[REDACTED] Dr. Warren offered no opinion that the [REDACTED] infringed the '789 Patent. *See generally*, Ex. 25 (Warren Inf. Rep.) at ¶¶ 68-69 (infringement opinions only apply to the Oura Gen. 4 ring).

Despite Dr. Warren not putting forth an infringement opinion, Oura's technical expert on noninfringement of the '789 Patent, Dr. Bobak Jack Mortazavi, relied on the documents produced by Oura to show that [REDACTED] do not meet the asserted claims of the '789 Patent. *See generally*, Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 651-80. Specifically, Dr. Mortazavi was able to use the produced discovery to show that [REDACTED] does not meet claim limitations [7.C][vi][a], [7.C][vi][c], and [7.C][vi][e] (*see id.* at ¶¶ 655-71) [REDACTED] does not meet claim limitations [7.C][i]-[iv] and [7.C][vi][c] (*see id.* at ¶¶ 672-80). Samsung then deposed Dr. Mortazavi regarding

his noninfringement opinions concerning [REDACTED]. See Ex. 27 (Mortazavi Tr.) at 532:15-569:20.

III. LEGAL STANDARD

“[T]he test for determining whether a respondent has met its burden for adjudication of a redesigned or alternative product includes four factors: (1) whether the product is within the scope of the investigation; (2) whether it has been imported; (3) whether it is sufficiently fixed in design; and (4) whether it has been sufficiently disclosed by respondent during discovery.” *Certain Human Milk Oligosaccharides and Methods of Producing the Same*, Inv. No. 337-TA-1120, Comm’n Op. at 18 (June 8, 2020) (EDIS Doc. ID 712205) (“*Oligosaccharides*”).

While this inquiry is typically undertaken in the context of redesigns, there is no requirement that a product for which adjudication is sought be a redesign—indeed, the test applies to both “a redesigned or alternative product.” *Id.* (emphasis added); see also *Certain Outdoor and Semi-Outdoor Electronic Displays, Products Containing Same, and Components Thereof*, Inv. No. 337-TA-1331, Init. Det. at 4-5 (June 20, 2023) (EDIS Doc. ID 804206) (“In the undersigned’s view, the *Oligosaccharides* framework is not limited to ‘redesigns,’ and Complainant identifies no meaningful difference between a redesigned product and an alternative non-accused product”) (extensively discussing caselaw and Commission policy); *Certain Two-Way Radio Equip. and Sys., Related Software, and Components Thereof*, Inv. No. 337-TA-1053, Comm’n Op. at 23 (Dec. 18, 2018) (EDIS Doc. ID 664543) (“the Commission has encouraged respondents under certain circumstances to put forward products (typically redesigned products) for adjudication even if they are not expressly accused by the complainant”) (emphasis added).

Once a product has been put forth for adjudication, it is the complainant’s burden to prove infringement. *Oligosaccharides*, Inv. No. 337-TA-1120 at 23; see also *Certain Crafting Machines and Components Thereof*, Inv. No. 337-TA-1426, Init. Det. at 40 (without allegations or evidence

that a product infringes a patent, the complainant cannot carry its burden of proof on infringement and summary determination of noninfringement is warranted).

IV. THE [REDACTED] HAVE MET THE TEST FOR ADJUDICATION

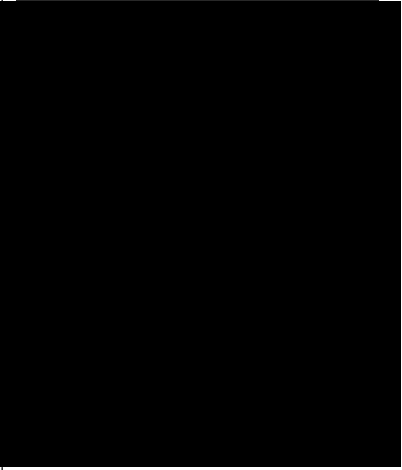
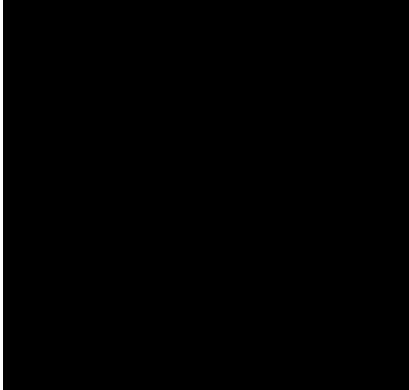
Applying the *Oligosaccharides* factors, both of the [REDACTED] are eligible for adjudication in this Investigation. They are indisputably within the scope of this Investigation, they have been imported, they are fixed in design (indeed, [REDACTED] had already been on sale since May), and sufficient discovery about each of these products was provided. A full analysis of each factor follows.

A. *The [REDACTED] are Within the Scope of this Investigation*

There can be no reasonable dispute that the [REDACTED] are within the scope of this Investigation. As set forth by the Notice of Institution, the scope of this investigation encompasses “wearable devices, specifically wearable biosignal interfaces and an operation method thereof as well as wearable electronic devices for detecting biometric information.” Notice of Institution of Investigation (Jan. 14, 2026) (EDIS Doc. ID 868875). As Samsung’s First Amended Complaint has set forth, “[t]he Accused Products include, but are not limited to, Respondents’ Oura Ring products.” Samsung Amended Complaint Under Section 337 of the Tariff Act of 1930, at ¶ 17. Furthermore, Samsung’s First Set of Interrogatories to Oura made clear that the “‘Accused ’789 Products’ means all of Respondents’ products that Samsung contends are covered by at least one Asserted Claim of the ’789 Patent... including yet to be released products with similar health tracking functionality as the ’789 accused features or functions.” Ex. 28 (Excerpts, Samsung First Set of Interrogatories) at Definitions, ¶ 13.

Like the Oura Gen. 4 ring that is the subject of Dr. Warren’s infringement opinions (*see* Ex. 25 at ¶¶ 68-69), the [REDACTED] are wearable electronic devices for detecting

biometric information: indeed, [REDACTED] is the Oura Gen. 5 ring that went on sale May 28, 2026, and [REDACTED]:

Oura Gen. 4 ring	[REDACTED]	[REDACTED]
 OURA1468_00096106 (Ex. 29)	 OURA1468_00255396 (Ex. 30)	 OURA1468_00255092 at - 092 (Ex. 31)
Biometric Sensors: • Temperature • Red, Green, and Infrared LEDs with photodetectors • Accelerometer OURA1468_00029590 at 607-610 (Ex. 32)	Biometric Sensors: • Temperature • Red, Green, and Infrared LEDs with photodetectors • Accelerometer OURA1468_00023857 at - 858 (Ex. 33)	[REDACTED] • [REDACTED] [REDACTED] [REDACTED] OURA1468_00255092 at - 137 (Ex. 31)

[REDACTED]

Ex. 31, OURA1468_00255092 at -097.

Finally, [REDACTED]

[REDACTED]

[REDACTED]

[REDACTED] Ex. 35 at 1, Exhibit A.

Accordingly, there can be no genuine dispute that the [REDACTED] are squarely within the scope of this Investigation.

B. The [REDACTED] Have Been Imported

While importation is not necessary for adjudication, importation is relevant to this inquiry (*see Oligosaccharides*, Inv. No. 337-TA-1120 at 18, n.21), and both the [REDACTED] have been imported. Oura made available to Samsung physical samples of the [REDACTED] in the United States on March 31, 2026, and these samples had previously been imported. *See* Ex. 18 (May 29, 2026 Production Email); Ex. 21 (Vincent Dep. Tr.) at 22:3-6 (“Oura is not currently manufacturing any products in the United States”); *see also Certain Electrical Connectors and Cages, Components, And Products Containing the Same Thereof*, Inv. No. 337-TA-1241, Init. Det. at 34-35 (March 11, 2022) (EDIS Doc. ID 767918) (noting that “[t]he

Commission has held that ‘importation’ in the context of section 337 means ‘importation by a respondent without regard to purpose,’” and the fact that “physical samples of certain Redesign Products were produced in the United States by Luxshare in discovery” supported a finding of importation); *Certain Portable Battery Jump Starters and Components Thereof (II)*, Inv. No. 337-TA-1359, Init. Det. at 27-28 (July 5, 2024) (EDIS Doc. ID 826438) (“Evidence that physical samples were imported for use as exhibits [in an ITC investigation]—as the redesigned products were here—has been found to be sufficient evidence of importation.”).

Indeed,

See Ex. 35 (Importation Stipulation) at Exhibit A; *see also id.* at ¶ 3 (“

”).

C. The [REDACTED] Are Sufficiently Fixed in Design

Both [REDACTED] are fixed in design. [REDACTED] went on sale in the United States on May 28, 2026 with shipments to consumers starting less than a week later by June 4, 2026. Ex. 21 (Vincent Dep. Tr.) at 63:18-64:6. [REDACTED]

[REDACTED] Ex. 21 (Vincent Dep. Tr.) at 21:13-20, 57:6-10, 63:18-64:6. Even before that, its relevant features were substantially fixed [REDACTED],

[REDACTED]

[REDACTED] See Ex. 36 (OURA1468_00255396). Given that the product is currently on the market, the [REDACTED] Ring is unquestionably fixed in design.

[REDACTED] is also fixed in design for purposes of this inquiry. The relevant question is not whether the products-at-issue are final in every possible respect, but rather, whether the products “are sufficiently fixed for adjudication.” *See Certain Photovoltaic Trunk Bus Cable Assemblies and Components Thereof*, Inv. No. 337-TA-1438, Init. Det. at 105-06 (Feb. 6, 2026) (EDIS Doc. ID 873421) (“Even if the 1438 ADs have not been certified and are not ready to sell today, the evidence supports that their structure is fixed, particularly with respect to the claimed features, so that the parties’ technical experts could (and did) understand and opine on them.”). As the experts of both parties agree, the ’789 Patent concerns a “rigid-flex PCB” design. *See* Ex. 25 (Warren Inf. Rep.) at ¶¶ 40-41, 47; Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 50-51. Thus, the inquiry focuses most particularly on the fixedness of the printed circuit board (“PCB”) design, which in the case of [REDACTED], is entirely fixed.

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED]
[REDACTED] *See generally*, Background Section, *supra*.

Indeed, further evidencing the fixed nature [REDACTED]
[REDACTED]

[REDACTED]; *Certain Audio Players and Controllers, Components Thereof, and Products Containing Same*, Inv. No. 337-TA-1191, Order No. 19 at 4 (Sept. 17, 2020) (EDIS Doc. ID 719871) (providing “physical devices” shows fixedness of design).
[REDACTED]

[REDACTED] *See Certain Electrical Connectors and Cages*, Inv. No. 337-TA-1241, Init. Det. at 35-36 (noting that products can be sufficiently fixed in design even without completing “testing necessary to be released to the market”); *Certain Video Security Equipment and Systems, Related Software, Components Thereof, and Products Containing Same*, Inv. No. 337-TA-1281, Init. Det. at 22 (Oct. 24, 2022) (EDIS Doc. ID 784112) (indicating that the fact “a change could be made in the future . . . is not sufficient for a finding that a redesign cannot be adjudicated”); *Certain Portable Battery Jump Starters*, Inv. No. 337-TA-1359, Init. Det. at 34, 41 (“The Commission, however, ‘does not require that a redesign involve a commercial product in order to be adjudicated.’ [] Nor is a commercially viable or ‘workable’ product required... the fact that a redesign has not been completely ‘finalized’ or that it requires further testing prior to commercial production is not required under Commission precedent for a design to be considered sufficiently fixed.”) (citations omitted).

Thus, given that [REDACTED] is on sale with a completely final design, and [REDACTED]
[REDACTED]

D. Sufficient Discovery Has Been Provided.

present a high bar: “the key question is whether Respondents provided ‘sufficient (not extensive) fact and expert discovery to put [complainant] on notice of that [product] and its relevant features.’” *Certain Outdoor and Semi-Outdoor Electronic Displays*, Inv. No. 337-TA-1331, Init.

technical documents for [REDACTED] on January 26, 2026, only 10 days after this Investigation was instituted. See Ex. 4 (Jan. 26, 2026 Investigation Production Letter). Oura began producing

technical documents for [REDACTED] a month later on February 25, 2026, after having identified [REDACTED] as domestic industry products that would be relied upon by Oura in

Investigation No. 337-TA-1468. Ex. 5 (1468 Inv. Initial Burden DI Contentions) at 10; Ex. 9 (Feb. 25, 2026 1468 Investigation Production Letter). On March 20, 2026, Samsung deposited Oura's VP

of Hardware Engineering extensively on technical details of both [REDACTED], including the relevant PCB structure:

[illegible]
$$[\dots]$$

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

Ex. 23 (Christophy Dep. Tr. (1478 Inv.)) at 150:8-15, 168:24-169:13; *see also id.* at 61:10-15 (confirming that [REDACTED]); 64:25-65:3 (confirming that [REDACTED]); 62:24-64:19 (explaining the hardware and sensors [REDACTED]); 65:10-72:10 (discussion of [REDACTED] hardware and design); 131:4-146:15 (testifying regarding the CAD and hardware [REDACTED]); 146:16-150:7 (manufacturing process of [REDACTED] Ring); 150:8-15 (testimony on the PCB structure of [REDACTED]); 154:9-155:15 (discussing the plan of record regarding [REDACTED]); 157:15-178:17 (testifying on physical design of [REDACTED], including hardware components).

On April 29, 2026, Oura's interrogatory responses identified the [REDACTED] as products for which Oura would seek a determination of noninfringement in this Investigation. *See* Ex. 15 (April 29 Interrogatory Responses) at 12-13, 79. Oura then continued to supplement its interrogatory responses to identify further relevant documents relating to [REDACTED], and noted Samsung's clear failure of proof with regard to any product besides the Oura Gen. 4 ring. *See* Ex. 19 (May 13, 2026 Supplemental Interrogatory Responses) at 13, 16, 18, 21, 23; Ex. 20 (May 27, 2026 Supplemental Interrogatory Responses) at 82-84. On May 15 and 19, 2026, Samsung deposed two Oura corporate witnesses designated on technical details of Accused Products, including [REDACTED]. Ex. 23 (Christophy Dep. Tr.) (May 15, 2026); Ex. 24 (Larson Dep. Tr.) (May 19, 2026). On May 21, 2026, Samsung deposed the senior director of sourcing, Mr. Vincent, who testified extensively about the development process of [REDACTED]

[REDACTED]

[REDACTED]. Ex. 21 (Vincent Dep. Tr.). In fact, Oura has complied with every discovery request from Samsung regarding the [REDACTED]. Not once did Samsung ever suggest that Oura's production, whether related to technical documents, native CAD files, product images, teardown images, physical samples, components, build of materials, or deposition witnesses was lacking in any capacity.³

The discovery provided by Oura (well over two hundred technical documents spanning several thousand pages, *see generally*, Background Section (detailing [REDACTED] discovery and Bates ranges) was more than sufficient for Samsung to be on notice of the relevant features of these products. In addition to the [REDACTED] CAD files produced on February 25, 2026 (OURA1468_00218121 - 137), Oura produced on the same date the [REDACTED] PCB schematic:

[REDACTED]

Ex. 41 (OURA1468_00219727 – 730); *see also* Ex. 42 (OURA1468_00219731) (similar schematics noting each “flex area” and “rigid area”). Oura also produced the PCB stackup diagram for [REDACTED] in the same production:

³ After Oura's rebuttal expert report was served, Oura also complied with two requests for native copies of certain earlier-produced documents cited by Dr. Mortazavi. The as-produced versions of these documents were neither insufficient for assessing the relevant features, nor was their existence essential in the first place, given the other document produced.

[REDACTED]

[REDACTED]

Ex. 43 (OURA1468_00219734 – 735).

On March 17, 2026, Oura produced teardown photos of both [REDACTED]:

[REDACTED]

Ex. 44 (OURA1468_00254628) ([REDACTED]); Ex. 45 (OURA1468_00254646) ([REDACTED]).

Within this production, Oura also produced the [REDACTED] PCB schematic and PCB stackup diagram:

[REDACTED]

[REDACTED]

Ex. 46 (OURA1468_00255339 – 341).

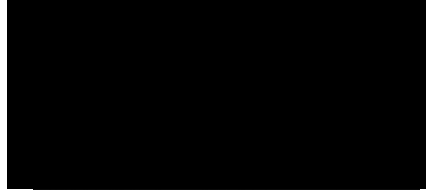
[REDACTED]

Ex. 47 (OURA1468_00255363 – 364).

Furthermore, on March 31, 2026, Oura made available for inspection physical samples of [REDACTED] (OURA_PHYS_0000006) [REDACTED] (OURA_PHYS_0000007) (Ex. 12 (1468 Inv. Production Email))—the products themselves were certainly sufficient to put Samsung on notice of these products and their properties. *See Certain Electrical Connectors and Cages*, Inv. No. 337-TA-1241, Init. Det. at 36–38 (“Luxshare’s production of samples and documentation for the Redesign Products should have been sufficient to facilitate an infringement analysis with respect to the ‘electrically lossy material’ limitation, although Amphenol did not perform this analysis”).

[REDACTED]

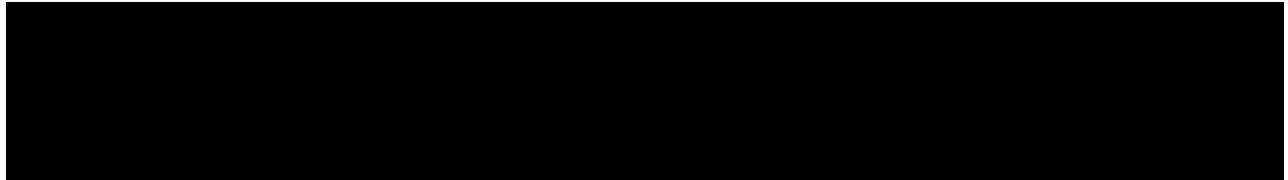
And on May 29, 2026, Oura produced extensive [REDACTED] testing photographs/documents, including cross-sectional views of their respective PCBs:



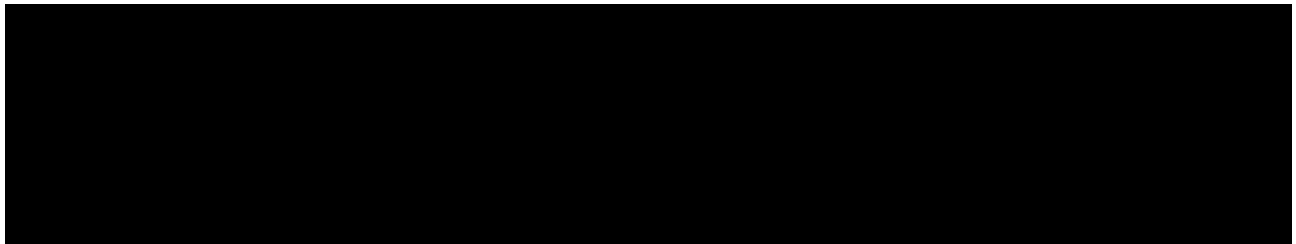
Ex. 48 (OURA1478_00014290) ([REDACTED])



Ex. 49 (OURA1478_00014241) ([REDACTED])



Ex. 50 (OURA1478_00014259) ([REDACTED])



Ex. 51 (OURA1478_00013220) ([REDACTED]). Dr. Mortazavi was then able to opine on noninfringement (Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 652-80) and answer deposition questions (Ex. 27 (Mortazavi Dep. Tr.) at 532:15-569:20) regarding [REDACTED].

Even Samsung's expert on the '789 Patent, Dr. Warren, provided testimony on redirect examination from Samsung's counsel that demonstrated that he understood the structure of [REDACTED]:



[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]

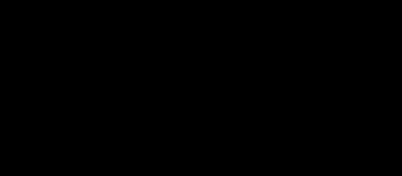
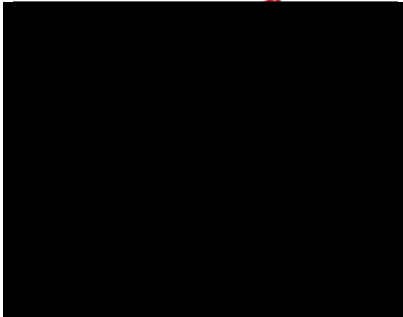
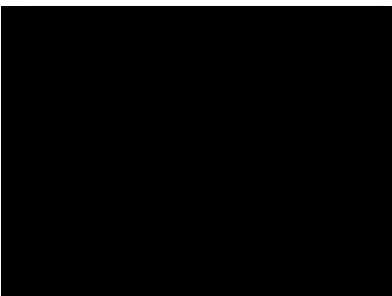
[REDACTED]

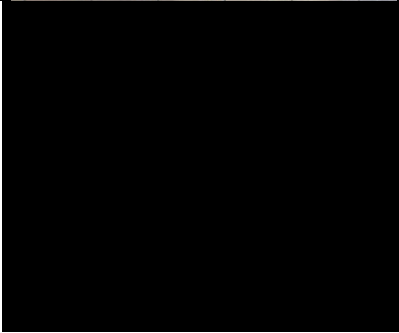
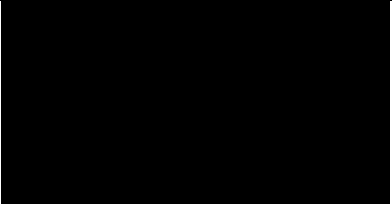
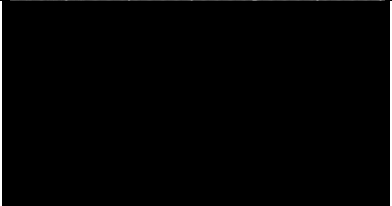
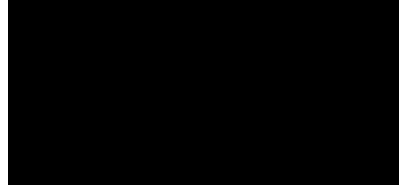
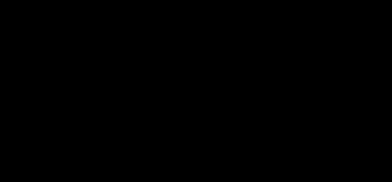
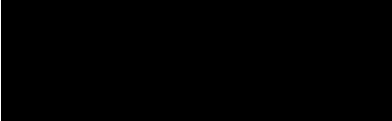
Ex. 40 (Warren Dep. Tr.) at 206:22-207:4; *see also id.* at 35:14-15, 41:5-10, 135:7-9, 200:10-215:2 (further testimony concerning [REDACTED]). This surprise testimony on redirect examination was inappropriate and should be struck—as will be set forth in a separate motion to strike—but nonetheless demonstrates that sufficient discovery had been provided.

The quantity and timing of discovery more than exceeds Commission precedent for providing sufficient discovery. *See Oligosaccharides*, Inv. No. 337-TA-1120, Comm’n Op. at 23, n. 25 (finding that respondent produced sufficient discovery when respondent produced some documentation and noting complainant “chose not to question witnesses or to test” the redesign); *Certain Mobile Devices with Multifunction Emulators*, Inv. No. 337-TA-1170, Init. Det. at 9-10 (March 16, 2021) (EDIS Doc. ID 738549) (“As an example of the flexibility on this issue, the Commission held in *Oligosaccharides* that two documents produced without explanation from the producing party during fact discovery (and one, a patent application containing no reference to the redesigned product name) qualified as ‘sufficient’ discovery under factor (4).”).

Indeed, the discovery produced was of the same type as relied upon by Samsung’s expert with regard to alleged infringement of the ’789 Patent by the Oura Gen. 4 ring. A brief comparison

of the principal evidence relied upon by Dr. Warren when opining on alleged infringement of the Oura Gen. 4 ring to the evidence produced by Oura for [REDACTED] follows:

PCB Schematics		
Oura Gen. 4 ring	[REDACTED]	[REDACTED]
 OURA1468_00219849 – 850 (Ex. 53) (relied upon at Ex. 25 (Warren Inf. Rep.) at ¶¶ 71, 104, 111, 122, 124, 128, 132)	 OURA1468_00219727 – 730 (Ex. 41)	 OURA1468_00255363 – 364 (Ex. 47)
PCB Stackup Diagram		
Oura Gen. 4 ring	[REDACTED]	[REDACTED]
 OURA1468_00219855 – 863 (Ex. 54) (relied upon at Ex. 25 (Warren Inf. Rep.) at ¶¶ 74, 107, 112, 113, 115, 137-38, and generally <i>passim</i>)	 OURA1468_00219734 – 735 (Ex. 43)	 OURA1468_00255339 – 341 (Ex. 46)

Cross-Section Teardowns		
Oura Gen. 4 ring		
	 OURA1468_00254646 (Ex. 45)	 OURA1468_00254628 (Ex. 44)
OURA1468_0028605 (Ex. 55) (relied upon at Ex. 25 (Warren Inf. Rep.) at ¶ 105)		
Scanning Electron Microscope Images		
Oura Gen. 4 ring		
	 OURA1478_00014290 (Ex. 48)	 OURA1478_00014241 (Ex. 49)
SAMS382-0034753 (Ex. 56) (relied upon at Ex. 25 (Warren Rep.) at ¶ 117, 119, 129, and generally <i>passim</i>)		

See generally, Ex. 25 (Warren Rep.) at ¶¶ 70-312 (providing infringement opinions). Indeed, Oura proactively provided extensive scanning electron microscope images of the PCBs of   (produced at OURA1478_00013076 - 4793), whereas it was Samsung's expert who obtained teardown photos for the Oura Gen. 4 ring. Oura did not withhold discovery, and should Samsung have believed that there were deficiencies, it could have raised any such concerns. *See Oligosaccharides*, Inv. No. 337-TA-1120, Comm'n Op. at 23 ("If Glycosyn and its expert deemed such evidence to be insufficient, Glycosyn could and should have taken available procedural steps,

such as a motion to reopen discovery or compel further discovery, because the burden of establishing infringement remains with Glycosyn.”).

Accordingly, the test for providing sufficient discovery has been met, and this factor also favors adjudication of the [REDACTED] products.

* * *

In sum, Oura more than meets its burden to show that the [REDACTED] are eligible for adjudication in this Investigation. Both the [REDACTED] are within the scope of this Investigation, have been imported, are sufficiently fixed in design, and discovery about the products has been provided by Oura throughout the Investigation.

V. SUMMARY DETERMINATION OF NO INFRINGEMENT OF THE '789 PATENT BY THE [REDACTED] SHOULD BE GRANTED

Given that Oura has met its burden to show that the [REDACTED] are eligible for adjudication, the burden flips to Samsung to prove infringement. *See Oligosaccharides*, Inv. No. 337-TA-1120 at 23. Samsung cannot carry its burden, and accordingly summary determination that the [REDACTED] do not infringe asserted claims 7-10 of the '789 Patent is warranted.

A. *Samsung Has Put Forth No Evidence of Infringement by [REDACTED]*

Despite Samsung bearing the proof to prove infringement, Samsung has failed to proffer any expert opinion on infringement regarding [REDACTED]—indeed, it has not even offered infringement contentions regarding [REDACTED].⁴ This fact alone is reason to grant summary judgment of noninfringement. *See Certain Crafting Machines and Components Thereof*, Inv. No. 337-TA-1426, Init. Det. at 40 (“Cricut has not provided any allegations or evidence that the redesigned products, which are ripe for adjudication, infringe the asserted patents. []

⁴ On the final day of expert discovery, Samsung attempted to have Dr. Warren put forth an infringement opinion through redirect deposition testimony. This was inappropriate, and is subject to a concurrently-filed motion to strike.

Consequently, Cricut cannot satisfy its burden of proof on infringement, and HTVRONT is entitled to summary determination of non-infringement.”); *see also Oligosaccharides*, Inv. No. 337-TA-1120, Comm’n Op. at 24 (finding no infringement after noting that “Glycosyn presented no expert evidence to establish the presence of a lacZ gene or lacZΩ fragment in the TTFL12 strain, and thereby infringement by that strain”) (citing *Centricut, LLC v. Esab Group, Inc.*, 390 F.3d 1361, 1370 (Fed. Cir. 2004) for its finding of no infringement where the patent owner did not support its contentions with an expert opinion).

B. Dr. Mortazavi’s Opinions of Noninfringement by [REDACTED] Stand Unrebutted

Additionally, as a separate basis to find noninfringement, Dr. Mortazavi has advanced unrebutted evidence and expert opinions that the [REDACTED] do not meet the asserted claims of the ’789 Patent. *See generally*, Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 651-80; *see also Certain Crafting Machines and Components Thereof*, Inv. No. 337-TA-1426, Init. Det. at 40 (noting that “undisputed evidence that the redesigned products do not infringe the asserted patents [] also justifies summary determination,” thereby recognizing that unrebutted evidence of noninfringement is merely an additional basis for noninfringement and is not a necessity).

1. Dr. Mortazavi’s Opinions of Noninfringement by [REDACTED]

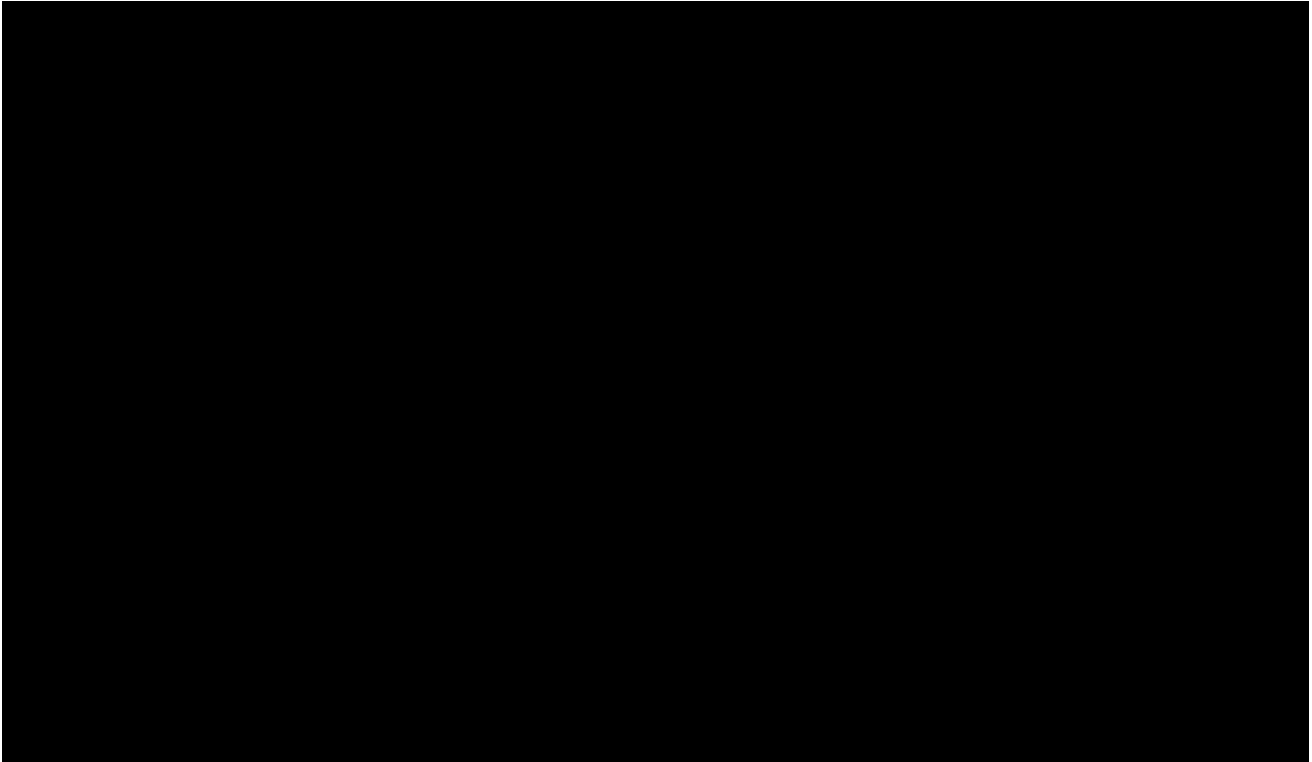
Dr. Mortazavi was able to use the produced discovery to show that [REDACTED] does not meet claim limitations [7.C][vi][a], [7.C][vi][c], and [7.C][vi][e]. *See* Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 655-71.

For claim limitation [7.C][vi][a], which requires a “second material different from the first material,” wherein the first material insulates the rigid PCB portions as a first exterior surface and the second material insulates the flexible PCB portions as a first exterior surface, Dr. Mortazavi showed that the respective first exterior surfaces of the rigid and flexible PCB portions is “[REDACTED]

[REDACTED]

[REDACTED]

[REDACTED]’ *Id.* at ¶¶ 656-59. Dr. Mortazavi annotated a [REDACTED] PCB stackup diagram to show this:



Id. at ¶ 658 (annotating a color version of OURA1478_00006316 to show [REDACTED]
[REDACTED]).⁵

This can also be seen in, for example, the [REDACTED] PCB teardown photos, which show [REDACTED]
[REDACTED]:

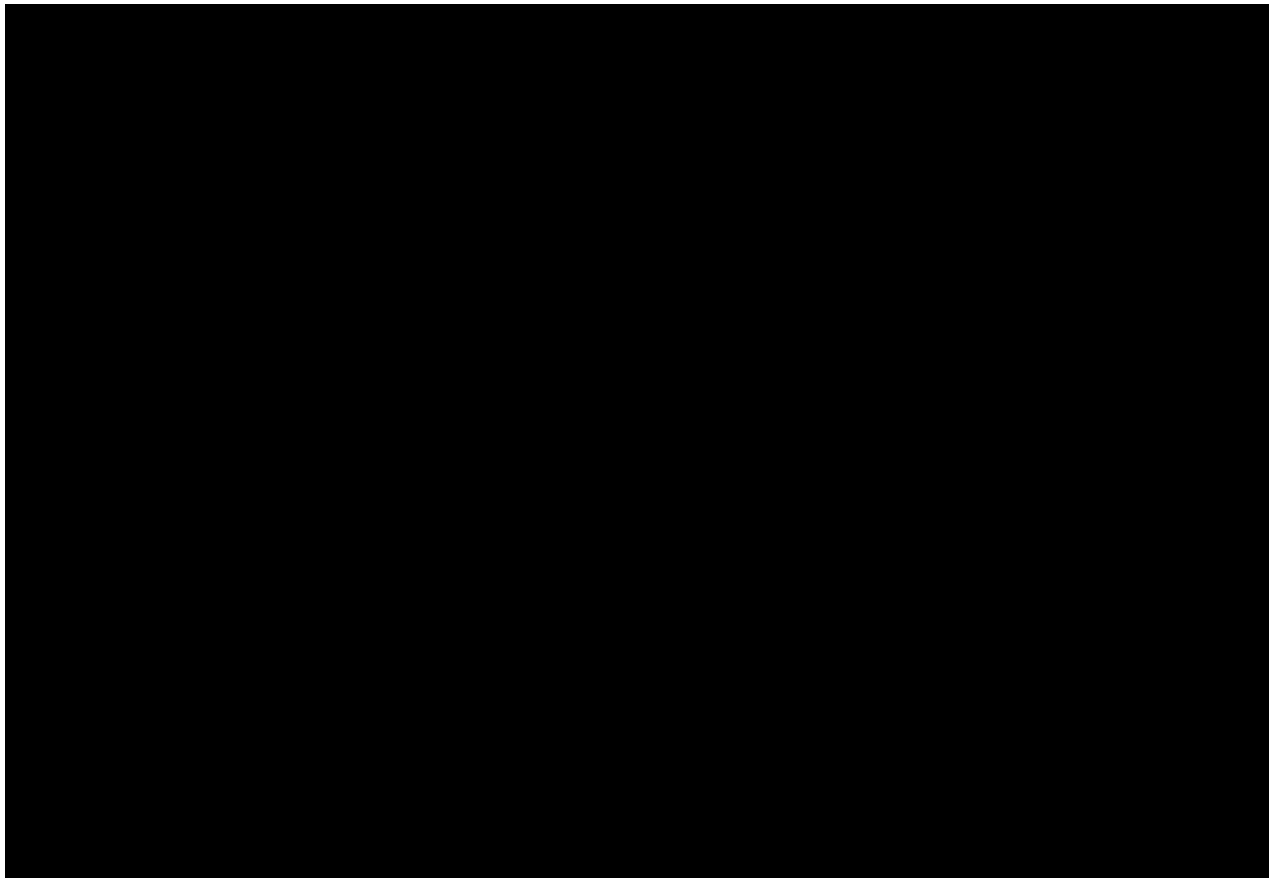
⁵ OURA1478_00006316 (Ex. 57) is substantively the same as OURA1468_00219734 (Ex. 43), produced February 25, 2026 and identified in Oura’s April 29, 2026 Interrogatory Responses regarding “components of the printed circuit board (PCB).” *See* Ex. 15 at 15-17. OURA1478_00006316 was also identified as relevant to this interrogatory (Rog. 3) on May 13, 2026. Ex. 19 at 18.

[REDACTED]

Ex. 50 (OURA1478_00014259) (zoomed in). This alone is sufficient for noninfringement by

[REDACTED].

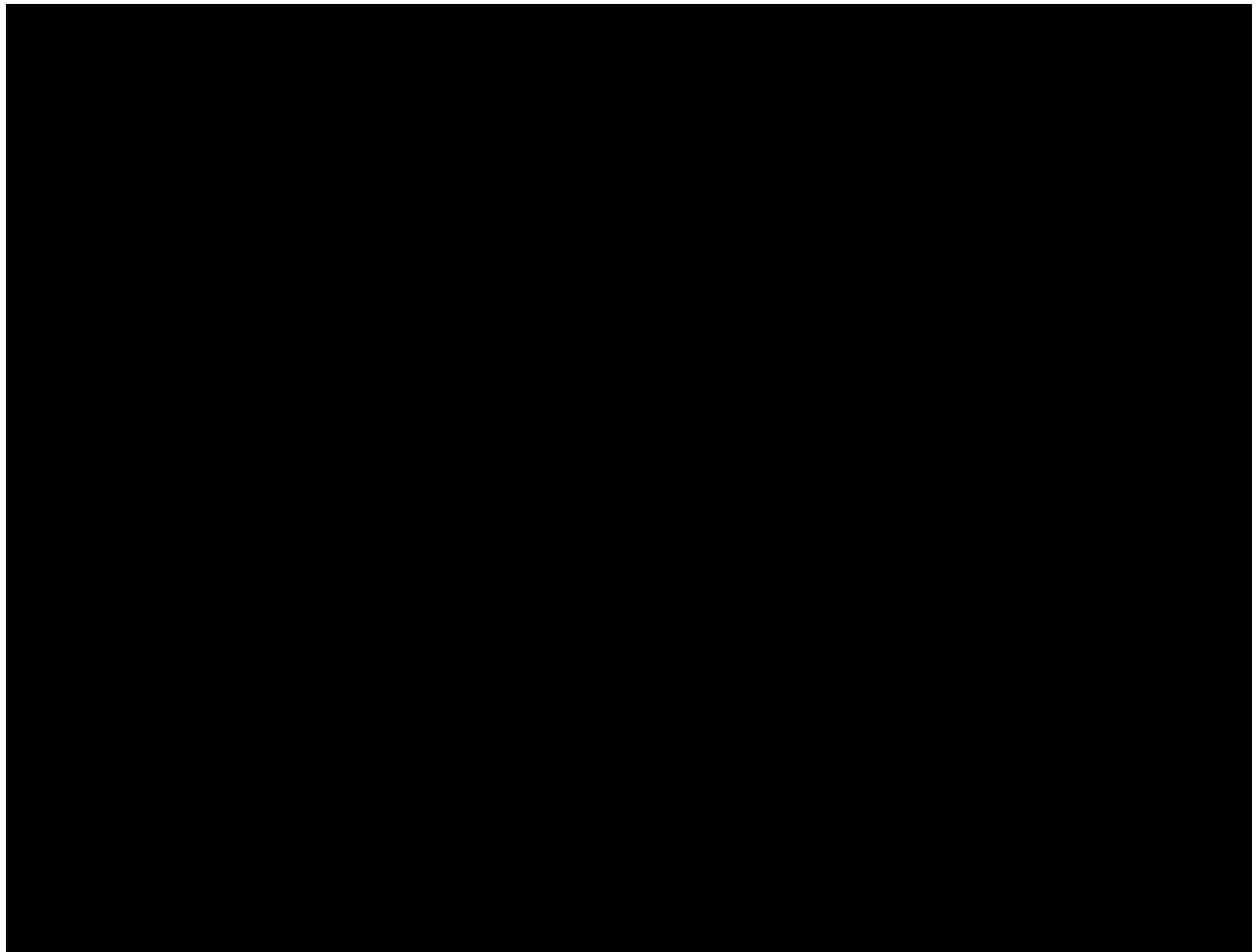
For claim limitation [7.C][vi][c], which requires that the second insulating layer for the flexible PCB portions have a “second thickness thinner than the first thickness” of the second insulating layer for the rigid PCB portions, Dr. Mortazavi showed that [REDACTED]. See Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 662-64. Dr. Mortazavi again showed this with an annotated version of the [REDACTED] PCB stackup diagram:



[REDACTED]

Id. at ¶ 663 (annotating a color version of OURA1478_00006316 to show [REDACTED]
[REDACTED]); Ex. 50 (OURA1478_00014259) (zoomed in image shown, *supra*) (showing [REDACTED]
[REDACTED]). This too is sufficient for noninfringement.

For claim limitation [7.C][vi][e], which requires a “second material” in the third insulating layer of the flexible PCB portions that is different from the “first material” in the third insulating layer of the rigid PCB portions, Dr. Mortazavi showed that [REDACTED]
[REDACTED] Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 667-70. Dr. Mortazavi again showed this with an annotated version of the [REDACTED] PCB stackup diagram:



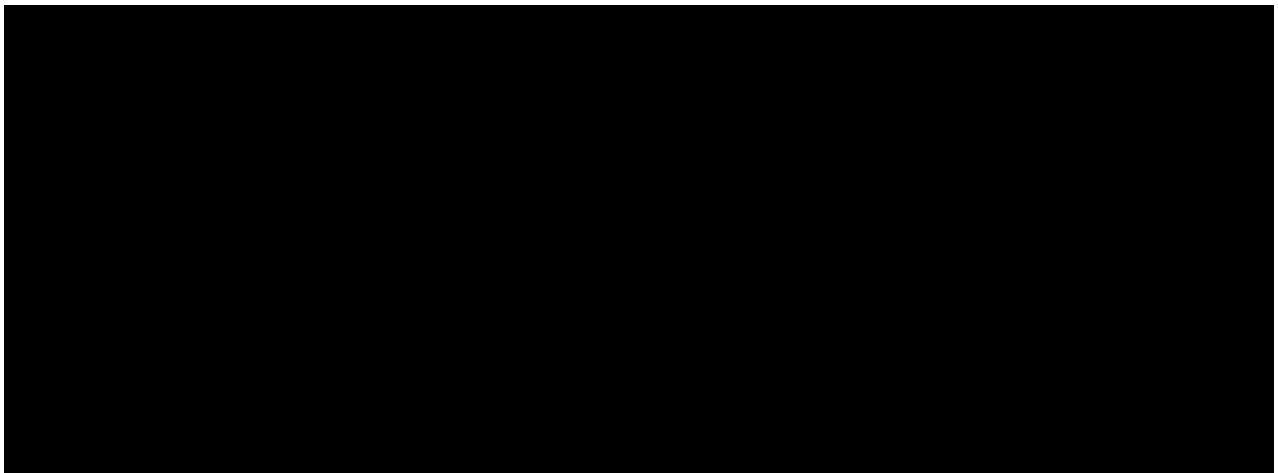
[REDACTED]

Id. at ¶ 669 (annotating a color version of OURA1478_00006316). This is a third reason for noninfringement by the [REDACTED] ring.

2. *Dr. Mortazavi's Opinions of Noninfringement by [REDACTED]*

Dr. Mortazavi was able to use the produced discovery to show that [REDACTED] does not meet claim limitations [7.C][i]-[iv] and [7.C][vi][c]. *See* Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 672-80.

For claim limitations [7.C][i]-[iv], which collectively require “a combination of connected and alternating rigid and flexible PCB portions that form a rigid-flexible-rigid-flexible structure,” Dr. Mortazavi showed [REDACTED]. *Id.* at ¶¶ 673-74. Dr. Mortazavi annotated a [REDACTED] PCB stackup diagram to show this:

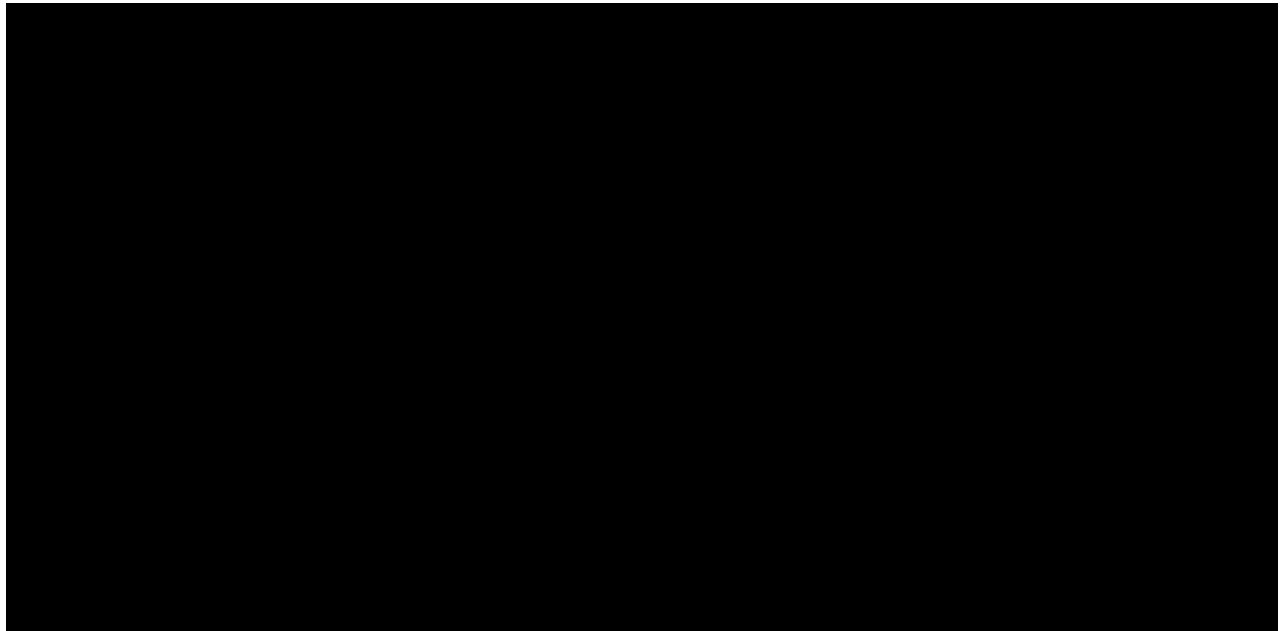


Id. at ¶ 673 (annotating OURA1468_00255354).⁶ This alone is sufficient for noninfringement by [REDACTED].

For claim limitation [7.C][vi][c], which requires that the second insulating layer for the flexible PCB portions have a “second thickness thinner than the first thickness” of the second insulating layer for the rigid PCB portions, Dr. Mortazavi showed [REDACTED].

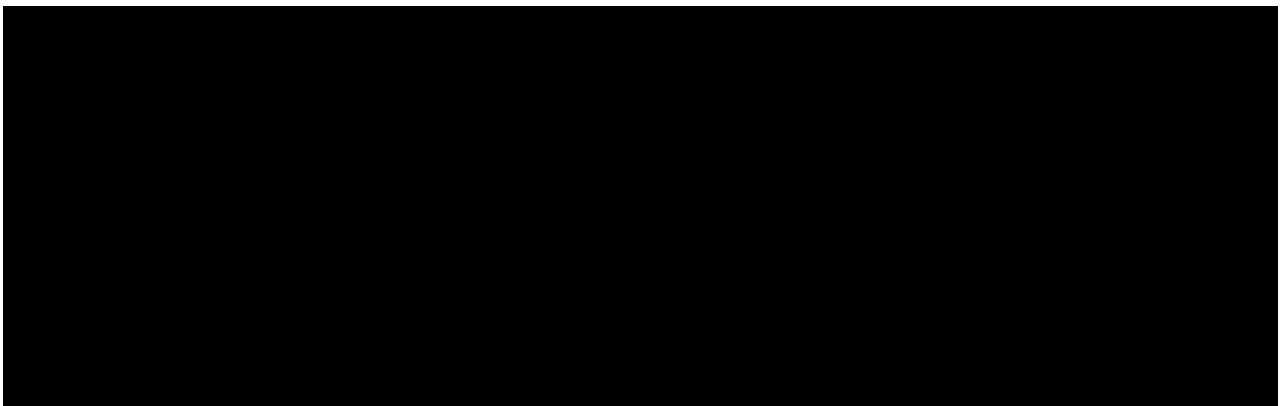
⁶ OURA1468_00255354 was specifically identified in Oura's April 29, 2026 interrogatory responses, including in response to Samsung's interrogatory about PCB structure. *See* Ex. 15 (April 29 Interrogatory Responses) at 13, 17.

See Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 677-79. Dr. Mortazavi showed this with an annotated version of the [REDACTED] PCB stackup diagram:



Id. at ¶ 678 (annotating OURA1468_00255354). The [REDACTED]

[REDACTED]



Ex. 58 (OURA1478_00013242). This too is sufficient for noninfringement by [REDACTED].

* * *

Thus, not only does Samsung offer no evidence with which to meet its burden of proof, the evidence actually shows that the [REDACTED] do not meet the limitations of any asserted claim of the '789 Patent. The unrebutted opinion by Dr. Mortazavi combined with the

[REDACTED]

evidence that he relies upon accordingly provides a second basis for granting summary determination that the [REDACTED] do not infringe asserted claims 7-10 of the '789 Patent.

VI. CONCLUSION

Samsung cannot simply refuse to have certain products adjudicated within an Investigation. As has been noted in nearly every one of the decisions cited above, the Commission has a well-reasoned “policy in favor of adjudicating redesigns to prevent subsequent and potentially burdensome proceedings that could have been resolved in the first instance in the original Commission investigation.” *See Oligosaccharides*, Inv. No. 337-TA-1120, Comm’n Op. at 18. Here, Oura’s next generation [REDACTED] were disclosed to Samsung during discovery and Oura stated in its contention in April that these products did not infringe the asserted patents; in response, Samsung took minor discovery regarding these products but otherwise put forth no allegations of infringement. There can thus be no genuine dispute that Samsung cannot carry its burden to prove infringement, and summary determination that the [REDACTED] do not infringe asserted claims 7-10 of the '789 Patent is therefore warranted.

Ground Rule 1.7 Certification

Pursuant to Ground Rule 1.7, the undersigned certifies that this response complies with the word limit requirements of Ground Rule 5.6.2.1 (under 10,000 words). Specifically, the word count of the word-processing system used to prepare this filing is 7,697 words.

Dated: July 13, 2026

Respectfully submitted,

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**U.S. INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In The Matter Of

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

**RESPONDENTS OURARING INC. AND OURA HEALTH OY'S STATEMENT OF
MATERIAL FACTS IN SUPPORT OF MOTION FOR SUMMARY DETERMINATION
OF NON-INFRINGEMENT BY OURA'S ALTERNATIVE PRODUCTS**

Pursuant to Ground Rule 5.6.2.1, Respondents Ouraring Inc. and Oura Health Oy (collectively, "Oura") hereby provides this statement of the material facts with which it contends there is no genuine issue, and which entitle Respondents to a summary determination as a matter of law that Oura's [REDACTED] do not infringe asserted claims 7-10 of U.S. Patent No. 10,978,789.



Item #	Factual Statement	Citation	Disputed?	Factual Rebuttal	Citation
Background					
1	On December 12, 2025, Samsung filed a complaint seeking 337 Investigation against Oura by alleging that Oura infringes four asserted patents: U.S. Patents Nos. 10,642,359 (“’359 patent”), 10,945,677 (“’677 patent”), 10,231,675 (“’675 patent”), and the ’789 patent (collectively, “Asserted Patents”).	Original Complaint ¶1 (EDIS Doc. ID 866474)			
2	In its complaint, Samsung identified the Accused Products as “certain wearable devices, such as smart rings.”	Original Complaint ¶3 (EDIS Doc. ID 866474)			
3	The infringement claim charts accompanying Samsung’s complaint alleged infringement of only Oura Ring Gen. 4.	Original Complaint at Exs. 13-16 (EDIS Doc. ID 866474)			
4	On January 14, 2026, the Commission instituted the Investigation defining the scope of the investigation as “wearable devices, specifically wearable biosignal interfaces and an operation method thereof as well as wearable electronic devices for detecting biometric information.”	Notice of Institution of Investigation (Jan. 14, 2026) (EDIS Doc. ID 868875)			
5	Prior to institution of this Investigation, Oura and Samsung were engaged in another ITC Investigation, <i>Certain Smart Wearable Devices, Systems, and Components Thereof</i> , Inv. No. 337-TA-1468 (“1468 Investigation”).	Ex. 2 (Oura Complaint, <i>Certain Smart Wearable Devices, Systems, and Components Thereof</i> , ITC Inv. No. 337-TA-1468)			

6	In the complaint for the 1468 Investigation, Ora identified Ora Ring Generation (“Gen.”) 3, Ora Ring Gen. 4, and “[REDACTED]” as Ora’s domestic industry products.	Ex. 2 (Ora Complaint, Certain Smart Wearable Devices, Systems, and Components Thereof, ITC Inv. No. 337-TA-1468), ¶ 5			
7	Samsung and Ora agreed to a discovery cross-use agreement between the 1468 and 1478 Investigations that obviated the need for re-production of the 1468 Investigation materials in this Investigation.	Ex. 3 (February 12, 2026 Correspondence between Ora and Samsung counsel); Joint Discovery Stipulation ¶ 29 (EDIS Doc. ID 878423)			
8	On January 26, 2026, Ora produced 99 technical documents comprising photographs of [REDACTED] (OURA1468_00023843 - 847).	Ex. 4 (Jan. 26, 2026 Ora Production Letter (1468 Investigation))			
9	On January 26, 2026, Ora produced [REDACTED] hardware tech specs (OURA1468_00023857 - 865).	Ex. 4 (Jan. 26, 2026 Ora Production Letter (1468 Investigation))			
10	On January 26, 2026, Ora produced [REDACTED] testing and teardown photographs (OURA1468_00028664 - 760).	Ex. 4 (Jan. 26, 2026 Ora Production Letter (1468 Investigation))			
11	On January 26, 2026, Ora produced a [REDACTED] design overview (OURA1468_00056564 - 575).	Ex. 4 (Jan. 26, 2026 Ora Production Letter (1468 Investigation))			
12	On February 9, 2026, Ora identified [REDACTED] as domestic industry products in its initial burden domestic industry contentions that Ora would rely on in Investigation No. 337-TA-1468 and served charts that included images of the physical samples, teardowns, and design files for [REDACTED].	Ex. 5 (Ora Feb. 9, 2026 First Supplemental Interrogatory Responses (1468 Investigation) (Excerpts)) at 10; Ex. 6 (Ora Feb. 9, 2026 Initial Burden DI Contentions: [REDACTED] – U.S. Patent No. 11,868,178 (1468 Investigation));			



		Ex. 7 (Oura Feb. 9, 2026 Initial Burden DI Contentions – [REDACTED] – U.S. Patent No. 11,868,178 (1468 Investigation))			
13	On February 11, 2026, in the 1468 Investigation, Oura made the source code for all its domestic industry products available for inspection by Samsung, including firmware and application code related to [REDACTED].	Ex. 8 (Feb. 11, 2026 Oura Source Code Production Email (1468 Investigation))			
14	On February 25, 2026, Oura produced 28 technical documents comprising [REDACTED] CAD files (OURA1468_00218121 - 131).	Ex. 9 (Feb. 25, 2026 Oura Production Letter (1468 Investigation))			
15	On February 25, 2026, Oura produced [REDACTED] CAD files (OURA1468_00218132 - 137).	Ex. 9 (Feb. 25, 2026 Oura Production Letter (1468 Investigation))			
16	On February 25, 2026, Oura produced [REDACTED] bill of materials, design schematics, and a PWB stackup specification (OURA1468_00219723 - 738).	Ex. 9 (Feb. 25, 2026 Oura Production Letter (1468 Investigation))			
17	On March 17, 2026, Oura produced 85 technical documents comprising [REDACTED] teardown documents (OURA1468_00254628 - 645).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
18	On March 17, 2026, Oura produced [REDACTED] teardown documents (OURA1468_00254646 - 659).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
19	On March 17, 2026, Oura produced [REDACTED] bill of materials (OURA1468_00254743 - 5091).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			

20	On March 17, 2026, Oura produced [REDACTED] feature and design documents (OURA1468_00255092 - 338).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
21	On March 17, 2026, Oura produced [REDACTED] schematic design documents and PCB stackup (OURA1468_00255339 - 375).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
22	On March 17, 2026, Oura produced [REDACTED] photographs (OURA1468_00255376 - 394).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
23	On March 17, 2026, Oura produced a [REDACTED] sensor overview (OURA1468_00255395).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
24	On March 17, 2026, Oura produced [REDACTED] technical features specifications (OURA1468_00255396 - 447).	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation))			
25	On March 26, 2026, Oura produced a product overview of [REDACTED] with a technical features specification (OURA1468_00256348 - 376).	Ex. 11 (Mar. 26, 2026 Oura Production Letter (1468 Investigation))			
26	On March 31, 2026, Oura made available for inspection physical samples of the [REDACTED] (OURA_PHYS_0000006) and the [REDACTED] (OURA_PHYS_0000007), including both new and teardown samples.	Ex. 12 (Mar. 31, 2026 Oura Production Email (1468 Investigation))			
27	In March 2026, in connection with the 1468 Investigation, Samsung deposed Miguel Christophy.	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026))			

28	Miguel Christophy is Vice President of Hardware Engineering at Oura, whose primary focus is the development and production of [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 28:13-20, 20:8-11, 26:1-27:16			
29	At his March 2026 deposition, Miguel Christophy confirmed that [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 61:10-15			
30	At his March 2026 deposition, Miguel Christophy confirmed that [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 64:25-65:3			
31	At his March 2026 deposition, Miguel Christophy explained the hardware and sensors of [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 62:24-64:19			
32	At his March 2026 deposition, Miguel Christophy discussed the [REDACTED] hardware and design.	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 65:10-72:10			
33	At his March 2026 deposition, Miguel Christophy testified regarding the CAD and hardware of [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 131:4-146:15.			
34	At his March 2026 deposition, Miguel Christophy discussed the manufacturing process of [REDACTED] Ring.	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 146:16-150:7			
35	At his March 2026 deposition, Miguel Christophy testified regarding the PCB structure of [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 150:8-15			
36	At his March 2026 deposition, Miguel Christophy discussed the plan of record regarding [REDACTED].	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 154:9-155:15			
37	At his March 2026 deposition, Miguel Christophy testified about the physical	Ex. 13 (M. Christophy Dep. Tr. (1468 Investigation) (Mar. 20, 2026)) at 157:15-178:17			

	design of the [REDACTED], including hardware components.				
38	On March 16, 2026, Samsung served its initial infringement contentions in the 1478 Investigation.	Ex. 14 (Samsung Initial Infringement Contentions (1478 Investigation) (Mar. 16, 2026) (Excerpts))			
39	Samsung's initial infringement contentions in the 1478 Investigation accused only Oura Ring Gen. 4 of infringing the Asserted Patents, stating: "Samsung accuses the following wearable devices and corresponding applications of infringement: Oura Ring 4 and its corresponding software application, including the Oura App."	Ex. 14 (Samsung Initial Infringement Contentions (1478 Investigation) (Mar. 16, 2026) (Excerpts)) at 1			
40	Samsung's initial infringement contentions in the 1478 Investigation made no mention of either [REDACTED].	Ex. 14 (Samsung Initial Infringement Contentions (1478 Investigation) (Mar. 16, 2026) (Excerpts))			
41	On April 29, 2026, Oura amended its interrogatory responses to affirmatively identify the [REDACTED] rings as "Accused Products."	Ex. 15 (Excerpts, Oura __ Suppl. Response to Samsung First Set of Interrogatories (Apr. 29, 2026)) at 12-13			
42	On April 29, 2026, Oura served an interrogatory response that stated the [REDACTED] "do not directly infringe, induce infringement of, or contribute to the infringement of the '359 Patent, the '677 Patent, the '675 Patent, and the '789 Patent."	Ex. 15 (Oura Third Suppl. Responses to Samsung's First Set of Interrogatories (1478 Investigation) (Apr. 29, 2026) (excerpts)) at 79			
43	In its April 29, 2026 interrogatory response regarding the identity of the Accused Products, Oura identified the technical documents previously produced in the	Ex. 15 (Oura Third Suppl. Responses to Samsung's First Set of Interrogatories (1478 Investigation) (Apr. 29, 2026) (excerpts)) at 13			

	1468 Investigation at OURA1468_00254743 – OURA1468_00255395, OURA1468_00255396 – OURA1468_00255447, and OURA1468_00256348 - OURA1468_00256376 as being responsive.				
44	In its April 29, 2026 interrogatory response regarding the identity of the Accused Products, Oura stated that “source code for the [REDACTED] rings” for both firmware and Oura Application code continued to be available for inspection on source code laptop since February 11 2026.”	Ex. 15 (Oura Third Suppl. Responses to Samsung’s First Set of Interrogatories (1478 Investigation) (Apr. 29, 2026) (excerpts)) at 12			
45	In its April 29, 2026 interrogatory responses, Oura identified technical documents related to [REDACTED] as responsive to Interrogatory Nos. 2-5, such as OURA1468_00219723 – 738.	Ex. 15 (Oura Third Suppl. Responses to Samsung’s First Set of Interrogatories (1478 Investigation) (Apr. 29, 2026) (excerpts)) at 15, 17, 19-21.			
46	On May 5, 2026, Oura produced a [REDACTED] BOM (OURA1478_00004838).	Ex. 16 (May 5, 2026 Oura Production Letter (1478 Investigation))			
47	On May 5, 2026, Oura produced 14 datasheets for [REDACTED] components (OURA1478_00004839 - 5180).	Ex. 16 (May 5, 2026 Oura Production Letter (1478 Investigation))			
48	On May 5, 2026, Oura produced 48 [REDACTED] BOMs (OURA1478_00006234 - 281).	Ex. 16 (May 5, 2026 Oura Production Letter (1478 Investigation))			
49	On May 5, 2026, Oura produced 7 [REDACTED] component specifications and a PCB stack-up diagram (OURA1478_00006283 - 316).	Ex. 16 (May 5, 2026 Oura Production Letter (1478 Investigation))			

50	On May 29, 2026, Oura produced [REDACTED] testing photographs/documents, including cross-sectional views of the PCB (OURA1478_00013076 - 4258).	Ex. 17 (May 29, 2026 Oura Production Letter (1478 Investigation))			
51	On May 29, 2026, Oura produced [REDACTED] photographs/testing documents, including cross-sectional views of the PCB (OURA1478_00014259 - 793).	Ex. 17 (May 29, 2026 Oura Production Letter (1478 Investigation))			
52	On May 29, 2026, Oura made available for inspection the physical samples of the rings, including [REDACTED], along with the new and tested PCB samples.	Ex. 18 (May 29, 2026 Oura Production Email (1478 Investigation))			
53	On May 13, 2026, Oura served supplemental responses to Interrogatory Nos. 1-5 that identified OURA1478_00004838 – 5180, OURA1478_00006234 – 281, OURA1478_00006283 – 316 as responsive documents.	Ex. 19 (Oura Fifth Suppl. Responses to Samsung’s First Set of Interrogatories (1478 Investigation) (May 13, 2026) (Excerpts)) at 13, 16, 18, 21, 23.			
54	In its May 27, 2026 final interrogatory responses, Oura stated Samsung’s failure to advance any infringement contentions for “any other Oura product” and stated that “it has produced documents from which Complainants may ascertain relevant information responsive to this Interrogatory, including without limitation those cited in Interrogatory Nos. 1-5.”	Ex. 20 (Oura Sixth Suppl. Responses to Samsung’s First Set of Interrogatories (1478 Investigation) (May 27, 2026) (Excerpts)) at 82-84			
55	In connection with this Investigation, Samsung deposed Mr. Christopher James Vincent, Senior Director of Sourcing at Oura.	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026))			

56	At his deposition in this Investigation, Mr. Vincent provided the following testimony: [REDACTED] [REDACTED] [...] [REDACTED] [REDACTED]	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 21:13-20, 57:6-10.			
57	At his deposition in this Investigation, Mr. Vincent testified that “[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 63:18-64:13			
58	In this Investigation, Oura designated three technical corporate witnesses to testify on the Accused Products: (1) Mr. Christophy, VP of Hardware Engineering at Oura, related to hardware of all accused products (2) Mr. Michael Larson, Director of Firmware, to testify on firmware of all accused products, and (3) Mr. Simon Ratner, VP of Software Engineering, to	Ex. 22 (2026.05.01 - Oura 30b6 Deposition Designations)			



	testify regarding software or application of all accused products.				
59	At his deposition in this Investigation, Mr. Christophy testified that “yeah, I’m prepared to answer regarding Gen. 1, Oura 4, as we call it, [REDACTED].”	Ex. 23 (M. Christophy Dep. Tr. (1478 Investigation) (May 15, 2026) (Excerpt)) at 10:16-11:21			
60	During the deposition of Mr. Larson, the following exchange occurred: [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]	Ex. 24 (M. Larson Dep. Tr. (1478 Investigation) (May 19, 2026 (Excerpts)) at 11:22-12:12:10.			
61	During the deposition of Mr. Larson, Mr. Larson answered questions regarding [REDACTED].	Ex. 24 (M. Larson Dep. Tr. (1478 Investigation) (May 19, 2026 (Excerpts)) at 26:15-27:5, 44:9-45:6, 61:6-9			
62	[REDACTED] launched as Oura Ring Gen. 5 on May 28, 2026 with consumers receiving their ordered shipments starting June 4, 2026 from the manufactured inventory.	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 63:18-64:6.			
63	Fact discovery in this Investigation ended on May 29, 2026.	Order No. 8 at 2.			

64	On June 10, 2026, Samsung's technical expert on infringement of the '789 Patent, Dr. Steven Warren, submitted his opening expert report.	Ex. 25 (Excerpts, Expert Report of Steven Warren, Ph.D. Concerning Infringement and Domestic Industry Technical Prong Issues related to U.S. Patent No. 10,978,789 (June 10, 2026)) ("Warren Inf. Rep.")			
65	In his opening expert report regarding the '789 Patent, [REDACTED]	Ex. 25 (Warren Inf. Rep.) ¶3			
66	In his opening expert report regarding the '789 Patent, [REDACTED]	Ex. 25 (Warren Inf. Rep.)			
67	On June 26, 2026, Oura's technical expert on non-infringement of the Asserted Patents submitted his rebuttal expert report.	Ex. 26 (Excerpts, Rebuttal Expert Report of Dr. Bobak Jack Mortazavi Regarding Noninfringement of U.S. Patent Nos. 10,642,359, 10,945,677, 10,231,675, and 10,978,789 (June 26, 2026))("Mortazavi Reb. Rep.")			
68	In his rebuttal expert report on non-infringement, Dr. Mortazavi relied on the documents produced by Oura to show that [REDACTED] do not meet the asserted claims of the '789 Patent.	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 651-80			
69	In his rebuttal expert report on non-infringement, Dr. Mortazavi used produced discovery to offer opinions that	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 655-71.			

	██████ does not meet claim limitations [7.C][vi][a], [7.C][vi][c], and [7.C][vi][e].				
70	In his rebuttal expert report on non-infringement, Dr. Mortazavi used produced discovery to offer opinions that ██████ does not meet claim limitations [7.C][i]-[iv] and [7.C][vi][c].	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 672-80			
71	Samsung deposed Dr. Mortazavi regarding his noninfringement opinions concerning ██████ on July 8, 2026.	Ex. 27 (. Mortazavi Dep. Tr. (1478 Investigation) (July 8, 2026) (Excerpts)) at 532:15-569:20.			
Eligibility for Adjudication					
72	Samsung's First Amended Complaint states that "[t]he Accused Products include, but are not limited to, Respondents' Oura Ring products."	Samsung Amended Complaint Under Section 337 of the Tariff Act of 1930, at ¶ 17.			
73	The ██████ is a wearable electronic device for detecting biometric information.	Ex. 30 (OURA1468_00255396); Ex. 33 (OURA1468_00023857) at OURA1468_00023858			
74	The ██████ is a wearable electronic device for detecting biometric information.	Ex. 31 (OURA1468_00255092) at OURA1468_00255092, OURA1468_00255137			
75	The Oura Gen. 4 ring, ██████ are the same type of product.	Ex. 31 (OURA1468_00255092) at OURA1468_00255097			
76	Oura and Samsung have stipulated that ██████ are "subject articles."	Ex. 35 (Excerpts, Joint Stipulation Regarding Importation and Inventory Between Complainants and Respondents) at Ex. A			
77	The ██████ rings have been imported.	Ex. 18 (May 29, 2026 Oura Production Email (1478 Investigation));			

		Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 22:3-6			
78	Mr. Vincent testified that “[REDACTED]”	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 22:3-6			
79	In the Importation Stipulation, the parties stipulated as follows: “[REDACTED]”	Ex. 35 (Excerpts, Joint Stipulation Regarding Importation and Inventory Between Complainants and Respondents) at ¶3			
80	In Confidential Exhibit A to the Importation Stipulation, [REDACTED] are listed as Subject Articles.	Ex. 35 (Excerpts, Joint Stipulation Regarding Importation and Inventory Between Complainants and Respondents) at Ex. A			
81	The Importation Stipulation defines “Subject Articles” as “wearable devices that allegedly infringe U.S. Patent Nos. 10,642,359 (‘the ’359 Patent’), 10,231,675 (‘the ’675 Patent’), 10,945,677 (‘the ’677 Patent’), and 10,978,789 (‘the ’789 Patent’).”	Ex. 35 (Excerpts, Joint Stipulation Regarding Importation and Inventory Between Complainants and Respondents) at 1			
82	Mr. Vincent testified at his May 21, 2026 deposition that [REDACTED]	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 21:13-20, 57:6-10, 63:18-64:6			
83	[REDACTED]	Ex. 36 (OURA1468_00255396)			

84	[REDACTED]	Ex. 37 (OURA1468_00256348)			
85	[REDACTED]	Ex. 37 (OURA1468_00256348)			
86	Mr. Vincent testified at his May 21, 2026 deposition that Oura had reached [REDACTED].	Ex. 21 (C.J. Vincent Dep. Tr. (1478 Investigation) (May 21, 2026)) at 57:6-10; 64:7-13			
87	[REDACTED]	Ex 38 (OURA1468_00255163) at -164, -165, -180; Ex. 39 (OURA1468_00255190) at -210			
88	[REDACTED]	Ex 38 (OURA1468_00255163) at -164, -165, -180			
89	[REDACTED]	Ex. 39 (OURA1468_00255190) at -210			
90	On March 31, 2026, Oura made available for inspection physical samples of the [REDACTED] (OURA_PHYS_0000006) and the [REDACTED] (OURA_PHYS_0000007).	Ex. 12 (Mar. 31, 2026 Oura Production Email (1468 Investigation))			
91	At his March 2026 deposition, Miguel Christophy testified as follows: [REDACTED]	Ex. 23 (M. Christophy Dep. Tr. (1478 Investigation) (May 15, 2026) (Excerpt)) at 150:8-15, 168:24-169:13			



	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]				
92	On February 25, 2026, Oura produced OURA1468_00219727 – 730, which is the [REDACTED] PCB schematic.	Ex. 9 (Feb. 25, 2026 Oura Production Letter (1468 Investigation)); Ex. 41 (OURA1468_00219727 – 730)			
93	On February 25, 2026, Oura produced OURA1468_00219731, which is a [REDACTED] PCB schematic that notes [REDACTED]	Ex. 9 (Feb. 25, 2026 Oura Production Letter (1468 Investigation)); Ex. 42 (OURA1468_00219731)			
94	On February 25, 2026, Oura produced OURA1468_00219734 – 735, which is the PCB stackup diagram for [REDACTED]	Ex. 9 (Feb. 25, 2026 Oura Production Letter (1468 Investigation)); Ex. 43 (OURA1468_00219734 – 735)			

95	On March 17, 2026, Oura produced teardown photos of [REDACTED] including OURA1468_00254628.	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation)); Ex. 44 (OURA1468_00254628)			
96	On March 17, 2026, Oura produced teardown photos of [REDACTED] including OURA1468_00254646.	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation)); Ex. 45 (OURA1468_00254646)			
97	On March 17, 2026, Oura produced OURA1468_00255363 – 364, which is the [REDACTED] PCB schematic.	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation)); Ex. 47 (OURA1468_00255363 – 364)			
98	On March 17, 2026, Oura produced OURA1468_00255339 – 341, which is the [REDACTED] PCB stackup diagram.	Ex. 10 (Mar. 17, 2026 Oura Production Letter (1468 Investigation)); Ex. 46 (OURA1468_00255339 – 341)			
99	On May 29, 2026, Oura produced [REDACTED] and testing photographs/documents, including cross-sectional views of their respective PCBs, such as OURA1478_00014290 and OURA1478_00014259.	Ex. 17 (May 29, 2026 Oura Production Letter (1478 Investigation)); Ex. 48 (OURA1478_00014290); Ex. 50 (OURA1478_00014259)			
100	On May 29, 2026, Oura produced [REDACTED] and testing photographs/documents, including cross-sectional views of their respective PCBs, such as OURA1478_00014241 and OURA1478_00013220.	Ex. 17 (May 29, 2026 Oura Production Letter (1478 Investigation)); Ex. 49 (OURA1478_00014241); Ex. 51 (OURA1478_00013220)			

101	Samsung's expert on the '789 Patent, Dr. Warren, [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]	Ex. 40 (S. Warren Dep. Tr. (1478 Investigation (July 9, 2026) (Excerpts)) at 206:22-207:4; <i>see also id.</i> at 35:14-15, 41:5-10, 135:7-9, 200:10-215:2			
102	Samsung's expert, Dr. Warren, [REDACTED] [REDACTED]	Ex. 25 (Warren Inf. Rep.) ¶¶ 71, 104, 111, 122, 124, 128, 132; Ex. 53 (OURA1468_00219849 – 850)			
103	Oura produced PCB schematics for the [REDACTED] rings.	Ex. 41 (OURA1468_00219727 – 730); Ex. 47 (OURA1468_00255363 – 364)			

104	Samsung's expert, Dr. Warren, [REDACTED]	Ex. 25 (Warren Inf. Rep.) at ¶¶ 74, 107, 112, 113, 115, 137-38, and generally <i>passim</i>); Ex. 54 (OURA1468_00219855 – 863)			
105	Oura produced PCB Stackup Diagrams for the [REDACTED] rings.	Ex. 43 (OURA1468_00219734 – 735); Ex. 46 (OURA1468_00255339 – 341)			
106	Samsung's expert, Dr. Warren, [REDACTED]	Ex. 25 (Warren Inf. Rep.) at ¶ 105; Ex. 55 (OURA1468_0028605)			
107	Oura produced cross-section teardowns for the [REDACTED] rings.	Ex. 45 (OURA1468_00254646); Ex. 44 (OURA1468_00254628)			
108	Samsung's expert, Dr. Warren, relied on scanning electronic microscope images, such as SAMS382-0034753, in support of his opinions regarding infringement of the '789 Patent by the Oura Gen. 4 ring.	Ex. 25 (Warren Inf. Rep.) at ¶¶ 117, 119, 129, and generally <i>passim</i>); Ex. 55 (OURA1468_0028605)			
109	Oura produced scanning electronic microscope images for the [REDACTED] rings.	Ex. 48 (OURA1478_00014290); Ex. 49 (OURA1478_00014241)			
Summary Determination of No Infringement					
110	Claim limitation [7.C][vi][a] of the '789 Patent requires a "second material different from the first material," wherein the first material insulates the rigid PCB portions	Ex. 1 ('789 Patent) at Claim 7			

	as a first exterior surface and the second material insulates the flexible PCB portions as a first exterior surface.				
111	[REDACTED]	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 656-59; Ex. 57 (OURA1478_00006316)			
112	[REDACTED]	Ex. 50 (OURA1478_00014259)			
113	OURA1478_00006316 is substantively the same as OURA1468_00219734.	Ex. 57 (OURA1478_00006316); Ex. 43 (OURA1468_00219734)			
114	OURA1468_00219734 was identified in Oura's April 29, 2026 Interrogatory Responses regarding "components of the printed circuit board (PCB)."	Ex. 15 (Oura Third Suppl. Responses to Samsung's First Set of Interrogatories (1478 Investigation) (Apr. 29, 2026) (excerpts)) at 15-17			
115	OURA1478_00006316 was identified in Oura's May 13, 2026 Interrogatory Responses regarding "components of the printed circuit board (PCB)."	Ex. 19 (Oura Fifth Suppl. Responses to Samsung's First Set of Interrogatories (1478 Investigation) (May 13, 2026) (Excerpts)) at 18			
116	Claim limitation [7.C][vi][c] of the '789 Patent requires that the second insulating layer for the flexible PCB portions have a "second thickness thinner than the first thickness" of the second insulating layer for the rigid PCB portions.	Ex. 1 ('789 Patent) at Claim 7			
117	[REDACTED]	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 662-64; Ex. 57 (OURA1478_00006316)			

118	Claim limitation [7.C][vi][e] of the '789 Patent requires a "second material" in the third insulating layer of the flexible PCB portions that is different from the "first material" in the third insulating layer of the rigid PCB portions.	Ex. 1 ('789 Patent) at Claim 7			
119	[REDACTED]	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 667-70; Ex. 57 (OURA1478_00006316);			
120	Claim limitations [7.C][i]-[iv] of the '789 Patent collectively require "a combination of connected and alternating rigid and flexible PCB portions that form a rigid-flexible-rigid-flexible structure."	Ex. 1 ('789 Patent) at Claim 7			
121	[REDACTED]	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 673-74			
122	OURA1468_00255354 was specifically identified in Oura's April 29, 2026 interrogatory responses, including in response to Samsung's interrogatory about PCB structure.	Ex. 15 (Oura Third Suppl. Responses to Samsung's First Set of Interrogatories (1478 Investigation) (Apr. 29, 2026) (excerpts)) at 13, 17			
123	[REDACTED]	Ex. 26 (Mortazavi Reb. Rep.) at ¶¶ 677-79; Ex. 58 (OURA1478_00013242)			

Dated: July 13, 2026

Respectfully submitted,

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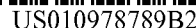
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Exhibit 1



(45) **Date of Patent:** Apr. 13, 2021

- USPC 361/790
See application file for complete search history.

- (56)
- References Cited**

- U.S. PATENT DOCUMENTS

- FOREIGN PATENT DOCUMENTS

- | | | |
|----|-----------|--------|
| EP | 2 180 769 | 4/2010 |
| KR | 101261983 | 5/2013 |
- (Continued)

- ## OTHER PUBLICATIONS

- International Search Report dated Sep. 19, 2017 issued in counterpart application No. PCT/KR2017/006070, 7 pages.

- (Continued)

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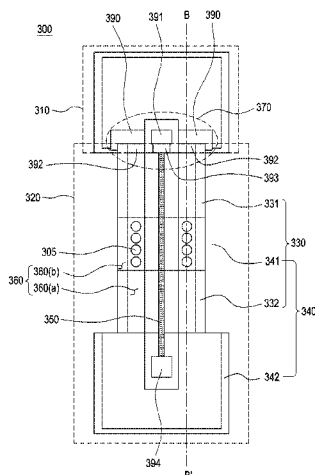
Related U.S. Application Data

- (57) **ABSTRACT**

- Disclosed is an electronic device including a housing, a printed circuit board (PCB) in the housing, wherein the PCB includes a plurality of layers with one or more conductive and insulation layers, a first electrical component formed as at least a part of or in the housing, a second electrical component above or near the PCB in the housing, wherein the first and second electrical components are separated, and at least one electrical path extending from the first electrical component to the second electrical component, wherein at least a portion of the electrical path runs on or inside the PCB, wherein the PCB includes a region including a pattern of conductive vias, wherein each of the vias extends through at least part of the plurality of layers to contact at least one

- (52) **U.S. Cl.**
CPC *H01Q 1/243* (2013.01); *H01Q 1/38*
(2013.01); *H01Q 5/35* (2015.01); *H01Q 21/28*
(2013.01);

- (Continued)



of the one or more conductive layers, and wherein the electrical path runs through the region without contacting the vias.

20 Claims, 14 Drawing Sheets

(51) **Int. Cl.**

H01Q 5/35 (2015.01)
H05K 1/02 (2006.01)
H01Q 21/28 (2006.01)
H01Q 1/38 (2006.01)
H01R 12/79 (2011.01)
H01R 13/66 (2006.01)
H05K 1/11 (2006.01)
H05K 1/14 (2006.01)
H05K 1/18 (2006.01)
H05K 7/14 (2006.01)

(52) **U.S. Cl.**

CPC *H01R 12/79* (2013.01); *H01R 13/6691* (2013.01); *H05K 1/028* (2013.01); *H05K 1/0218* (2013.01); *H05K 1/0243* (2013.01); *H05K 1/115* (2013.01); *H05K 1/142* (2013.01); *H05K 1/148* (2013.01); *H05K 1/18* (2013.01); *H05K 7/1427* (2013.01); *H05K 2201/0999* (2013.01); *H05K 2201/09609* (2013.01); *H05K 2201/10098* (2013.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

8,022,309 B2 9/2011 Pai et al.
 8,143,526 B2 3/2012 Hsu

8,263,873 B2 9/2012 Choi et al.
 8,633,394 B2 1/2014 Lai et al.
 2006/0270472 A1 11/2006 Chen et al.
 2009/0260859 A1 10/2009 Pai et al.
 2009/0294155 A1 12/2009 Suzuki et al.
 2011/0094784 A1 4/2011 Muro et al.
 2012/0229346 A1 9/2012 Chen et al.
 2013/0063914 A1 3/2013 Lin et al.
 2013/0257659 A1 10/2013 Darnell et al.
 2014/0147128 A1* 5/2014 Han H05K 1/142 398/135
 2014/0225806 A1 8/2014 Lee et al.
 2015/0223323 A1 8/2015 O'Flynn
 2015/0234426 A1 8/2015 Bailey

FOREIGN PATENT DOCUMENTS

KR 1020130081450 7/2013
 KR 101305518 9/2013
 KR 20140136634 12/2014
 KR 20150062059 6/2015

OTHER PUBLICATIONS

Samsung Galaxy et al: "Samsung Galaxy S7 Teardown Geschrieben von: Jeff Suovanen", IFIXIT Guide, XP055578568, Mar. 8, 2016, 19 pages.

European Search Report dated Apr. 17, 2019 issued in counterpart application No. 17813538.0-1203, 11 pages.

European Search Report dated Feb. 5, 2020 issued in counterpart application No. 17813538.0-1203, 12 pages.

* cited by examiner

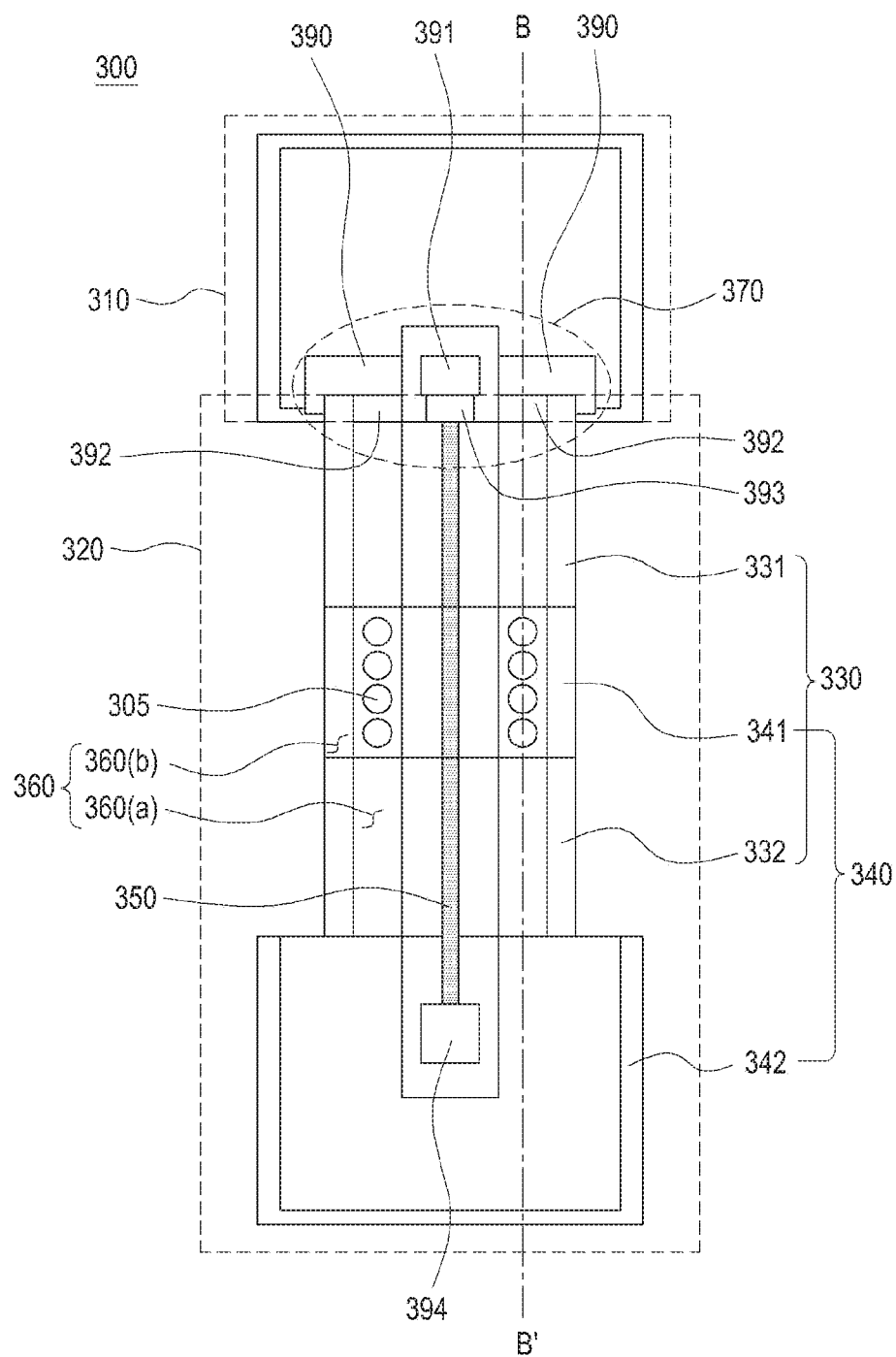


FIG.1

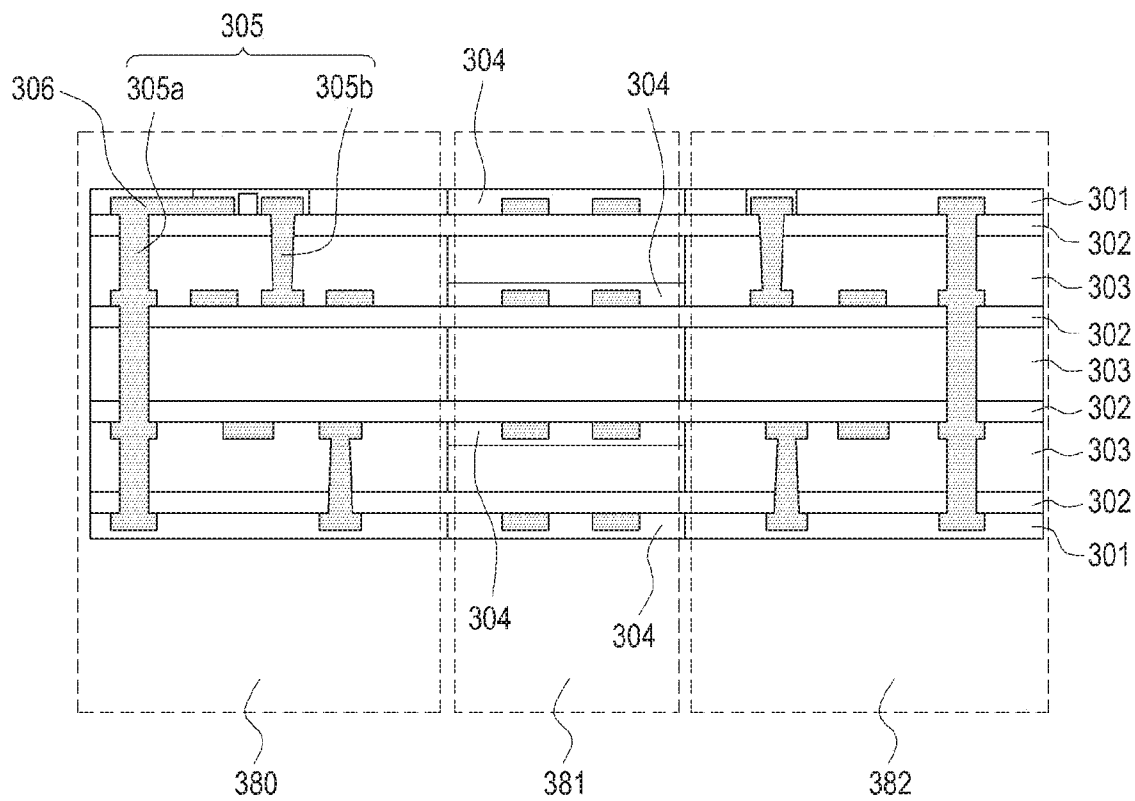


FIG.2

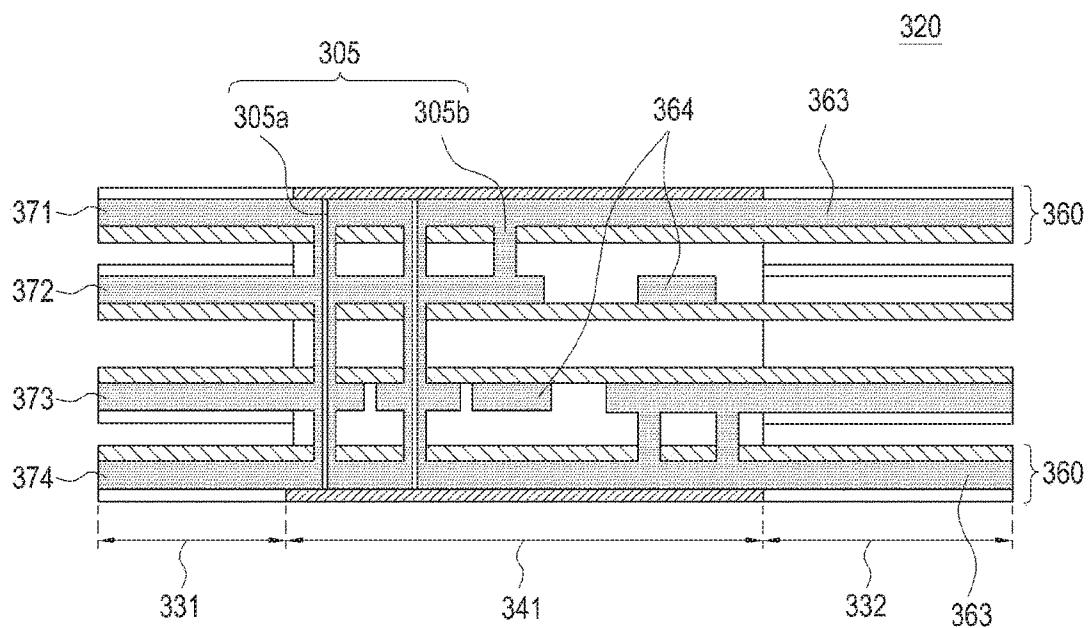


FIG.3

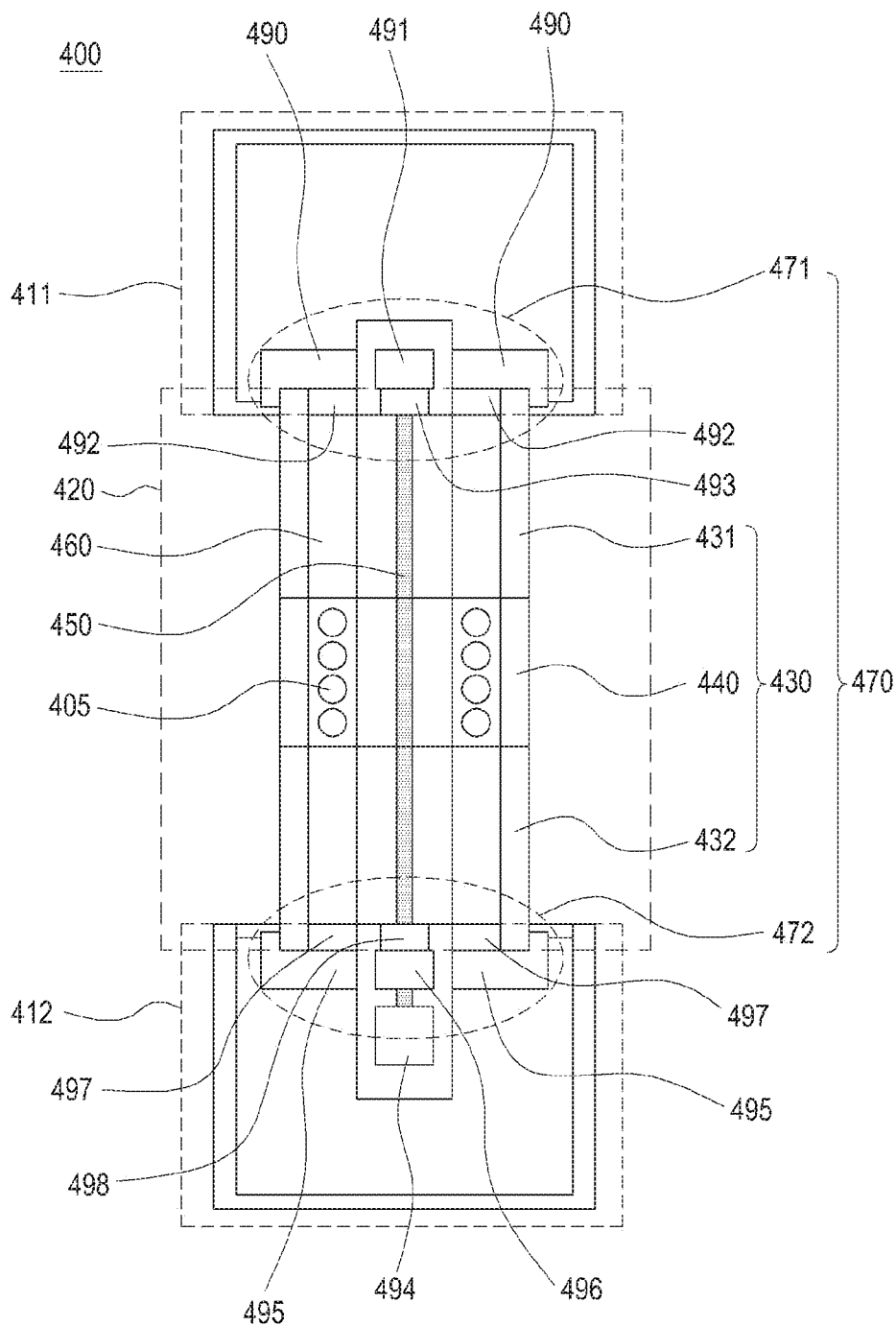


FIG. 4

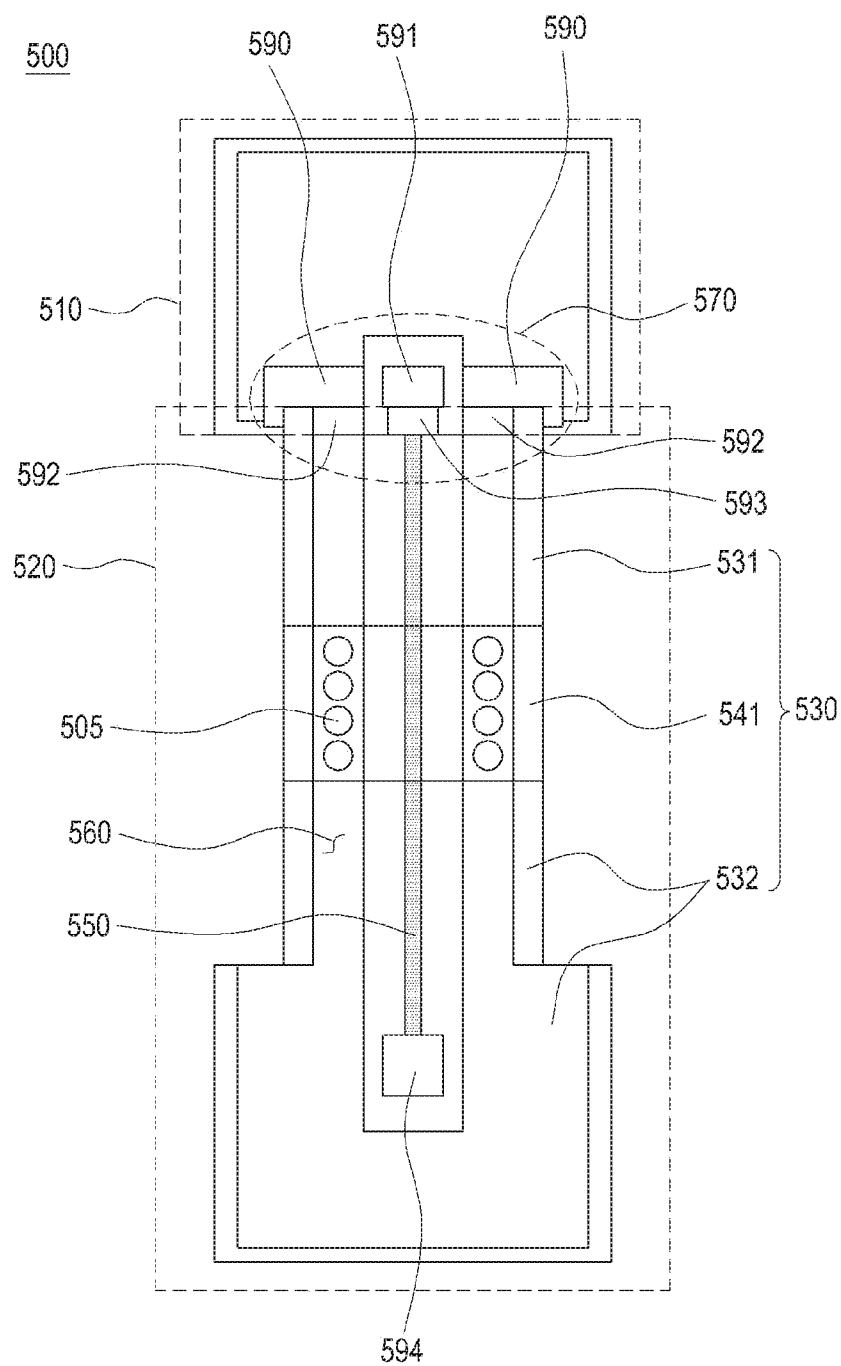


FIG.5

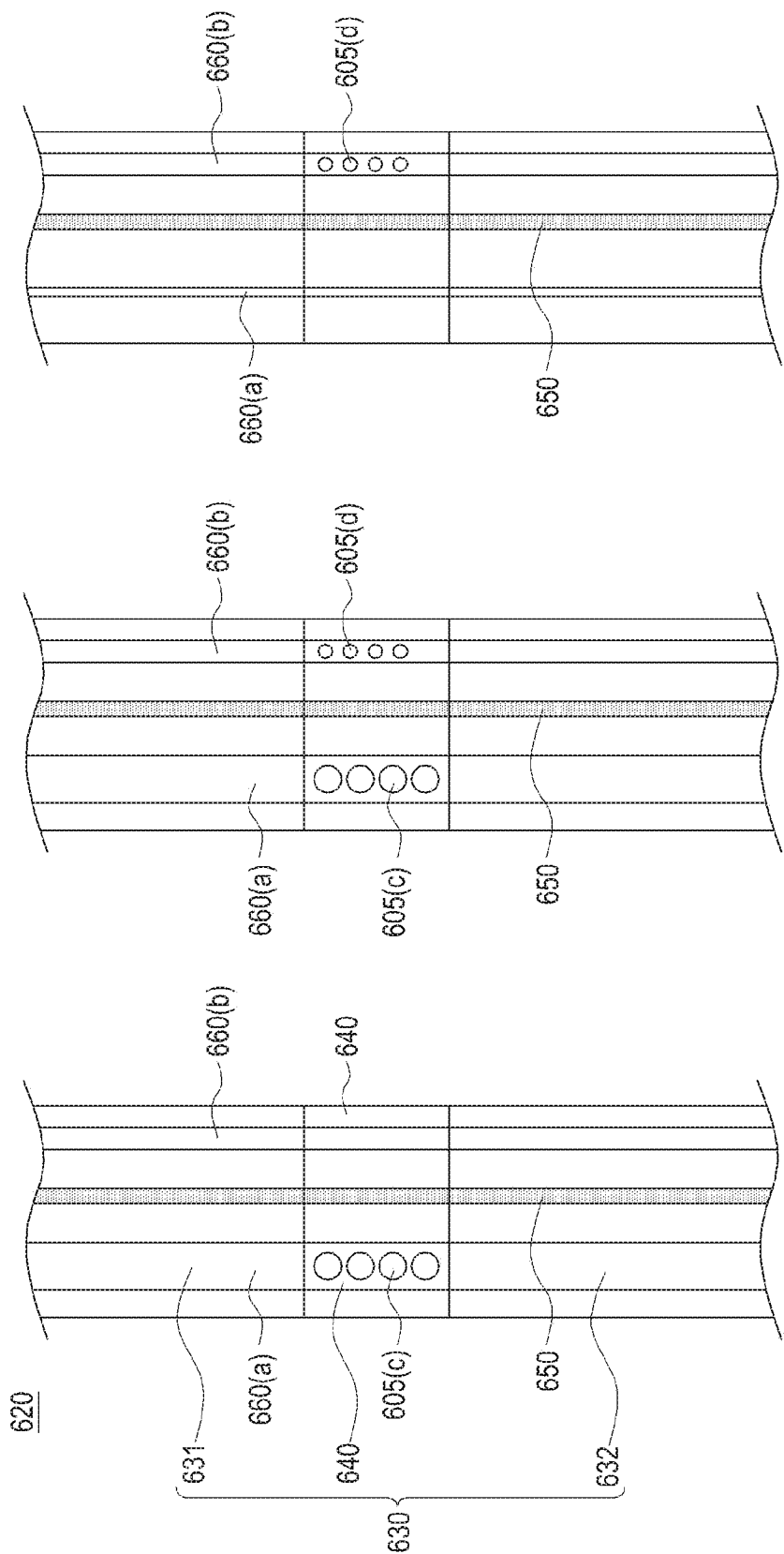


FIG. 6C

FIG. 6B

FIG. 6A

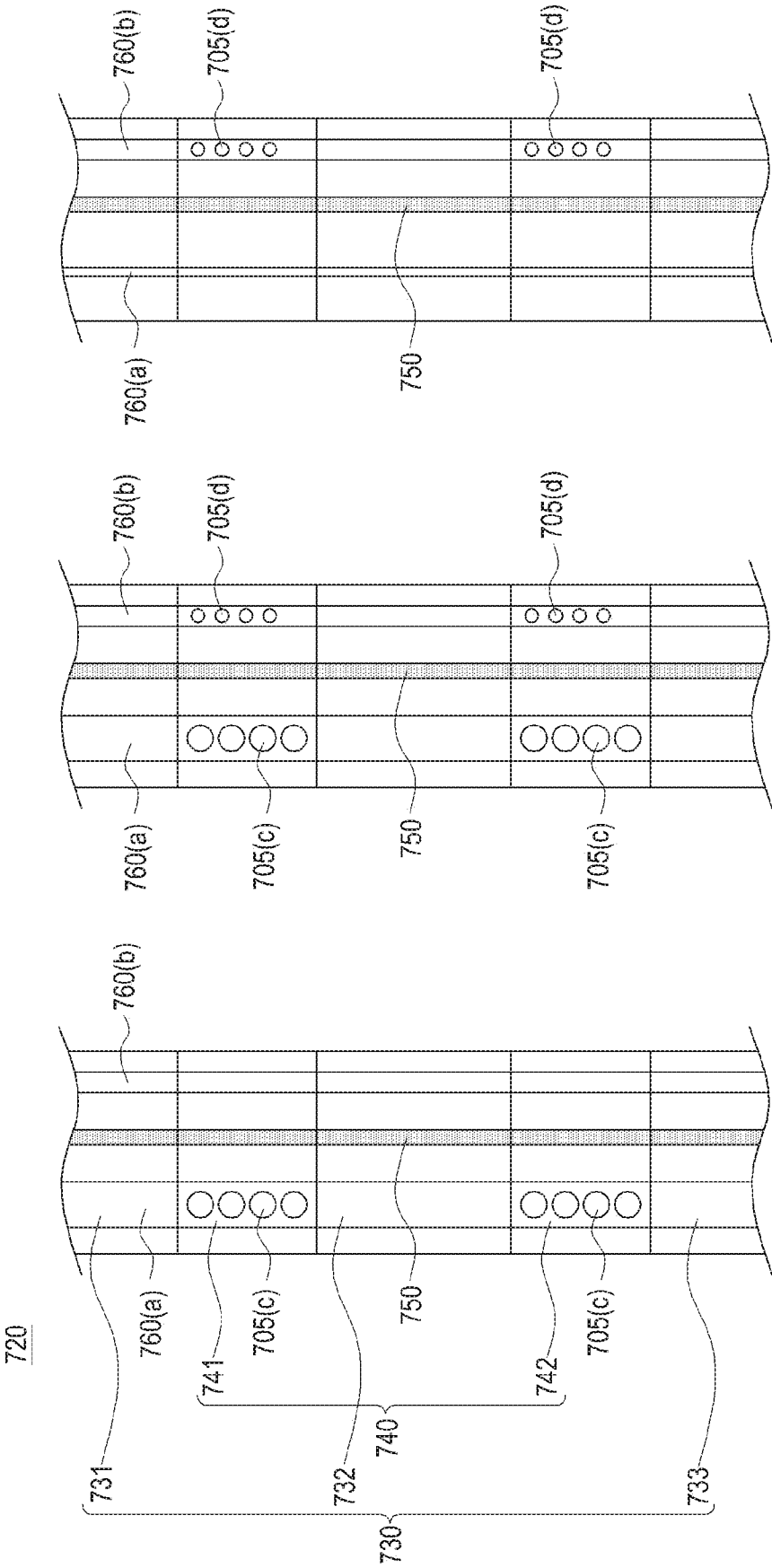


FIG. 7C

FIG. 7B

FIG. 7A

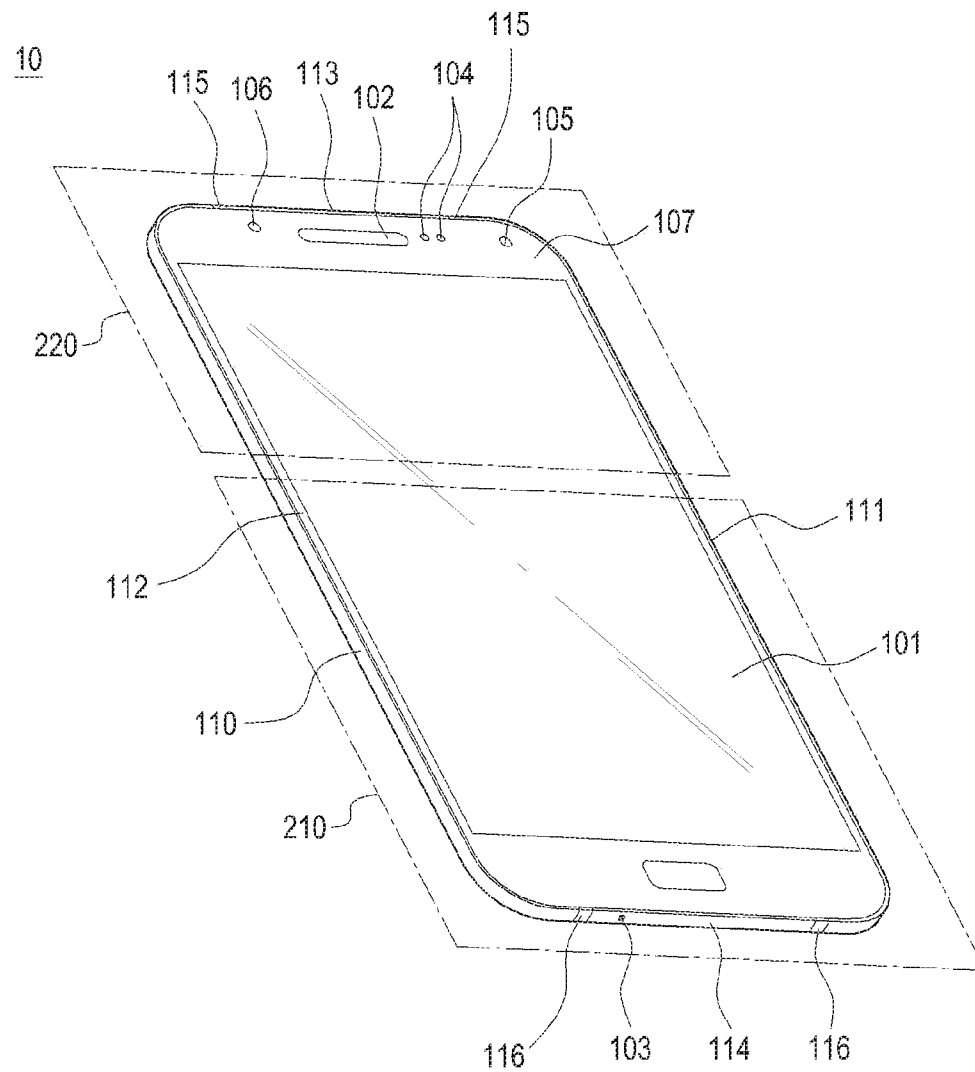


FIG. 8

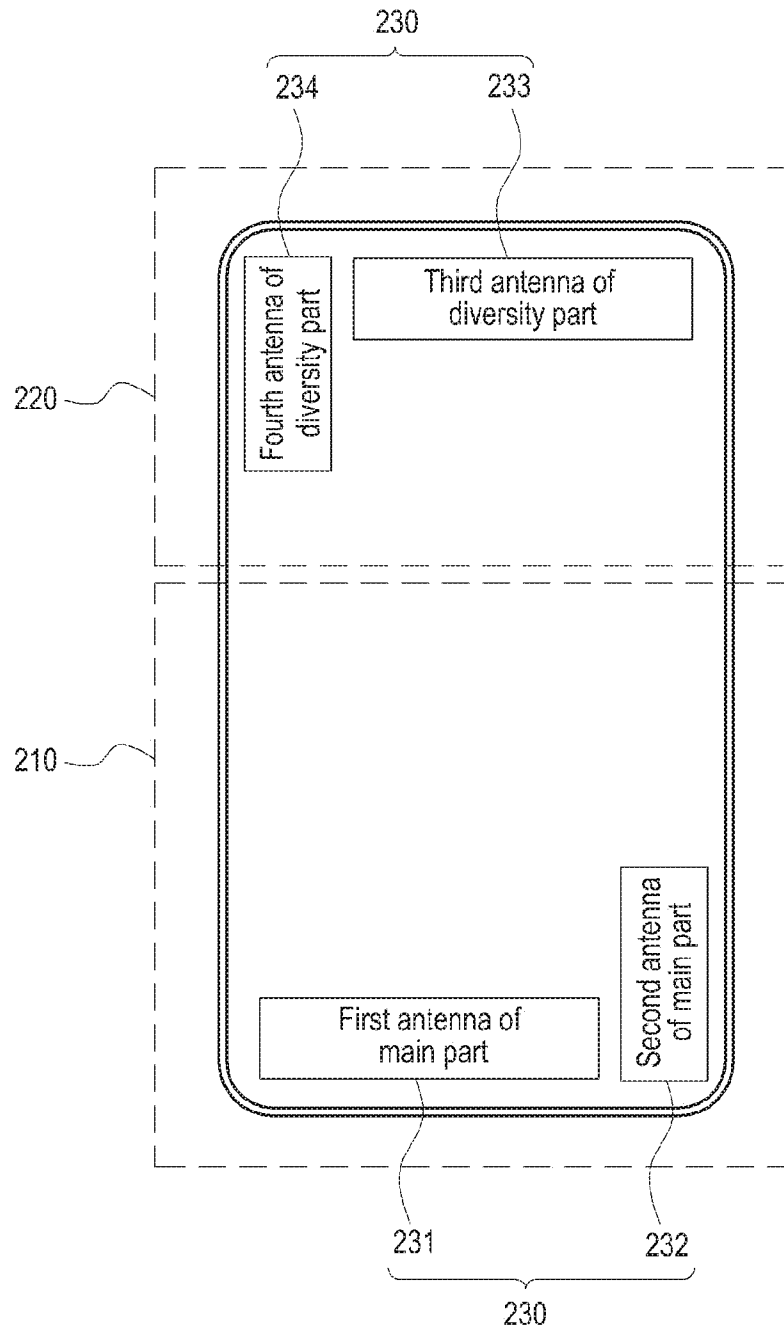


FIG.9

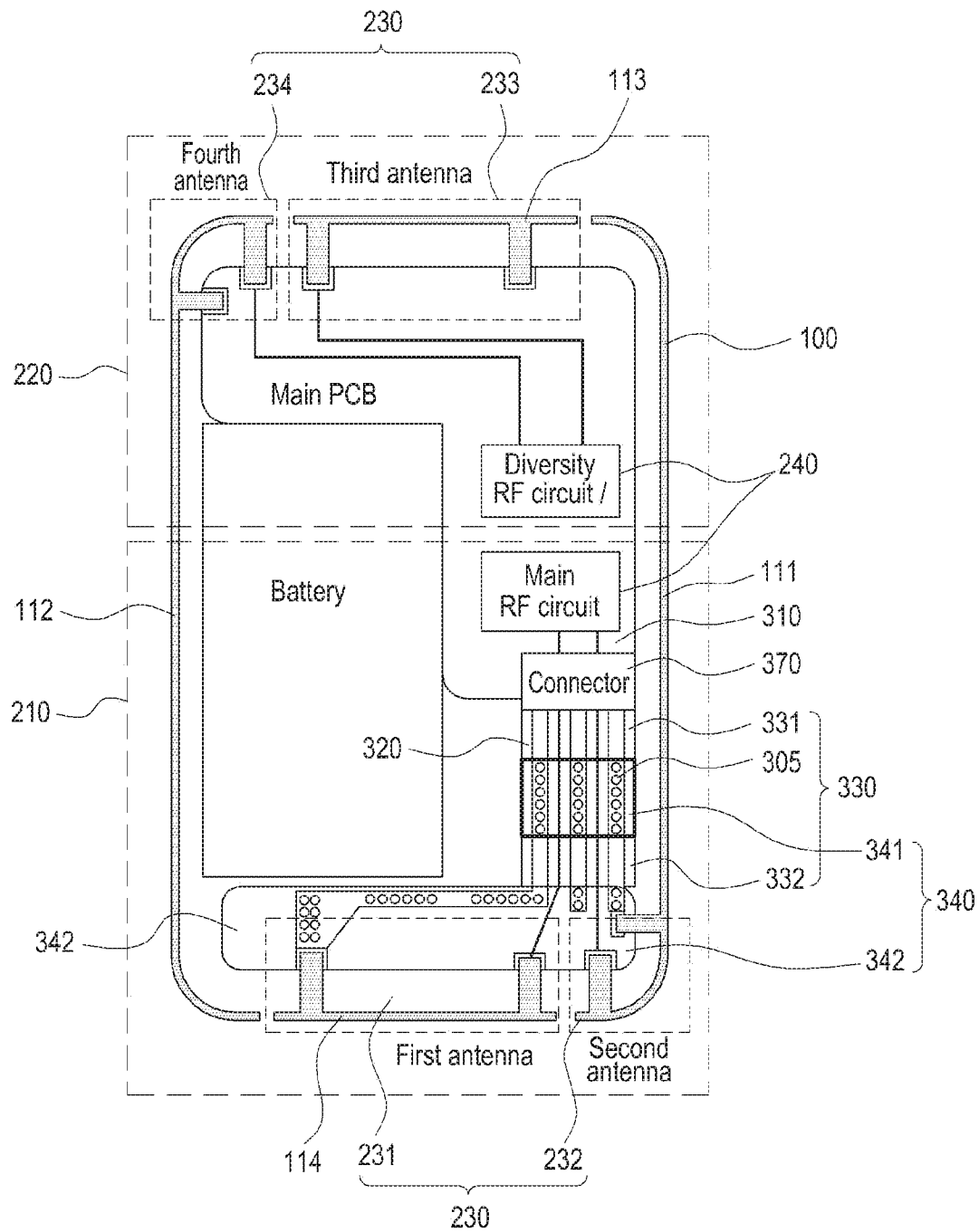


FIG. 10

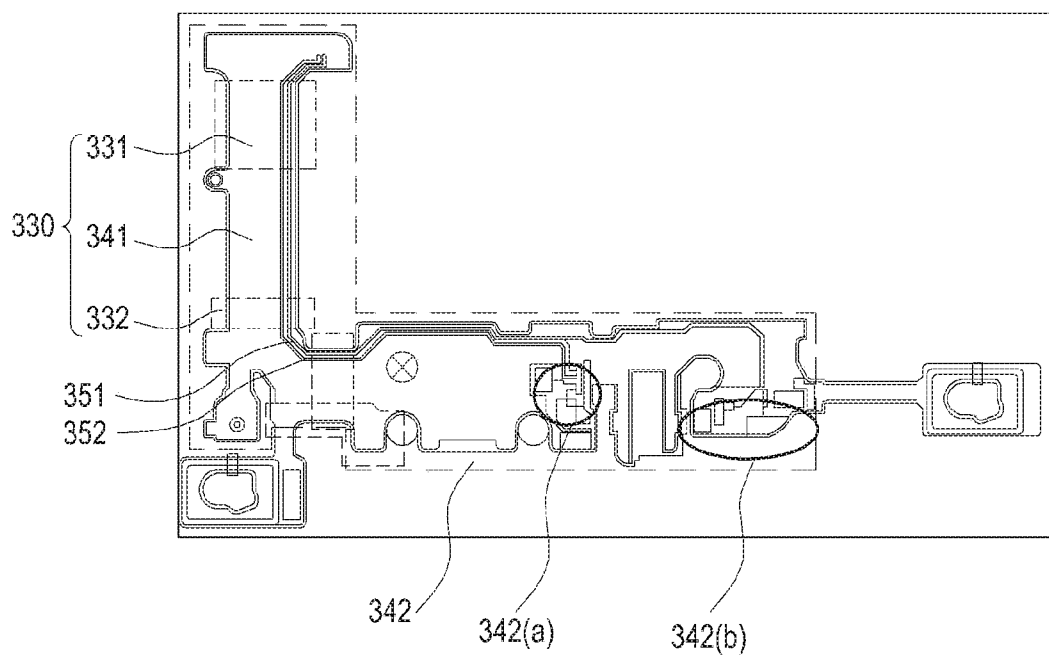


FIG. 11

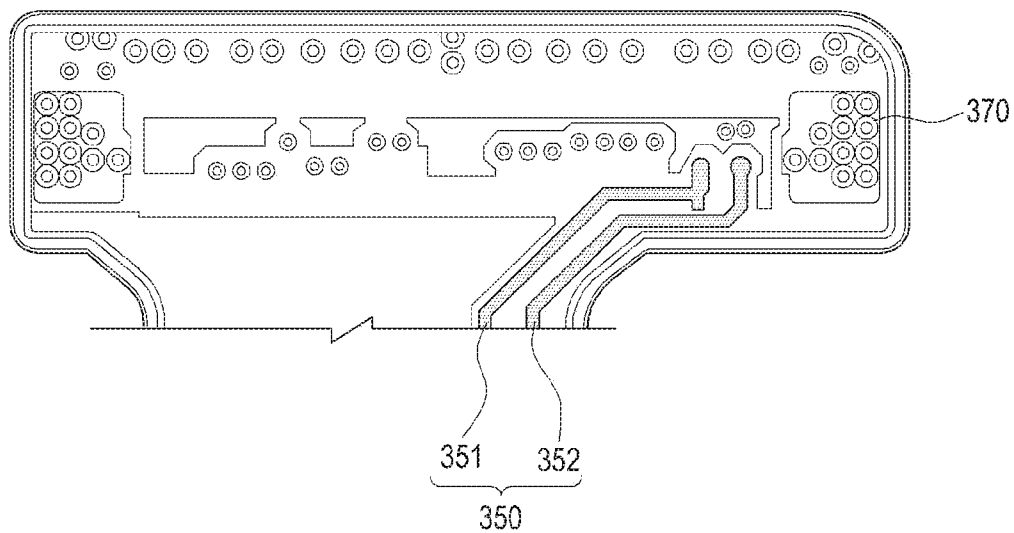


FIG. 12

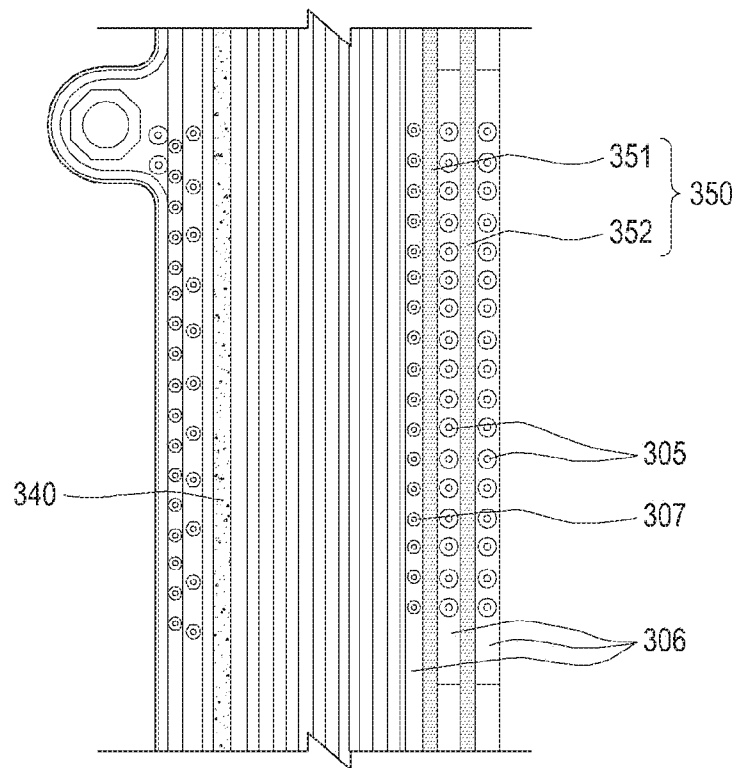


FIG.13

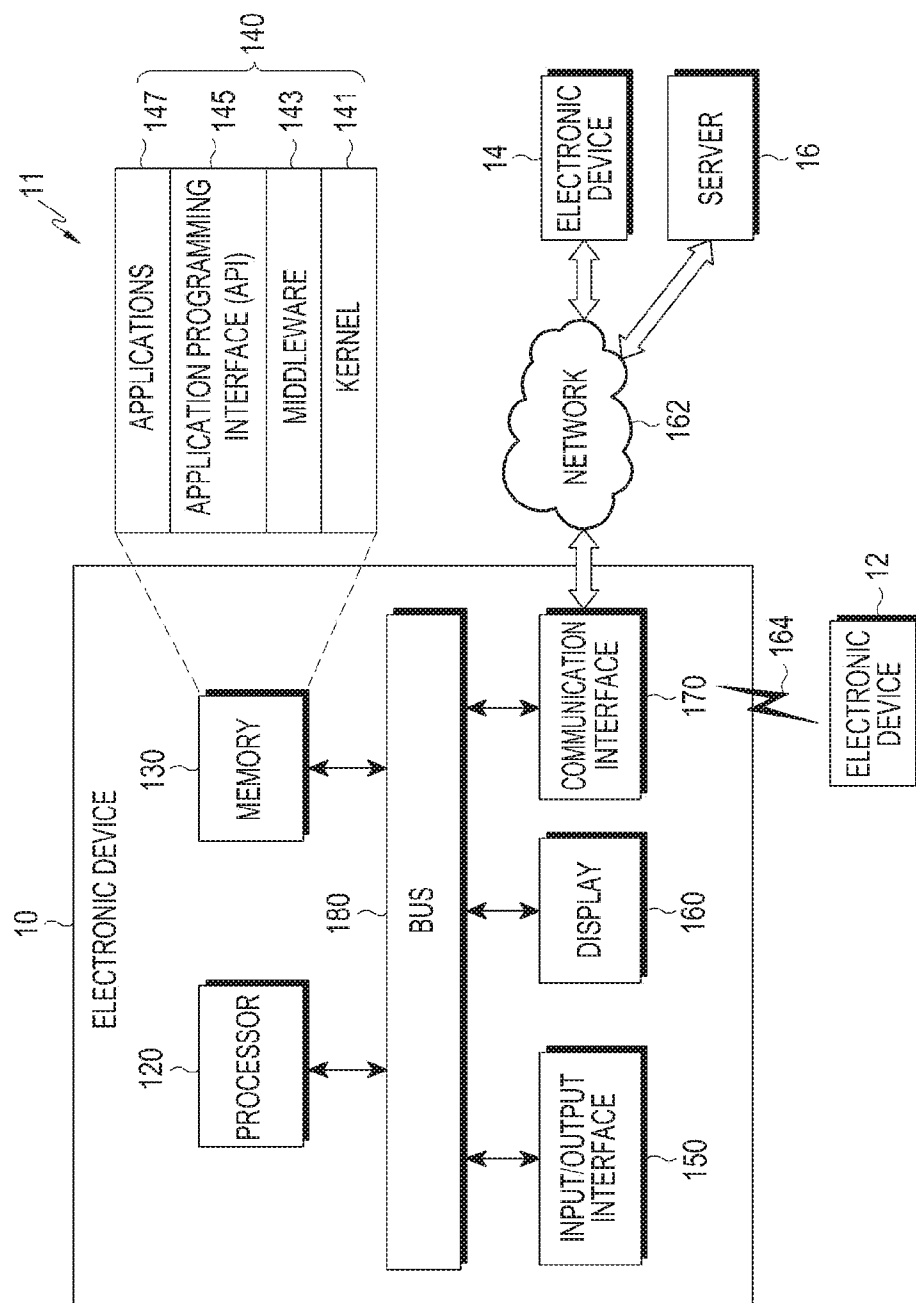


FIG.14

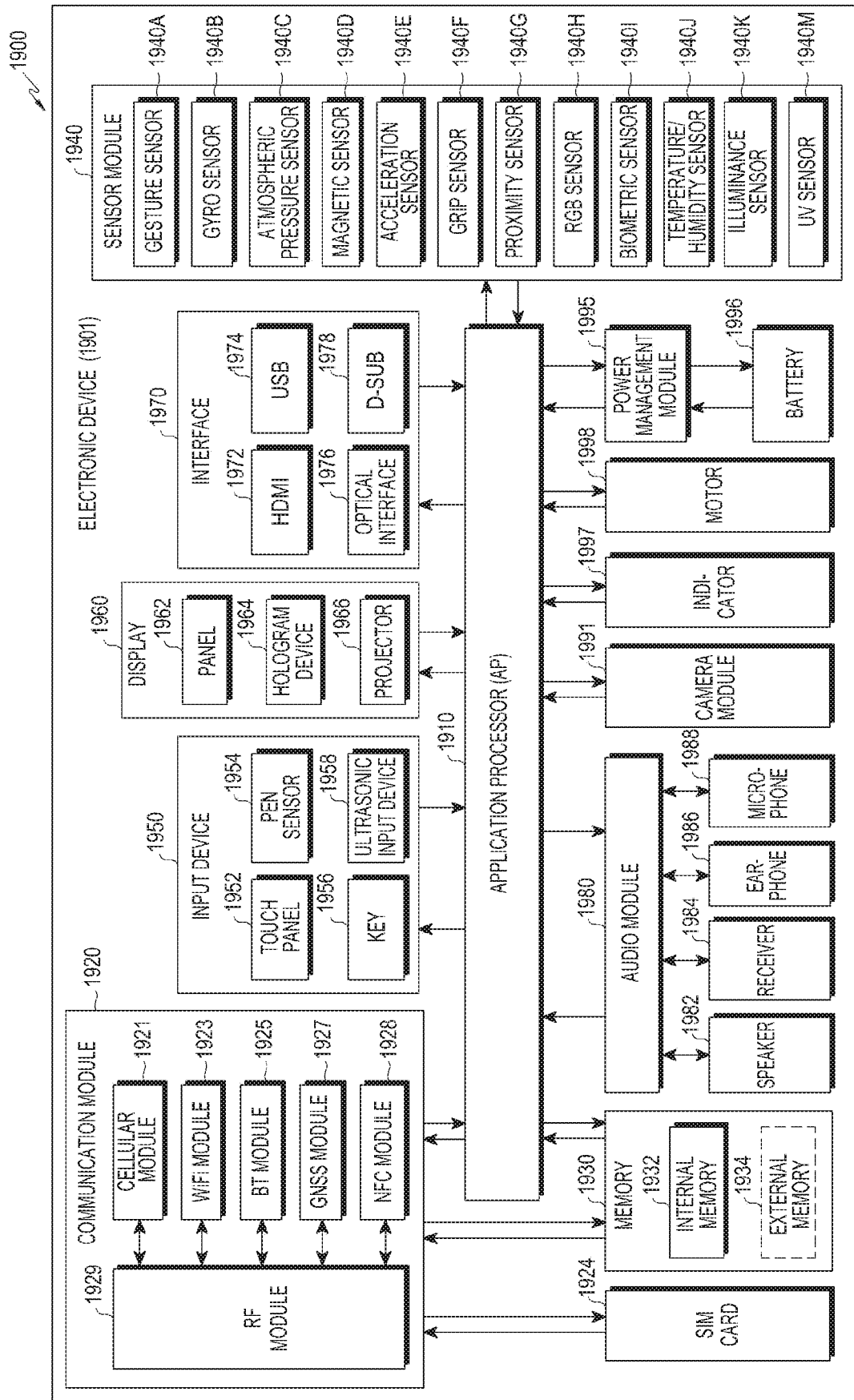


FIG.15

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ELECTRONIC DEVICE INCLUDING HIGH-FREQUENCY TRANSMISSION CIRCUIT

PRIORITY

This application is a Continuation Application of U.S. patent application Ser. No. 15/621,687, filed on Jun. 13, 2017, and claims priority under 35 U.S.C. § 119(a) to a Korean Patent Application filed on Jun. 13, 2016 in the Korean Intellectual Property Office and assigned Serial No. 10-2016-0073355, the content of each of which is incorporated herein by reference.

BACKGROUND

1. Field of the Disclosure

The present disclosure relates generally to a high-frequency transmission circuit included in an electronic device, and more particularly, to a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.

2. Description of the Related Art

Recently, portable terminals having new functions have been rapidly developed and have accounted for a greater part of people's lives with the common use thereof.

Internal circuits of portable terminals are generally implemented on printed circuit boards (PCBs). PCB technologies have been rapidly developing, and flexible PCBs (FPCBs) that can be bent easily are currently used widely, as well as conventional rigid PCBs.

In addition, coaxial cables are typically used for high-frequency transmission lines (especially, radio frequency (RF) lines) that are applied to wireless terminals. A large amount of additional material cost may be required to produce coaxial cables used for electronic devices as high-frequency transmission lines, and additional labor cost may be incurred by an increase in the number of assembly steps. In another example, electronic devices have narrow internal spaces available on account of various types of circuit modules mounted therein, and due to the narrow internal spaces, a defect rate may increase when small connectors used for coaxial cables are assembled.

SUMMARY

An aspect of the present disclosure provides a transmission line capable of effectively transmitting a noiseless high-frequency signal in a narrow space without affecting other modules.

Another aspect of the present disclosure provides that a coaxial cable used as a transmission line in a portable terminal can be replaced by a circuit board that includes an FPCB, and thus it is possible to eliminate material costs associated with the manufacture of a coaxial cable.

Another aspect of the present disclosure provides a rigid PCB that is disposed between transmission lines of an FPCB, whereby it is possible to reduce line loss occurring in the transmission lines of the FPCB and to provide a high-frequency signal without loss.

Another aspect of the present disclosure provides that it is possible to efficiently use a mounting space in a wireless

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terminal, which enables diversification of the external design of the wireless terminal to improve customer satisfaction.

In accordance with an aspect of the present disclosure, an electronic device is provided. The electronic device includes a housing; a PCB disposed in the housing, wherein the PCB includes a plurality of layers that include one or more conductive layers and one or more insulation layers; a first electrical component formed to be at least a part of the housing or disposed in the housing; a second electrical component disposed above or near the PCB in the housing, wherein the second electrical component is separated from the first electrical component; and at least one electrical path extending from the first electrical component to the second electrical component, wherein at least a portion of the electrical path runs on or inside the PCB, wherein the PCB includes a region including a pattern of conductive vias, wherein each of the conductive vias extends through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path may run through the region without making contact with the conductive vias.

In accordance with another aspect of the present disclosure, an electronic device is provided. The electronic device includes a first PCB; a second PCB connected to the first PCB through a connector and including at least one FPCB and at least one rigid PCB that are integrated with each other, wherein the at least one FPCB flexible and the at least one rigid PCB each include a plurality of layers; an electrical path that transmits a high-frequency signal through the connector and passes over or through the second PCB; and at least one conductive via disposed in the at least one rigid PCB to connect grounds between the plurality of layers.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects, features, and advantages of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB in an electronic device is used for the high-frequency transmission line;

FIG. 2 is a partial sectional view of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 2 is taken along line B-B' of FIG. 1;

FIG. 3 is a sectional view of an interlayer configuration of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 3 is taken along line B-B' of FIG. 1;

FIG. 4 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line;

FIG. 5 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line;

FIGS. 6A, 6B, 6C, 7A, 7B, and 7C are front views of conductive vias according to an embodiment of the present disclosure, the conductive vias being arranged in PCBs in electronic devices;

FIG. 8 is a perspective view of an electronic device according to an embodiment of the present disclosure;

FIG. 9 is a schematic of an antenna arrangement in an electronic device according to an embodiment of the present disclosure;

FIG. 10 illustrates an antenna device and a PCB in an electronic device according to an embodiment of the present disclosure;

FIG. 11 is an illustration of a signal transmission using a second PCB according to an embodiment of the present disclosure;

FIGS. 12 and 13 are enlarged views of a first FPCB and a first rigid PCB;

FIG. 14 is a block diagram of an electronic device according to an embodiment of the present disclosure in a network environment; and

FIG. 15 is a block diagram of an electronic device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE PRESENT DISCLOSURE

Hereinafter, various embodiments of the present disclosure are described with reference to the accompanying drawings. However, it should be understood that there is no intent to limit the present disclosure to the particular forms disclosed herein; rather, the present disclosure is intended to be construed to cover various modifications, equivalents, and/or alternatives of embodiments of the present disclosure. In the description of the accompanying drawings, similar reference numerals may be used to designate similar elements.

As used herein, the expressions “have”, “may have”, “include”, or “may include” refer to the existence of a corresponding feature (e.g., numeral, function, operation, or element such as a component), and do not exclude one or more additional features.

In the present disclosure, the expressions “A or B”, “at least one of A and/or B”, and “one or more of A and/or B” may include all possible combinations of the items listed. For example, the expressions “A or B”, “at least one of A and B”, and “at least one of A or B” refer to all of (1) including at least one A, (2) including at least one B, and (3) including all of at least one A and at least one B.

The expressions “a first”, “a second”, “the first”, or “the second” used in various embodiments of the present disclosure may modify various components regardless of the order and/or the importance of the components but are not intended to limit the corresponding components. For example, a first user device and a second user device indicate different user devices although both of them are user devices. For example, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element without departing from the scope of the present disclosure.

It should be understood that when an element (e.g., a first element) is referred to as being (operatively or communicatively) “connected,” or “coupled,” to another element (e.g., a second element), the element may be directly connected or directly coupled to the other element or any other element (e.g., a third element) may be interposer between them. In contrast, it may be understood that when an element (e.g., a first element) is referred to as being “directly connected,” or “directly coupled” to another element (e.g., a second element), there is no element (e.g., a third element) interposed between them.

The expression “configured to” used in the present disclosure may be used interchangeably with the expressions, for example, “suitable for”, “having the capacity to”,

“designed to”, “adapted to”, “made to”, and “capable of” according to the situation. The term “configured to” may not necessarily imply “specifically designed to” in hardware. Alternatively, in some situations, the expression “device configured to” may indicate that the device, together with other devices or components, “is able to”. For example, the expression “processor adapted (or configured) to perform A, B, and C” may indicate a dedicated processor (e.g., an embedded processor) only for performing the corresponding operations or a general purpose processor (e.g., a central processing unit (CPU) or an application processor (AP)) that can perform the corresponding operations by executing one or more software programs stored in a memory device.

The terms used herein are merely used for the purpose of describing particular embodiments but are not intended to limit the scope of the present disclosure. A singular expression may include a plural expression unless they are definitely different in a context. Unless defined otherwise, all terms used herein, have the same meanings as those commonly understood by a person skilled in the art to which the present disclosure pertains. Such terms as those defined in a generally used dictionary may be interpreted to have the meanings equal to the contextual meanings in the relevant field of art, and are not intended to be interpreted to have ideal or excessively formal meanings unless clearly defined in the present disclosure. In some cases, even the terms defined in the present disclosure are not intended to be interpreted to exclude embodiments of the present disclosure.

An electronic device according to various embodiments of the present disclosure may include at least one of, for example, a smart phone, a tablet personal computer (PC), a mobile phone, a video phone, an electronic book reader (e-book reader), a desktop PC, a laptop PC, a netbook computer, a workstation, a server, a personal digital assistant (PDA), a portable multimedia player (PMP), a moving picture experts group (MPEG-1) audio layer-3 (MP3) player, a mobile medical device, a camera, and a wearable device. According to various embodiments of the present disclosure, the wearable device may include at least one of an accessory type (e.g., a watch, a ring, a bracelet, an anklet, a necklace, glasses, a contact lens, or a head-mounted device (HMD)), a fabric or clothing integrated type (e.g., electronic clothing), a body-mounted type (e.g., a skin pad, or tattoo), and a bio-implantable type (e.g., an implantable circuit).

According to an embodiment of the present disclosure, the electronic device may be a home appliance. A home appliance may include at least one of, for example, a television, a digital video disk (DVD) player, an audio player, a refrigerator, an air conditioner, a vacuum cleaner, an oven, a microwave oven, a washing machine, an air cleaner, a set-top box, a home automation control panel, a security control panel, a TV box (e.g., Samsung HomeSync®, Apple TV®, or Google TV™), a game console (e.g., Xbox® and PlayStation®), an electronic dictionary, an electronic key, a camcorder, and an electronic photo frame.

According to an embodiment of the present disclosure, the electronic device may include at least one of various medical devices (e.g., various portable medical measuring devices (a blood glucose monitoring device, a heart rate monitoring device, a blood pressure measuring device, a body temperature measuring device, etc.)), a magnetic resonance angiography (MRA), a magnetic resonance imaging (MRI), a computed tomography (CT) machine, and an ultrasonic machine), a navigation device, a global positioning system (GPS) receiver, an event data recorder (EDR), a flight data recorder (FDR), a vehicle infotainment device, an

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electronic device for a ship (e.g., a navigation device for a ship, and a gyro-compass), avionics, security devices, an automotive head unit, a robot for home or industry, an automated teller machine (ATM), point of sales (POS) device in a shop, or an Internet of Things (IoT) device (e.g., a light bulb, various sensors, electric or gas meter, a sprinkler device, a fire alarm, a thermostat, a streetlamp, a toaster, sporting goods, a hot water tank, a heater, a boiler, etc.).

According to an embodiment of the present disclosure, an electronic device may include at least one of a part of furniture or a building/structure, an electronic board, an electronic signature receiving device, a projector, and various types of measuring instruments (e.g., a water meter, an electric meter, a gas meter, and a radio wave meter). An electronic device may be a combination of one or more of the aforementioned various devices. An electronic device may also be a flexible device. Further, an electronic device is not intended to be limited to the aforementioned devices, but may include an electronic device to be developed in the future.

Hereinafter, an electronic device according to various embodiments of the present disclosure is described with reference to the accompanying drawings. In the present disclosure, the term “user” may indicate a person using an electronic device or a device (e.g., an artificial intelligence electronic device) using an electronic device.

FIG. 1 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB 300 in an electronic device is used for the high-frequency transmission line. The electronic device may be a smart phone or a wearable device.

Referring to FIG. 1, the electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB 300 in the housing; and an electrical path 350 passing over or through the printed circuit board 300.

The PCB 300 according to an embodiment of the present disclosure may be a first PCB 310 and a second PCB 320. For example, the first PCB 310 may be a main board on which electronic components included in the electronic device are mounted, the main board being capable of transmitting signals between the electronic components, and the second PCB 320 may be a sub-board that transmits signals generated by the electronic components. The sub-board may further include the electronic components.

The PCB 300 according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a communication processor (CP), a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).

The first and second PCBs 310 and 320 according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers 306 in FIG. 2 described below and one or more insulation layers 303.

The first PCB 310 in FIG. 1 according to an embodiment of the present disclosure may be a rigid PCB. The first PCB 310 may be electrically connected to the second PCB 320 through a connector 370. The connector 370 may include a ground pad 390 and a signal pad 391 of the first PCB 310 and a ground pad 392 and a signal pad 393 of the second PCB 320. The ground pad 390 of the first PCB 310 and the ground pad 392 of the second PCB 320 may be connected together by soldering. The signal pad 391 of the first PCB 310 and the signal pad 393 of the second PCB 320 may be connected together by soldering. The second PCB 320 may

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include an interconnection pad 394. An antenna contact member (for example, a C-clip) or an RF component (for example, a matching circuit) may be mounted on the interconnection pad 394.

The second PCB 320 according to an embodiment of the present disclosure may be connected to the first PCB 310 and may include an FPCB 330 or a rigid PCB 340. The FPCB 330 and the rigid PCB 340 of the second PCB 320 may be integrated with each other. For example, a first FPCB 331, a first rigid printed circuit board 341, a second flexible printed circuit board 332, and a second rigid PCB 342 may be arranged in sequence. The second rigid PCB 342 may include an antenna connector and a universal serial bus (USB) connector, for example.

The second PCB 320 according to an embodiment of the present disclosure may have the electrical path 350 therein through which electrical signals can be transmitted. The electrical path 350 may be connected with the first PCB 310 through the FPCB 330 and the rigid PCB 340, which are integrated with each other.

The electrical path 350 according to an embodiment of the present disclosure may extend from a first electrical component 230 in FIG. 10 described below to a second electrical component 240 and may be disposed on or in the PCB 300 in FIG. 1. The electrical path 350 may be implemented as a transmission line of a side-guarded micro-strip-line type or a strip-line type, where the transmission line is shielded by a ground line 360 in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path 350 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 320, may receive a high-frequency signal through the connector 370 and may forward the received signal to antenna radiators 231 and 232 in FIG. 10 described below. Furthermore, a plurality of electrical paths 350 in FIG. 1 may be provided, and a ground line 360 may be disposed between the electrical paths 350 so that the electrical paths 350 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground line 360 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 350, and conductive vias 305 may be arranged in a portion of the area of a ground line 360(a). Accordingly, the electrical path 350 can pass through the PCB 300 without making contact with the conductive vias 305.

According to an embodiment of the present disclosure, the connector 370 for connecting the electrical path 350 may be disposed between the first and second PCBs 310 and 320 in order to transmit a high-frequency signal therebetween. The connector 370 may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the connector 370, and, for example, a B-to-B-type connector may be mounted on the rigid printed circuit board, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the connector 370 may be implemented in the sequence of a ground, an RF signal, and a ground.

Hereinafter, the plurality of layers and the conductive vias 305 of the PCB 300 is described in detail.

FIG. 2 is a partial sectional view of the plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 2 is taken along line B-B' of FIG. 1. Hereinafter, the PCB is described as the second PCB 320. However, the following description may be applied to second printed circuit boards 420, 520, 620, and 720 in other embodiments, which will be described below.

Referring to FIG. 2, the second printed circuit board 320 according to an embodiment of the present disclosure may be formed of a plurality of layers, each including at least one conductive layer 306, at least one first insulation layer 302, and at least one second insulation layer 303. The conductive layer 306 and the first insulation layer 302 may be included in a flexible copper clad laminated (FCCL) layer. Further, the first insulation layer 302 may include polyamide.

The conductive layers 306 according to an embodiment of the present disclosure are arranged in the stacking direction, and the first insulation layer 302 is disposed on the rear or front surface of each conductive layer 306. The individual layers may be alternately arranged. One or more conductive vias 305 may be formed to electrically connect the conductive layers 306.

According to an embodiment of the present disclosure, the second insulation layer 303 is provided between the conductive layers 306 to prevent the conductive layers 306 from being electrically connected together through contact therebetween. For example, the plurality of second insulation layers 303 disposed between the plurality of conductive layers 306 may function to insulate the layers and may include a resin and a glass fabric.

According to an embodiment of the present disclosure, a plurality of electronic components may be disposed on and electrically connected to the conductive layers 306. The electronic components may be electrically connected to the antenna radiators 231 and 232 in FIG. 10 described below provided in the electronic device. For example, the electronic components may include an antenna clip, a connector, and a switch device.

Without being limited thereto, however, various materials capable of passing electricity therethrough, other than metal, may be applied to the conductive layers 306 in FIG. 2.

According to an embodiment of the present disclosure, the conductive layers 306 may be electrically connected to any of a plurality of signal lines and ground lines.

The conductive layers 306 according to an embodiment of the present disclosure may be provided, on the top and bottom thereof, with a protective layer 301 capable of protecting the conductive layers 306. The protective layers 301 may be formed of an insulating coating material, such as solder resist, and, for example, a photo imageable solder resist ink (PSR ink) may be used for the solder resist. The PSR ink may be left on a product because it provides insulation and protection even after a component is mounted. Accordingly, the protective layers 301 coated on the exterior of the conductive layers 306 can prevent a short circuit, a connection, corrosion, and contamination of a circuit during a manufacturing process and can protect the conductive layers 306 from external shock, humidity, and chemical substances after the manufacturing process.

The conductive vias 305 according to an embodiment of the present disclosure, which electrically connect the conductive layers 306, may include a first conductive via 305(a) capable of electrically connecting all the layers and a second conductive via 305(b) capable of electrically connecting adjacent layers. For example, the first conductive via 305(a) may be formed by computerized numerical control (CNC) hole machining and copper plating, and the second conductive via 305(b) may be formed by laser hole machining and copper plating.

The conductive vias 305 according to an embodiment of the present disclosure may include a region that forms a predetermined pattern within the PCB 300. The conductive

vias 305 may be configured to make contact with at least one of the conductive layers 306 and to pass through at least one of the multiple layers.

A pattern cover layer 304 according to an embodiment of the present disclosure may be disposed in a single layer on the bottom or top of the conductive layers 306 while surrounding a pattern and thus can protect the internal pattern. For example, the pattern cover layer 304 may be formed of polyimide (PI) and an adhesive.

According to an embodiment of the present disclosure, rigid PCBs 380 and 382 and a FPCB 381 may be integrated with one another. The FPCB 381 may not include the second insulation layer 303, but may include the pattern cover layer 304 instead of the protective layer 301.

FIG. 3 is a sectional view of an interlayer configuration of a plurality of layers of the PCB of FIG. 1, according to an embodiment of the present disclosure, where FIG. 3 is taken along line B-B' of FIG. 1.

Referring to FIG. 3, the second PCB 320 may include the first FPCB 331, the first rigid PCB 341, and the second FPCB 332, which are integrally connected together.

Further, the second PCB 320 may include first to fourth ground pattern layers 371, 372, 373, and 374 that are connected by the conductive vias 305 in order to form a ground. The entirety of the first and second ground pattern layers 371 and 372 forms a ground. Only a part of the second and third ground pattern layers 372 and 373 forms a ground. According to an embodiment of the present disclosure, there may be a layer having no ground layer.

Signal line patterns 364 not connected to the ground may be provided. These signal line patterns 364 may be used as RF signal lines.

The ground pattern layers according to an embodiment of the present disclosure may be connected by the first conductive via 305(a) capable of electrically connecting all of the layers and the second conductive via 305(b) capable of electrically connecting adjacent layers. For example, all or some of the ground pattern layers may be connected using the first conductive via 305(a) passing through all of the layers. For example, the first to fourth ground pattern layers 371, 372, 373, and 374 may be connected by the first conductive via 305(a). The adjacent ground pattern layers may be connected to each other by the second conductive via 305(b). For example, the first and second ground pattern layers 371 and 372 may be connected to each other by the second conductive via 305(b).

According to an embodiment of the present disclosure, a PCB (for example, the PCB 300 of FIG. 1) may include the first PCB 310 and the second PCB 320 including a rigid PCB and a FPCB. In this case, the PCB is described as the PCB 300 of FIG. 1 as an example, but may be applied to the PCBs 400, 500, 600, and 700 of FIGS. 4, 5, 6A, 6B, 6C, 7A, 7B, and 7C.

According to an embodiment of the present disclosure, the high-frequency transmission line provided by the PCB in the electronic device may be configured such that the FPCB 330 having advantages of reducing a material cost and ensuring a mounting space can substitute for a coaxial cable.

For example, the first and second FPCBs 331 and 332 may be used between the second rigid PCB 342 and the first PCB 310 in order to transmit a high-frequency signal from the second rigid PCB 342 to the first PCB 310.

In another example, in order to eliminate loss that occurs when a high-frequency signal is transmitted using the FPCB 330, the first rigid PCB 341 may be disposed between the first and second FPCB 331 and 332, and the ground pattern layers may be connected by the conductive vias 305.

According to an embodiment of the present disclosure, in order to mitigate an increase in line loss, the electronic device may further include a section containing the first rigid PCB **341** in the middle of the FPCB **330**. If the FPCB **330** has a length of 10 mm or more, the transmission line in the section may cause a great line loss. For example, the transmission line may have a change in the strip line characteristics due to the inductance generated by the ground lines **360** disposed on the opposite sides thereof, and thus the line loss generated by the electrical path **350** may increase markedly.

According to an embodiment of the present disclosure, the electronic device may have the one or more vias **305** arranged on the first rigid PCB **341**. When the grounds are connected by the conductive vias **305** arranged on the first rigid PCB **341**, the inductance generated by the ground lines **360** can be reduced, for example, by virtue of the interlayer ground connection on the ground lines **360**, thereby reducing adverse effects, such as line loss. Furthermore, an all-ground layer (ground line) is disposed between the layers providing the second PCB **320**, and the entire ground line **360** may be connected to the all-ground layer through the conductive vias **305**, whereby it is possible to have a much greater line loss reduction effect.

According to an embodiment of the present disclosure, a rigid PCB may be disposed between the FPCBs, thereby preventing the layers from being separated or twisted as described above. In the case where the existing FPCBs are installed on a drive unit, the layers of the FPCBs may be separated or twisted. Even in this case, there may be a change in the characteristics of a strip line, which may cause line loss in the transmission line or other adverse property changes.

However, an electronic device having the aforementioned arrangement is advantageous in that the side effects mentioned above are not exhibited even when the FPCBs move, whereby it is possible to reduce line loss in the transmission line and to alleviate other adverse characteristics.

In addition, in the case where a transmission line has a greater length, it is possible to solve shortcomings, such as line loss, by additionally placing rigid PCBs in several positions on the FPCBs and using a connection between the ground line **360**, other ground pattern layers **363**, and the conductive vias **305**. This is described below in greater detail with reference to FIG. 4.

FIG. 4 is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, in which a PCB in an electronic device is used for the high-frequency transmission line. An electronic device may be a smart phone or a wearable device.

According to an embodiment of the present disclosure, an electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB **400** in the housing; and an electrical path **450** passing over or through the PCB **400**.

Referring to FIG. 4, the PCB **400** according to an embodiment of the present disclosure may include a second PCB **420**, a third PCB **411**, and a fourth PCB **412**. For example, the third PCB **411** may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components. Furthermore, for example, the fourth PCB **412** may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components. Moreover, for example, the second PCB **420** may be a sub-board that

transmits signals generated by electronic components. The sub-board may further include the electronic components.

The PCB **400** according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a CP, a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).

The second to fourth PCBs **420**, **411**, and **412** according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers and one or more insulation layers.

The third and fourth PCBs **411** and **412** according to an embodiment of the present disclosure may be a rigid PCB. The third or fourth PCB **411** and **412** may be electrically connected to the second PCB **410** through a connector **470**.

The third PCB **411** may be disposed on one end of the second PCB **420**, and the fourth PCB **412** may be disposed on an opposite end of the second PCB **420**. Hereinafter, the third and fourth printed circuit boards **411** and **412** are described as third and fourth rigid PCBs **411** and **412**, respectively.

The third rigid PCB **411** according to an embodiment of the present disclosure may be connected with the second PCB **420**, and a first connector **471** may be disposed to transmit a high-frequency signal through the electrical path **450**. The first connector **471** may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the first connector **471**, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the first connector **471** may be implemented in the sequence of a ground, an RF signal, and a ground.

The first connector **471** according to an embodiment of the present disclosure may include a ground pad **490** and a signal pad **491** of the third rigid PCB **411** and a ground pad **492** and a signal pad **493** of the second PCB **420**. The ground pad **490** of the third rigid PCB **411** and the ground pad **492** of the second PCB **420** may be connected together by soldering. The signal pad **491** of the third rigid PCB **411** and the signal pad **493** of the second PCB **420** may be connected by soldering.

The fourth rigid PCB **412** according to an embodiment of the present disclosure may be connected to the second PCB **420**, and a second connector **472** may be disposed to transmit a high-frequency signal through the electrical path **450**. The second connector **472** may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the second connector **472**, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB. Furthermore, for example, the pin map of a transmission line included in the first connector **472** may be implemented in the sequence of a ground, an RF signal, and a ground.

The second connector **472** according to an embodiment of the present disclosure may include a ground pad **495** and a signal pad **496** of the fourth rigid PCB **412** and a ground pad **497** and a signal pad **498** of the second PCB **420**. The ground pad **495** of the fourth rigid PCB **412** and the ground pad **497** of the second PCB **420** may be connected by soldering. The signal pad **496** of the fourth rigid PCB **412** and the signal pad **498** of the second PCB **420** may be connected by soldering. The second PCB **420** may include an interconnection pad **494**. An antenna contact member (for example, a C-clip) or

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an RF component (for example, a matching circuit) may be mounted on the interconnection pad **494**.

The second PCB **420** according to an embodiment of the present disclosure may be disposed between the third rigid PCB **411** and the fourth rigid PCB **412**, and may include an FPCB **430** or a rigid PCB **440**. The flexible printed circuit board **430** and the rigid PCB **440** of the second PCB **420** may be integrated with each other. For example, a first FPCB **431**, the first rigid PCB **440**, and a second flexible PCB **432** may be arranged in sequence. The first rigid PCB **440** may include an antenna connector and a USB connector, for example.

The electrical path **450** according to an embodiment of the present disclosure may extend from a first electrical component including an antenna radiator to a second electrical component including a communication circuit, and may be disposed on or in the PCB **400**. The electrical path **450** may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type, the transmission line being shielded by a ground line in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path **450** according to an embodiment of the present disclosure, which is disposed on or in the second PCB **420**, may receive a high-frequency signal through the connector **470** and may forward the received signal to antenna radiators **231** and **232** in FIG. **10** described below. Furthermore, a plurality of electrical paths **450** in FIG. **4** may be provided, and a ground line **460** may be disposed between the electrical paths **450** so that the electrical paths **450** can independently transmit a high-frequency signal without mutually affecting the characteristics thereof.

The ground line **460** according to an embodiment of the present disclosure may be disposed adjacent to the electrical path **450**, and conductive vias **405** may be arranged in a portion of the area of the ground line **460**. Accordingly, the electrical path **450** can pass through the PCB **400** without making contact with the conductive vias **405**.

A plurality of layers of the PCB **400** and the conductive vias **405** are the same as those in the above-described embodiment with reference to FIGS. **1** to **3**, and therefore descriptions thereof are omitted here.

FIG. **5** is a front view of a high-frequency transmission line according to an embodiment of the present disclosure, where a PCB in an electronic device is used for the high-frequency transmission line. The electronic device may be a smart phone or a wearable device.

Referring to FIG. **5**, an electronic device may include a housing corresponding to a case; a printed circuit unit including at least one PCB **500** in the housing; and an electrical path **550** passing over or through the PCB **500**.

The PCB **500** according to an embodiment of the present disclosure may include a first PCB **510** and a second PCB **520**. For example, the first PCB **510** may be a main board on which electronic components included in an electronic device are mounted, the main board being capable of transmitting signals between the electronic components, and the second PCB **520** may be a sub-board on which the electronic components are mounted, the sub-board being capable of transmitting signals generated by the electronic components. The sub-board may further include the electronic components.

The PCB **500** according to an embodiment of the present disclosure may be provided with a circuit of a terminal, which may be, for example, at least one of an AP, a CP, a memory, and an RF transceiver, and may include a signal line (for example, an RF signal line).

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The first and second PCBs **510** and **520** according to an embodiment of the present disclosure may be formed of a plurality of layers, and the plurality of layers may include one or more conductive layers and one or more insulation layers.

The first PCB **510** according to an embodiment of the present disclosure may be a rigid PCB. The first PCB **520** may be electrically connected to the second PCB **510** through a connection unit **570**. The first PCB **510** may be connected to the second PCB **520**, and the connector **570** may be disposed to transmit a high-frequency signal through the electrical path **550**. The connector **570** may include a soldering pad, a zip-type connector, or a B-to-B-type connector. A rigid PCB may be employed for the connector **570**, and, for example, a B-to-B-type connector may be mounted on the rigid PCB, or a soldering pad may be located on the rigid PCB.

The connector **570** according to an embodiment of the present disclosure may include a ground pad **590** and a signal pad **591** of the first PCB **510** and a ground pad **592** and a signal pad **593** of the second PCB **520**. The ground pad **590** of the first PCB **510** and the ground pad **592** of the second PCB **520** may be connected by soldering. The signal pad **591** of the first PCB **510** and the signal pad **593** of the second PCB **520** may be connected together by soldering. The second PCB **520** may include an interconnection pad **594**. An antenna contact member (for example, a C-clip) or an RF component (for example, a matching circuit) may be mounted on the interconnection pad **594**.

The second PCB **520** according to an embodiment of the present disclosure may include an FPCB **530** and a rigid PCB **540** that are integrated with each other. For example, the second PCB **520** may include a first FPCB **531**, a first rigid PCB **541**, and a second FPCB **532**, which are arranged in sequence. The second FPCB **532** may include an antenna connector and a USB connector. Unlike in an embodiment described above with reference to FIG. **1**, the second FPCB **532** in FIG. **5** is configured to perform the function of the second rigid PCB **342** in FIG. **1** as well.

The electrical path **550** in FIG. **5** according to an embodiment of the present disclosure may extend from a first electrical component including an antenna radiator to a second electrical component including a communication circuit, and may be disposed on or in the PCB **500**. The electrical path **550** may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type, the transmission line being shielded by a ground line in order to transmit a high-frequency signal, which may be, for example, an RF signal.

The electrical path **550** according to an embodiment of the present disclosure, which is disposed on or in the second PCB **520**, may receive a high-frequency signal through the connector **570** and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths **550** may be provided, and a ground line **560** may be disposed between the electrical paths **550** so that the electrical paths **550** can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground line **560** according to an embodiment of the present disclosure may be disposed adjacent to the electrical path **550**, and conductive vias **505** may be arranged in a portion of the area of the ground line **560**. Accordingly, the electrical path **550** can pass through the PCB **500** without making contact with the conductive vias **505**.

A plurality of layers of the PCB **500** and the conductive vias **505** are the same as those in an above-described

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embodiment with reference to FIGS. 1 to 3, and therefore descriptions thereof are omitted here.

FIGS. 6A, 6B, and 6C are front views of conductive vias according to an embodiment of the present disclosure, the conductive vias being arranged in electronic device PCBs. FIGS. 6A, 6B, and 6C illustrate various forms of PCBs.

Referring to FIG. 6A, a second PCB 620 according to an embodiment of the present disclosure may include an FPCB 630 and a rigid PCB 640, which are integrated with each other. For example, the second PCB 620 may include a first FPCB 631, the first rigid PCB 640, and a second FPCB 632, which are arranged in sequence.

An electrical path 650 according to an embodiment of the present disclosure may be disposed on or in the second PCB 620 and may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type in order to transmit a high-frequency signal.

The electrical path 650 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 620, may receive a high-frequency signal through a connector and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 650 may be provided, and a ground line 660 may be disposed between the electrical paths 650 so that the electrical paths 650 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground lines 660 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 650 and may be connected with one or more conductive layers through conductive vias 605(c) in the PCB. Since the conductive vias 605(c) are arranged in one ground line 660, the electrical path 650 may pass through the PCB without making contact with the conductive vias 605(c).

The electrical path 650 according to an embodiment of the present disclosure is disposed to pass through the flexible PCB 630 and the rigid PCB 640, and thus the ground lines 660 disposed on opposite sides of the electrical path 650 may also be disposed to pass through the FPCB 630 and the rigid PCB 640. While two ground lines 660 are provided in FIG. 6A, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

For example, the first ground line 660(a) disposed on the left side may have a greater width than the second ground line 660(b) disposed on the right side, and the plurality of conductive vias 605(c) may be formed in the region where the rigid PCB 640 and the ground line 660(a) cross each other. The second ground line 660(b) may not include the conductive vias 605(c) since the second ground line 660(b) has a lesser width than the first ground line 660(a).

The PCB is illustrated as having the conductive vias 605(c) arranged in the first ground line 660(a) having a great width. The conductive vias 605(c) arranged in the first ground line 660(a) may include vias passing through all layers. Ground patterns in all or some of the layers and the ground line on a side of the electrical path may be connected by the conductive vias 605(c) passing through all the layers.

Referring to FIG. 6B, ground lines 660 may be separately disposed on opposite sides of an electrical path 650 in the second PCB 620 and may be formed to pass through the flexible PCB and the rigid PCB along the lengthwise direction of the electrical path 650. While the two ground lines 660 are provided in FIG. 6B, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 660(a) disposed on the left side may

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have a greater width than the second ground line 660(b) disposed on the right side, and a plurality of conductive vias 605(c) may be formed in the region where the first ground line 660(a) and the rigid PCB 640 cross each other. The second ground line 660(b) on the right side may not include the conductive vias 605(c) having the size since the second ground line 660(b) has a lesser width than the first ground line 660(a). Instead, the second ground line 660(b) on the right side may have conductive vias 605(d) having a lesser size than the conductive vias 605(c).

For example, the rigid PCB 640 formed of a plurality of layers may include the third conductive vias 605(c) passing through all of the layers in the PCB and the fourth conductive vias 605(d) passing through some of the layers. The third conductive vias 605(c) may have a greater hole size than the fourth conductive vias 605(d) passing through some of the layers, and may be arranged in the ground line 660(a) having a great width. However, the fourth conductive vias 605(d) may be arranged in the second ground line 660(b) having a lesser width and disposed on the right side, since the fourth conductive vias 605(d) have a lesser hole size than the third conductive vias 605(c). Even when the ground lines 660 are connected with a ground in a different layer through some conductive vias 605, it is possible to reduce line loss.

Referring to FIG. 6C, ground lines 660 may be separately disposed on opposite sides of an electrical path 650 and may be formed in the lengthwise direction of the electrical path 650. While two ground lines 660 are provided in FIG. 6C, the number of ground lines 660 may be diversely varied so as to avoid contact with the electrical path 650, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 660(a) disposed on the left side may have a lesser width than the second ground line 660(b) disposed on the right side, and a plurality of conductive vias 605(d) may be formed in the region where the second ground line 660(b) and the rigid PCB 640 cross each other. The second ground line 660(b) on the right side has a greater width than the first ground line 660(a), but is too narrow to have the great conductive vias 605(c). Instead, the conductive vias 605(d) having a lesser width may be arranged in the second ground line 660(b) on the right side.

Therefore, according to an embodiment of the present disclosure, the second ground line 660(b) having a lesser width and disposed on the right side may electrically connect the PCB using the fourth conductive vias 605(d) passing through some layers. Even when the ground lines 660 are connected with a ground in a different layer through some of the fourth conductive vias 605(d), it is possible to reduce line loss.

FIGS. 7A, 7B, and 7C are top views illustrating conductive vias arranged in PCBs according to an embodiment of the present disclosure. FIGS. 7A, 7B, and 7C illustrate various forms of PCBs.

Referring to FIG. 7A, a second PCB 720 may include a plurality of FPCBs and/or a plurality of rigid PCBs. For example, the second PCB 720 may be configured such that FPCBs 730 and rigid PCBs 740 are alternately disposed and are integrated with each other.

The FPCBs 730 according to an embodiment of the present disclosure may include a first FPCB 731, a second FPCB 732, and a third FPCB 733. The rigid PCBs 740 may include a first rigid PCB 741 and a fifth rigid PCB 742.

According to an embodiment of the present disclosure, at least two rigid PCBs 740 may be provided between the FPCBs 730. For example, the second PCB 720 may include

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the first FPCB 731, the first rigid PCB 741, the second FPCB 732, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

The first and fifth rigid PCBs 741 and 742 may be disposed to be spaced apart from each other and may include a plurality of conductive vias 705(c) and ground lines 760.

An electrical path 750 according to an embodiment of the present disclosure may be disposed on or in the second PCB 720 and may be implemented to be a transmission line of a side-guarded micro-strip-line type or a strip-line type in order to transmit a high-frequency signal.

The electrical path 750 according to an embodiment of the present disclosure, which is disposed on or in the second PCB 720, may receive a high-frequency signal through a connector and may forward the received signal to an antenna radiator. Furthermore, a plurality of electrical paths 750 may be provided, and the ground line 760 may be disposed between the electrical paths 750 so that the electrical paths 750 can independently transmit the high-frequency signal without mutually affecting the characteristics thereof.

The ground lines 760 according to an embodiment of the present disclosure may be disposed adjacent to the electrical path 750 and may be connected with one or more conductive layers through the conductive vias 705(c) in the PCB 720.

The ground lines 760 according to an embodiment of the present disclosure may be formed to pass through the FPCB 730 and the rigid PCBs 740 along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the present disclosure, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

For example, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a greater width than the second ground line 760(b) disposed on the right side, and the plurality of conductive vias 705(c) may be formed in the regions where the first ground line 760(a) crosses the first and fifth rigid PCBs 741 and 742. In another example, the second ground line 760(b) on the right side may not include the conductive vias 705(c) having the size since the second ground line 760(b) has a lesser width than the first ground line 760(a). The PCB is illustrated as having the conductive vias 705(c) arranged in the first ground line 760(a) having a great width.

Referring to FIG. 7B, the second PCB 720 may include the FPCBs 730 and the rigid PCBs 740 that are integrated with each other. For example, the second PCB 720 may be configured such that the FPCBs 730 and the rigid PCBs 740 are alternately disposed and are integrated with each other.

The second PCB 720 according to an embodiment of the present disclosure may include the first FPCB 731, the first rigid PCB 741, the second FPCB 732, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

Ground lines 760 according to an embodiment of the present disclosure may be separately disposed on opposite sides of the electrical path 750 and may be formed along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the FIG. 7B, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a greater width than the second ground line 760(b) disposed on the right side, and a plurality of conductive vias 705(c) may be formed in the regions where the first ground line 760(a)

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crosses the rigid PCBs 740. However, the second ground line 760(b) on the right side may not include the conductive vias 705(c) having the size since the second ground line 760(b) has a lesser width than the first ground line 760(a). Instead, the second ground line 760(b) on the right side may have conductive vias 705(d) having a lesser size than the conductive vias 705(c).

For example, the rigid PCBs 740 formed of a plurality of layers may include the third conductive vias 705(c) passing through all the layers in the PCB and the fourth conductive vias 705(d) passing through some of the layers. The third conductive vias 705(c) may have a greater hole size than the fourth conductive vias 705(d) passing through some of the layers, and may be arranged in the ground line 760(a) having a great width. However, the fourth conductive vias 705(d) may be arranged in the second ground line 760(b) having a lesser width and disposed on the right side, since the fourth conductive vias 705(d) have a lesser hole size than the third conductive vias 705(c). Even when the ground lines 760 are connected with a ground in a different layer through some conductive vias 705, it is possible to reduce a line loss.

Referring to FIG. 7C, the second PCB 720 may include the FPCBs 730 and the rigid PCBs 740 that are alternately disposed and are integrated with each other. In this embodiment, at least two rigid PCBs 740 may be provided between the FPCBs 730. For example, the second PCB 720 may include the first FPCB 731, the second FPCB 732, the first rigid PCB 741, the fifth rigid PCB 742, and the third FPCB 733, which are arranged in sequence.

Ground lines 760 according to an embodiment of the present disclosure may be separately disposed on opposite sides of the electrical path 750 and may be formed along the lengthwise direction of the electrical path 750. While two ground lines 760 are provided in the present disclosure, the number of ground lines 760 may be diversely varied so as to avoid contact with the electrical path 750, without being limited thereto.

According to an embodiment of the present disclosure, the first ground line 760(a) disposed on the left side of the first and fifth rigid PCBs 741 and 742 may have a lesser width than the second ground line 760(b) disposed on the right side, and a plurality of conductive vias 705(d) may be formed in the regions where the second ground line 760(b) crosses the rigid PCBs 740. The second ground line 760(b) on the right side has a greater width than the first ground line 760(a), but is too narrow to have the great conductive vias 705(c). Instead, the conductive vias 705(d) having a lesser width may be arranged in the second ground line 760(b) on the right side.

Therefore, according to an embodiment of the present disclosure, even when the ground lines 760 are connected to a ground in a different layer through some of the fourth conductive vias 705(d) arranged in the second ground line 760(b), it is possible to reduce a line loss.

FIG. 8 is a perspective view of an electronic device according to an embodiment of the present disclosure. The electronic device 10 may be a smart phone or a wearable device. The electronic device 10 may be an electronic device that includes the PCBs described above with reference to FIGS. 1 to 7.

Referring to FIG. 8, a display 101 may be mounted on the front 107 of the electronic device 10. A speaker device 102 for receiving a counterpart's speech may be provided on the upper side of the display 101. A microphone device 103 for transmitting the speech of a user of the electronic device 10 to the counterpart may be provided on the lower side of the display 101.

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According to an embodiment of the present disclosure, one or more components for performing various functions of the electronic device **10** may be arranged around the speaker device **102**. For example, the one or more arranged components may include at least one sensor module **104**. The sensor module **104** may include, for example, at least one of an illuminance sensor (e.g., an optical sensor), a proximity sensor, an infrared sensor, and an ultrasonic sensor. The components may also include a camera device **105**. The components may also include a light emitting diode (LED) indicator **106** for informing a user of status information of the electronic device **10**.

According to an embodiment of the present disclosure, the electronic device **10** may include a metal bezel **110** (for example, capable of serving as at least a part of a metal housing). The metal bezel **110** may be disposed along the outer periphery of the electronic device **10** and may extend to at least one area of the back of the electronic device **10**, the at least one area being connected to the outer periphery. The metal bezel **110** may define the thickness of the electronic device **10** along the outer periphery of the electronic device **10** and may be formed in a loop shape. Without being limited thereto, however, the metal bezel **110** may also be formed in such a manner that the metal bezel contributes to at least part of the thickness of the electronic device **10**. The metal bezel **110** may also be disposed only in at least one area of the outer periphery of the electronic device **10**. The metal bezel **110** may include one or more cut-off portions **115** and **116**. Unit bezel parts separated by the cut-off portions **115** and **116** may be used as an antenna radiator that operates in at least one frequency band.

According to an embodiment of the present disclosure, the metal bezel **110** may have a loop shape along the outer periphery of the electronic device and may be disposed to contribute to the entirety or a part of the thickness of the electronic device **10**. When the electronic device **10** is viewed from the front, the metal bezel **110** may include a right bezel part **111**, a left bezel part **112**, an upper bezel part **113**, and a lower bezel part **114**. In this case, the lower bezel part **114** described above may serve as a unit bezel part formed by a pair of cut-off portions **116**.

According to an embodiment of the present disclosure, a main antenna device may be disposed in the lower region (an antenna region of a main part) of the electronic device **10**. The lower bezel part **114** may be used as a main antenna radiator by virtue of the pair of cut-off portions **116**. The lower bezel part **114** may serve as an antenna radiator that operates in at least two operating frequency bands depending on the feeding position. For example, the lower bezel part **114** may be a part of an antenna that supports a low band (LB) and a high/middle band (H/MB), or a part of an antenna that supports an H/MB.

According to an embodiment of the present disclosure, the antenna device of the present disclosure is simply for illustrative purpose, and the aforementioned functions of the lower bezel part **114** may be performed by the upper bezel part **113** separated by the other cut-off portions **115**, or may be performed by the upper and lower bezel parts.

According to an embodiment of the present disclosure, an antenna region of a diversity part may be used as an antenna for diversity/multiple input multiple output (MIMO). For example, the upper bezel part **113** may be a part of a diversity antenna that supports LB and H/MB.

According to an embodiment of the present disclosure, at least one of the right and left bezel parts **111** and **112** may be supplied with electrical power to operate as an antenna. For example, the right or left bezel part **111** or **112** may be

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a part of an antenna that supports an H/MB band or LB and H/MB bands. For example, an antenna including the right or left bezel part **111** or **112** included in the antenna region of the main part may operate as a main antenna. An antenna including the right or left bezel part **111** or **112** included in the antenna region of the diversity part may operate as a diversity antenna. Hereinafter, the configuration of an antenna is described in detail.

FIG. 9 is a schematic of an antenna arrangement in an electronic device **10** according to an embodiment of the present disclosure. The electronic device **10** may be a smart phone or a wearable device. The electronic device **10** may be an electronic device that includes the PCBs described above with reference to FIGS. 1 to 7.

The electronic device **10** according to an embodiment of the present disclosure may include a plurality of antenna radiators **230**. For example, the electronic device **10** may include first and second antennas **231** and **232** of a main part **210** and third and fourth antennas **233** and **234** of a diversity part **220**.

The first antenna **231** may support LB and H/MB bands, and the second antenna **232** may support an H/MB band. According to an embodiment of the present disclosure, an LB band has a relatively great wavelength so that an antenna may have a relatively great size, and there may be a mounting space limitation by adding an antenna supporting multiple LBs to the main part **210** of the terminal. Therefore, only one antenna may be configured to support an LB band.

In the case where the electronic device **10** additionally has the diversity part **220** according to an embodiment of the present disclosure, the electronic device **10** may include the third and fourth antennas **233** and **234**. In order to enhance correlation/isolation characteristics, the antenna radiators **230** of the main part **210** may be included in the lower end portion of the terminal, and the antenna radiators **230** of the diversity part **220** may be included in the upper end portion of the terminal. For example, in the case where the antenna radiators **230** of the main part **210** are included in the lower end portion of the terminal, and the antenna radiators **230** of the diversity part **220** are included in the upper end portion of the terminal, it is possible to ensure an antenna separation distance available in the terminal even when a plurality of LB band antennas are included.

By virtue of the structure described above, the electronic device **10** can receive signals using four antennas in an H/MB band and can receive signals using two antennas in an LB band. That is, 4th order diversity/MIMO can be performed in an H/MB band, and 2nd order diversity/MIMO can be performed in an LB band.

FIG. 10 illustrates an antenna device and a PCB of an electronic device according to an embodiment of the present disclosure. Referring to FIG. 10, a main RF circuit in the main part **210** may supply electrical power to the first and second antennas **231** and **232** connected thereto through the connector **370** and the second PCB **320** using RF signal lines. The second PCB **320** may include the first FPCB **331**, the first rigid PCB **341**, the second FPCB **332**, and the second rigid PCB **342**. A ground may be removed from an antenna feeding part or from the surroundings of an antenna in order to enhance antenna characteristics.

According to an embodiment of the present disclosure, the metal bezel **110** may include the right bezel part **111**, the left bezel part **112**, the upper bezel part **113**, and the lower bezel part **114** when viewed from the front. The upper bezel part **113** may be maintained separately from the right and left bezel parts **111** and **112** by a pair of cut-off portions that are formed with a predetermined interval therebetween. The

lower bezel part **114** may be maintained separately from the right and left bezel parts **111** and **112** by a pair of cut-off portions that are formed with a predetermined interval therebetween. The pair of cut-off portions may be formed of a dielectric material. The pair of cut-off portions may be formed by double-injection molding or insert molding a synthetic resin into the metal bezel. Without being limited thereto, however, various types of electrical insulating materials may be applied to the pair of cut-off portions.

According to an embodiment of the present disclosure, a first feeding piece may be integrally formed with the lower bezel part **114** and may be supplied with electrical power by a first feeding part of the PCB. The first feeding piece of the lower bezel part **114** may be connected to the first feeding part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a first feeding pad may be disposed on the PCB and may be electrically connected to the first feeding piece of the lower bezel part **114**. A first electrical path (e.g., an interconnection wire line) may be formed from the first feeding pad to the first feeding part. The first feeding piece may be integrally formed with the lower bezel part **114** and may be supplied with electrical power by the first feeding part of the PCB. The first feeding piece of the lower bezel part **114** may be connected to the first feeding part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, the first feeding pad may be disposed on the PCB and may be electrically connected to the first feeding piece of the lower bezel part **114**. The first electrical path (e.g., an interconnection wire line) may be formed from the first feeding pad to the first feeding part.

The lower bezel part may be a portion of the first antenna **231** of the main part **210** that supports an LB or H/MB band. The left and right bezel parts **112** and **111** may also be supplied with electrical power by the same method. The left and right bezel parts **112** and **111** may be portions of the second antenna **231** of the main part **210** that supports an LB or H/MB band.

According to an embodiment of the present disclosure, a first electrical connection piece may be integrally formed with the lower bezel part **114** so as to be located at a position separate from the first feeding piece by a predetermined distance, and may be grounded to a first ground part of the PCB. The first electrical connection piece of the lower bezel part **114** may be grounded to the first ground part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the first ground part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a first ground pad may be disposed on the PCB and may be electrically connected to the first electrical connection piece of the lower bezel part **114**. A second electrical path (e.g., an interconnection wire line) may be formed from the first ground pad to the first ground part.

According to an embodiment of the present disclosure, a second feeding piece may be integrally formed with the lower bezel part **114** and may be supplied with electrical power by a first feeding part of the PCB. The second feeding piece of the lower bezel part **114** may be connected to a second feeding part of the PCB merely by installing the PCB

in the electronic device, or may be electrically connected to the second feeding part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a second feeding pad may be disposed on the PCB and may be electrically connected to the second feeding piece of the lower bezel part **114**. A third electrical path (e.g., an interconnection wire line) may be formed from the second feeding pad to the second feeding part. The lower bezel part **114** may be a portion of the second antenna **232** of the main part **210** that supports an LB or H/MB band.

According to an embodiment of the present disclosure, a second electrical connection piece may be integrally formed with the right bezel part **111** so as to be located at a position separate from the cut-off portions by a predetermined distance, and may be grounded to a second ground part of the PCB. The second electrical connection piece of the right bezel part **111** may be grounded to a second ground part of the PCB merely by installing the PCB in the electronic device, or may be electrically connected to the second ground part by a separate electrical connection member (e.g., a C-clip, etc.).

According to an embodiment of the present disclosure, a second ground pad may be disposed on the PCB and may be electrically connected to the second electrical connection piece of the right bezel part **111**. A fourth electrical path (e.g., an interconnection wire line) may be formed from the second ground pad to the second ground part.

For example, the feeding parts, the feeding pads, the ground parts, and the ground pads of the main part **210** may be disposed on the second PCB **320**. The main part **210** and an RF circuit of the diversity part **220** may be disposed on the first PCB (main PCB) **310**. The first and second PCBs may be connected by an FPCB. The first and second PCBs may be integrated with each other.

According to an embodiment of the present disclosure, the second rigid PCB **342** included in the second PCB **320** may be disposed at a lower vertical position than the first PCB (main PCB) **310**. Accordingly, components included in the second rigid PCB **342** may be additionally spaced apart from the antenna. Further, relatively thick components, such as a USB connector and a speaker, can also be disposed on the second rigid PCB **342**.

According to an embodiment of the present disclosure, a transmitted or received signal of the RF circuit of the main part **210** may be forwarded to the first and second feeding parts of the second PCB **320**.

The diversity part **220** may include the third and fourth antennas **233** and **234**. The third antenna **233** may include a part of the upper bezel part **113**, and the fourth antenna **234** may include the left or right bezel part **112** or **111**.

The third antenna **233** may support LB and H/MB bands, and the fourth antenna **234** may support an H/MB band. Feeding parts, feeding pads, ground parts, and ground pads of the diversity part **220** may be disposed on the first PCB (main PCB) **310**. Electrical paths connecting the feeding parts and the feeding pads and electrical paths connecting the ground parts and the ground pads in the diversity part **220** may be disposed on the first PCB (main PCB) **310**.

According to an embodiment of the present disclosure, when the main part **210** uses the right bezel part **111** as the second antenna **232** for the purpose of signal separation between the antennas, the diversity part **220** may use the opposite left bezel part **112** as the fourth antenna **234**. Alternately, when the main part **210** uses the left bezel part

112 as the second antenna 232, the diversity part 220 may use the opposite right bezel part 111 as the fourth antenna 234.

FIG. 11 is an illustration of a signal transmission using the second PCB 320 according to an embodiment of the present disclosure. The second PCB 320 may be the same as the second PCBs 320, 420, and 520 described above with reference to FIGS. 1 to 7.

Referring to FIG. 11, the second PCB 320 may include the FPCB 330 and the rigid PCB 340, which are integrated with each other. The second PCB 320 may include the first FPCB 331, the second FPCB 332, the first rigid PCB 341, and the second rigid PCB 342.

According to an embodiment of the present disclosure, the electrical path 350 for transmitting a high-frequency signal may also transmit signals other than RF signals using the second PCB 320. The other signals may include, for example, a USB signal, an ear-jack signal, a microphone (MIC) signal, a speaker signal, and a touch key signal.

According to an embodiment of the present disclosure, an antenna contact to which an RF signal is connected, and a USB connector, an ear-jack connector, a microphone (MIC) connector, and a speaker contact to which the other signals are connected may be disposed on the second rigid PCB 342. For example, RF signals may be transmitted through two electrical paths 350: the first electrical path 351 and the second electrical path 352. The ground line 360 may be disposed between the first and second electrical paths 351 and 352 to prevent the first and second electrical paths 351 and 352 from mutually affecting the characteristics thereof. The conductive vias 305 may be arranged in the region where the ground line 360 and the first rigid PCB 341 cross each other, and may be connected to a ground in a different layer. A rigid PCB connected to the rear end of the second rigid PCB 342 may be various types of modules of the electronic device, and may function as a touch key, for example.

According to an embodiment of the present disclosure, a first connector 342(a) connected to the first electrical path 351 may be a first antenna contact, a first USB connector, a first ear-jack connector, a first MIC connector, or a first speaker contact, and a second connector 342(b) connected to the second electrical path 352 may be a second antenna contact, a second USB connector, a second ear-jack connector, a second MIC connector, and a second speaker contact.

FIGS. 12 and 13 are enlarged views of a first FPCB and a first rigid PCB. The second PCB 320 may be one of the second PCBs 320, 420, 520, 620, and 720 described above with reference to FIGS. 1 to 7.

Referring to FIGS. 12 and 13, a high-frequency signal may be connected from a B-to-B-type connector A of the connector 370 to antenna (ANT) feeding B and C by a micro strip line or a strip line on the FPCB. Further, the ground lines 306 may exist on the left and right sides of the electrical path 350 through which an RF signal is transmitted on the connector, and may be formed parallel to the electrical path 350 along the lengthwise direction of the electrical path 350 so as to be connected to the ANT feeding B and C. In addition, the conductive vias 305 may be formed in the ground lines 306. Conductive vias 307 lesser in size than the conductive vias 305 may be formed on the right side of the electrical path 351 so as to prevent contact between the ground lines and a signal line on one side.

An electronic device according to an embodiment of the present disclosure includes a housing; a PCB disposed in the housing, the PCB including a plurality of layers that include

one or more conductive layers and one or more insulation layers; a first electrical component formed to be at least a part of the housing or disposed in the housing; a second electrical component disposed above or near the PCB in the housing, the second electrical component being separated from the first electronic component; and at least one electrical path extending from the first electrical component to the second electronic component, wherein at least a portion of the electrical path runs on or inside the PCB, wherein the PCB may include a region including a pattern of conductive vias, each of the vias extending through at least a part of the plurality of layers to contact at least one of the one or more conductive layers, and wherein the electrical path may run through the region without making contact with the conductive vias.

According to an embodiment of the present disclosure, the first electrical component may include an antenna radiator and the second electrical component may include a communication circuit.

According to an embodiment of the present disclosure, the antenna radiator may form at least one part of the housing.

According to an embodiment of the present disclosure, the pattern of the conductive vias may be aligned in one or more rows or columns along the electrical path when viewed from above the region of the PCB.

According to an embodiment of the present disclosure, the PCB may include an FPCB as at least a part thereof, where the FPCB may transmit a signal through the electrical path.

According to an embodiment of the present disclosure, the PCB may include a first PCB; and a second PCB connected to the first PCB through a connector, where the second PCB includes at least one FPCB and at least one rigid PCB, and where the electrical path may transmit a high-frequency signal through the connector and may pass over or through the second PCB.

According to an embodiment of the present disclosure, the electronic device may further include a ground line disposed around the electrical path so as to be adjacent thereto, where the ground line may be connected to the one or more conductive layers through the conductive vias in the rigid PCB.

According to an embodiment of the present disclosure, the second PCB may include a first FPCB connected to the first PCB through a communication circuit; a second PCB disposed to face the first flexible PCB with a separation distance therebetween; and a first rigid PCB disposed between the first and second FPCBs, where the first rigid PCB includes the pattern of the conductive vias.

According to an embodiment of the present disclosure, the electronic device may further include a second rigid PCB connected to the second FPCB, wherein a part of the antenna radiator that forms at least one portion of the housing may be disposed on the second rigid PCB.

According to an embodiment of the present disclosure, at least one of the plurality of layers may form a ground pattern in the ground line of the second PCB, where the entire line of the ground pattern includes a ground.

According to an embodiment of the present disclosure, the conductive vias of the first rigid PCB may include a first conductive via passing through all of the plurality of layers; and a second conductive via passing through some of the plurality of layers, and where the first conductive via may connect the ground pattern disposed on all or some of the layers and the ground line.

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According to an embodiment of the present disclosure, a plurality of ground lines may be formed on the rigid PCB so as to be spaced apart from the electrical path, and may have different widths.

According to an embodiment of the present disclosure, the conductive vias formed along the width of the ground line may include a third conductive via having a first hole size; and a fourth via having a second hole size that differs from the first hole size.

According to an embodiment of the present disclosure, a space for mounting an electronic component may be formed on or in the rigid PCB of the second PCB.

According to an embodiment of the present disclosure, the connector may be disposed on one side of the first or second PCB, where the electrical path passing over or through the second PCB may be connected to the first PCB.

According to an embodiment of the present disclosure, the connector may include a first connector disposed on one end of the first FPCB of the second PCB; and a second connector disposed on one end of the second FPCB of the second PCB.

According to an embodiment of the present disclosure, the electronic device may include a plurality of electrical paths and a plurality of antenna radiators, wherein the electrical paths may be connected to the respective antenna radiators corresponding thereto, and the ground line may be disposed between the plurality of electrical paths.

According to an embodiment of the present disclosure, the first PCB may include at least a part of the first electrical component and may be connected to the antenna radiator, and the second PCB may have the second electrical component disposed thereon.

According to an embodiment of the present disclosure, the FPCB and the rigid PCB of the second PCB may be integrated with each other.

According to an embodiment of the present disclosure, the region of the PCB may be a rigid PCB region, and the electronic device may include a first FPCB region disposed between the first electrical component and the region; and a second FPCB region disposed between the second electrical component and the region.

An electronic device according to an embodiment of the present disclosure includes a first PCB; a second PCB connected with the first PCB through a connector and including at least one FPCB and at least one rigid PCB that are integrated with each other, where the flexible and rigid PCBs include a plurality of layers; an electrical path that transmits a high-frequency signal through the connector and passes over or through the second PCB; and at least one conductive via disposed in the rigid PCB to connect grounds between the plurality of layers.

FIG. 14 is a block diagram of an electronic device 10, according to an embodiment of the present disclosure, in a network environment 11.

Referring to FIG. 14, the electronic device 10 may include a bus 180, a processor 120, a memory 130, an input/output interface 150, a display 160, and a communication interface 170. The electronic device 10 may omit at least one of the elements, or may further include other elements.

The bus 180 may include, for example, a circuit that interconnects the elements 120 to 170 and delivers communication (for example, a control message and/or data) between the elements.

The processor 120 may include one or more of a CPU, an AP, and a CP. For example, the processor 120 may carry out

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operations or data processing relating to the control and/or communication of at least one other element of the electronic device 10.

The memory 130 may include a volatile and/or non-volatile memory. For example, the memory 130 may store instructions or data relevant to at least one other element 120 to 170 of the electronic device 10. The memory 130 may store software and/or a program 140. The program 140 may include, for example, a kernel 141, middleware 143, an application programming interface (API) 145, and/or application programs (or applications) 147. At least some of the kernel 141, the middleware 143, and the API 145 may be referred to as an operating system (OS).

For example, the kernel 141 may control or manage system resources (for example, the bus 180, the processor 120, and the memory 130) that are used to execute operations or functions implemented by the other programs (for example, the middleware 143, the API 145, and the applications 147). Furthermore, the kernel 141 may provide an interface by which the middleware 143, the API 145, or the applications 147 access the individual elements of the electronic device 10 to control or manage the system resources.

For example, the middleware 143 may serve as an intermediary to allow the API 145 or the applications 147 to exchange data with the kernel 141 through communication.

Furthermore, the middleware 143 may process one or more task requests received from the applications 147 according to the priorities of the requests. For example, the middleware 143 may assign, to at least one of the applications 147, a priority to use the system resources (for example, the bus 180, the processor 120, and the memory 130) of the electronic device 10. For example, the middleware 143 may perform scheduling or loading balancing on the one or more task requests by processing the one or more task requests according to the priority assigned to the at least one of the applications.

The API 145 is, for example, an interface by which the applications 147 control functions provided by the kernel 141 or the middleware 143, and the API 145 may include at least one interface or function (for example, an instruction) for file control, window control, image processing, or text control, for example.

The input/output interface 150 may serve as, for example, an interface that can forward instructions or data input from a user or an external device to the other element(s) 120 to 170 of the electronic device 10. Furthermore, the input/output interface 150 may output instructions or data received from the other element(s) of the electronic device 10 to the user or the external device.

Examples of the display 160 may include a liquid crystal display (LCD), an LED display, an organic LED (OLED) display, a microelectromechanical systems (MEMS) display, and an electronic paper display. For example, the display 160 may display various types of content (for example, text, images, videos, icons, or symbols) for a user. The display 160 may include a touch screen, and may receive, for example, a touch, gesture, a proximity input, or a hovering input using an electronic pen or a part of a user's body.

For example, the communication interface 170 may establish communication between the electronic device 10 and a first external electronic device 12, a second external electronic device 14, or a server 16. For example, the communication interface 170 may be connected to a network 162 through wireless or wired communication to communicate with the second external electronic device 14 or the server 16.

The wireless communication may use, for example, at least one of long term evolution (LTE), LTE-advanced (LTE-A), code division multiple access (CDMA), wideband CDMA (WCDMA), universal mobile telecommunications system (UMTS), wireless broadband (WiBro), and global system for mobile communications (GSM), for example, as a cellular communication protocol. Furthermore, the wireless communication may include short range communication **164**, for example. The short range communication **164** may include, for example, at least one of wireless fidelity (WiFi), Bluetooth (BT), near field communication (NFC), and global navigation satellite system (GNSS). Depending on the service area and the bandwidth, the GNSS may include, for example, at least one of a global positioning system (GPS), a global navigation satellite system (Glonass), a Beidou navigation satellite system (Beidou), and a European global satellite-based navigation system (Galileo). Hereinafter, "GPS" may be interchangeably used with "GNSS." The wired communication may include, for example, at least one of a USB, a high definition multimedia interface (HDMI), recommended standard 232 (RS-232), and a plain old telephone service (POTS). The network **162** may include at least one of a communication network such as a computer network (e.g., a local area network (LAN) or a wide area network (WAN)), the Internet, and a telephone network.

Each of the first and second external electronic devices **12** and **14** may be of the same or a different type from the electronic device **10**. According to an embodiment of the present disclosure, the server **16** may include a group of one or more servers. All or some of the operations executed in the electronic device **10** may be executed in the electronic device **102**, the electronic device **104**, or the server **16**. When the electronic device **10** must perform a function or service automatically or in response to a request, the electronic device **10** may request the electronic device **12**, the electronic device **14**, or the server **16** to perform at least some functions relating thereto, instead of or in addition to performing the function or service by itself. The electronic device **102**, the electronic device **104**, or the server **106** may execute the requested function or additional functions and then deliver the result to the electronic apparatus **10**. The electronic device **10** may provide a received result as is, or may further process the received result to provide the requested function or service. To this end, for example, cloud computing, distributed computing, or client-server computing technology may be used.

The electronic device **10**, which is described below in greater detail, may be one of a wearable device, a notebook computer, a net-book computer, a smart phone, a tablet PC, a Galaxy tab, an i-Pad, and a wireless charging device mentioned above. In this embodiment, the electronic device **10** may be a smart phone.

The wireless charging device, according to an embodiment of the present disclosure, refers to a device that recharges the electronic device by wirelessly transmitting and receiving power in a short range.

In addition, the display **160** of the electronic device **10** may increase in size and may implement a luxurious design by minimizing the bezel area thereof, or may be implemented to be flexible, convex, or concave.

Namely, the peripheral portion of the display **160** may be bent to allow a view area to extend to the lateral portion. As the view area of the display **160** is bent to extend to the lateral, the view area may be enlarged, a separate screen may be used on the lateral portion, or a luxurious design may be

implemented. In other words, the display **160** may include a first view area and second view areas on opposite sides of the first view area.

FIG. **15** is a block diagram of an electronic device **1901** according to an embodiment of the present disclosure. The electronic device **1901** may include, for example, the entirety or a part of the electronic device **10** described above with reference to FIG. **14**. The electronic device **1901** in FIG. **15** may include at least one processor (e.g., an AP) **1910**, a communication module **1920**, a subscriber identification module card **1924**, a memory **1930**, a sensor module **1940**, an input device **1950**, a display **1960**, an interface **1970**, an audio module **1980**, a camera module **1991**, a power management module **1995**, a battery **1996**, an indicator **1997**, and a motor **1998**.

The processor **1910** may operate, for example, an OS or an application program to control a plurality of hardware or software elements connected to the processor **1910** and to perform various types of data processing and operations. The processor **1910** may be implemented to be a system on chip (SoC), for example. According to an embodiment of the present disclosure, the processor **1910** may further include a graphics processing unit (GPU) and/or an image signal processor. The processor **1910** may also include at least some (for example, a cellular module **1921**) of the elements illustrated in FIG. **15**. The processor **1910** may load, in a volatile memory, instructions or data received from at least one of the other elements (for example, a non-volatile memory) to process the loaded instructions or data, and may store various types of data in the non-volatile memory.

The configuration of the communication module **1920** may be the same as or similar to that of the communication interface **170** of FIG. **14**. The communication module **1920** may include, for example, a cellular module **1921**, a WiFi module **1923**, a BT module **1925**, a GNSS module **1927** (for example, a GPS module, a Glonass module, a Beidou module, or a Galileo module), an NFC module **1928**, and an RF module **1929**.

The cellular module **1921** may provide, for example, a voice call, a video call, a text message service, or an Internet service through a communication network. According to an embodiment of the present disclosure, the cellular module **1921** may identify and authenticate the electronic device **1901** in a communication network using the subscriber identification module card (SIM card) **1924**. The cellular module **1921** may perform at least some of the functions that the processor **1910** can provide. The cellular module **1921** may include a CP.

The WiFi module **1923**, the BT module **1925**, the GNSS module **1927**, and the NFC module **1928** may each include, for example, a processor for processing data transmitted and received through the corresponding module. According to an embodiment of the present disclosure, at least some (for example, two or more) of the cellular module **1921**, the WiFi module **1923**, the BT module **1925**, the GNSS module **1927**, and the NFC module **1928** may be included in one integrated circuit (IC) or IC package.

The RF module **1929** may transmit and receive, for example, a communication signal (for example, an RF signal). The RF module **1929** may include, for example, a transceiver, a power amplifier module (PAM), a frequency filter, a low noise amplifier (LNA), or an antenna. At least one of the cellular module **1921**, the WiFi module **1923**, the BT module **1925**, the GNSS module **1927**, and the NFC module **1928** may transmit and receive an RF signal through a separate RF module.

The subscriber identification module card **1924** may include, for example, an embedded SIM, and may include unique identification information (for example, an integrated circuit card identifier (ICCID)) or subscriber information (for example, an international mobile subscriber identity (IMSI)).

The memory **1930** (for example, the memory **130** in FIG. **14**) may include, for example, an internal memory **1932** or an external memory **1934**. The internal memory **1932** may include, for example, at least one of a volatile memory (for example, a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous DRAM (SDRAM)) and a non-volatile memory (for example, a one-time programmable read only memory (OTPROM), a programmable ROM (PROM), an erasable and programmable ROM (EPROM), an electrically erasable PROM (EEPROM), a mask ROM, a flash ROM, a flash memory (for example, a NAND flash memory or a NOR flash memory), a hard disc drive, or a solid state drive (SSD)).

The external memory **1934** may further include a flash drive, which may be, for example, a compact flash (CF) drive, a secure digital (SD) memory card, a micro SD (Micro-SD) memory card, a mini SD (Mini-SD) memory card, an extreme digital (xD) memory card, a multi-media card (MMC), or a memory stick. The external memory **1934** may be functionally and/or physically connected to the electronic device **1901** through various interfaces.

For example, the sensor module **1940** may measure a physical quantity, or may sense an operating state of the electronic device **1901**, to convert the measured or sensed information into an electrical signal. The sensor module **1940** may include, for example, at least one of a gesture sensor **1940A**, a gyro sensor **1940B**, an atmospheric pressure sensor **1940C**, a magnetic sensor **1940D**, an acceleration sensor **1940E**, a grip sensor **1940F**, a proximity sensor **1940G**, a color sensor **1940H** (for example, a red, green, blue (RGB) sensor), a biometric sensor **1940I**, a temperature/humidity sensor **1940J**, an illuminance sensor **1940K**, and a ultraviolet (UV) light sensor **1940M**. Additionally or alternatively, the sensor module **1940** may include, for example, an electronic nose (E-nose) sensor, an electromyography (EMG) sensor, an electroencephalogram (EEG) sensor, an electrocardiogram (ECG) sensor, an infrared (IR) sensor, an iris sensor, and/or a fingerprint sensor. The sensor module **1940** may further include a control circuit for controlling one or more sensors included therein. In an embodiment of the present disclosure, the electronic device **1901** may further include a processor configured to control the sensor module **1940** as a part of the processor **1910** or separately from the processor **1910** to control the sensor module **1940** while the processor **1910** is in a reduced power or sleep state.

The input device **1950** may include, for example, a touch panel **1952**, a (digital) pen sensor **1954**, a key **1956**, or an ultrasonic input device **1958**. The touch panel **1952** may use, for example, at least one of a capacitive type panel, a resistive type panel, an infrared type panel, and an ultrasonic type panel. Furthermore, the touch panel **1952** may further include a control circuit. The touch panel **1952** may further include a tactile layer to provide a tactile reaction to a user.

The (digital) pen sensor **1954** may include, for example, a recognition sheet that is a part of, or separate from, the touch panel. The key **1956** may include, for example, a physical button, an optical key, or a keypad. The ultrasonic input device **1958** may sense ultrasonic waves generated by an input tool through a microphone **1988** and may identify data corresponding to the sensed ultrasonic waves.

The display **1960** (for example, the display **160** in FIG. **14**) may include a panel **1962**, a hologram device **1964**, or a projector **1966**. The panel **1962** may have a configuration that is the same as, or similar to, that of the display **160** illustrated in FIG. **1**. The panel **1962** in FIG. **15** may be implemented to be, for example, flexible, transparent, or wearable. The panel **1962**, together with the touch panel **1952**, may be implemented to be one module. The hologram device **1964** may show a three-dimensional image in the air by using an interference of light. The projector **1966** may project light onto a screen to display an image. The screen may be located, for example, inside or outside of the electronic device **1901**. According to an embodiment of the present disclosure, the display **1960** may further include a control circuit for controlling the panel **1962**, the hologram device **1964**, and the projector **1966**.

The interface **1970** may include, for example, an HDMI **1972**, a USB **1974**, an optical interface **1976**, or a D-sub-miniature (D-sub) connector **1978**. The interface **1970** may be included, for example, in the communication interface **170** illustrated in FIG. **3**. Additionally or alternatively, the interface **1970** in FIG. **15** may include, for example, a mobile high-definition link (MHL) interface, an SD card/MMC interface, or an Infrared Data Association (IrDA) standard interface.

For example, the audio module **1980** may convert a sound into an electrical signal, and vice versa. At least some elements of the audio module **1980** may be included, for example, in the input/output interface **145** illustrated in FIG. **1**. The audio module **1980** in FIG. **15** may process sound information that is input or output, for example, through a speaker **1982**, a receiver **1984**, an earphone **1986**, or the microphone **1988**.

The camera module **1991** is a device that can photograph a still image and a dynamic image. According to an embodiment of the present disclosure, the camera module **1991** may include one or more image sensors (for example, a front sensor or a rear sensor), a lens, an image sensor processor (ISP), or a flash (for example, an LED or a xenon lamp).

The power management module **1995** may manage, for example, power of the electronic device **1901**. According to an embodiment of the present disclosure, the power management module **1995** may include a power management IC (PMIC), a charger IC, or a battery gauge. The PMIC may use a wired and/or wireless charging method. Examples of the wireless charging method may include a magnetic resonance method, a magnetic induction method, and an electromagnetic wave method. An additional circuit (for example, a coil loop, a resonance circuit, or a rectifier) for wireless charging may be further included. The battery gauge may measure, for example, a residual amount of the battery **1996** and a voltage, current, or temperature while charging. The battery **1996** may include, for example, a rechargeable battery and/or a solar battery.

The indicator **1997** may indicate a particular state (for example, a booting state, a message state, or a charging state) of the electronic device **1901** or a part thereof (for example, the processor **1910**). The motor **1998** may convert an electrical signal into a mechanical vibration and may generate a vibration or a haptic effect. The electronic device **1901** may include a processing unit (for example, a GPU) for supporting mobile TV. The processing unit for supporting mobile TV may process media data according to a standard, such as digital multimedia broadcasting (DMB), digital video broadcasting (DVB), or MediaFlo™.

Each of the above-described component elements of hardware according to the present disclosure may be con-

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figured with one or more components, and the names of the corresponding component elements may vary based on the type of electronic device. The electronic device according to various embodiments of the present disclosure may include at least one of the aforementioned elements. Some elements may be omitted or other additional elements may be further included in the electronic device. In addition, some of the hardware components according to various embodiments may be combined into one entity, which may perform functions identical to those of the relevant components before the combination.

While the present disclosure has been shown and described with reference to certain embodiments thereof, it will be apparent to those skilled in the art that the camera lens module according to the present disclosure is not limited to these embodiments, and various changes in form and details may be made therein without departing from the scope of the present disclosure as defined by the appended claims and their equivalents.

What is claimed is:

1. A portable communication device comprising:

a housing including a first conduction portion forming a first portion of a side surface of the portable communication device and a second conduction portion forming a second portion of the side surface, the first conduction portion operating as a first antenna and the second conduction portion operating as a second antenna;

a wireless communication circuitry; and

a first printed circuit board (PCB) including:

a first PCB portion;

a second PCB portion in contact with the first PCB portion and being more flexible than the first PCB portion;

a third PCB portion in contact with the second PCB portion and being less flexible than the second PCB portion;

a fourth PCB portion in contact with the third PCB portion and being more flexible than the third PCB portion; and

a plurality of lines located from the first PCB portion to the fourth PCB portion through the second PCB portion and the third PCB portion, the plurality of lines including a first ground line, a second ground line at least partially spaced apart from the first ground line, a first signal line located between the first ground line and the second ground line, and a second signal line located between the first signal line and the second ground line, the first signal line electrically connected with the first antenna and the wireless communication circuitry, and the second signal line electrically connected with the second antenna and the wireless communication circuitry.

2. The portable communication device of claim 1, wherein the plurality of lines includes a third ground line located between the first signal line and the second signal line, and

wherein the first PCB further includes an insulating layer on which the first ground line, the second ground line, the third ground line, the first signal line, and the second signal line are located, and a ground plate located below the insulating layer.

3. The portable communication device of claim 2, wherein the ground plate is electrically connected with the first ground line through a first conduction via formed in a first portion of the insulating layer, the second ground line through a second conduction via formed in a second portion of the insulating layer, and the third

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ground line through a third conduction via formed in a third portion of the insulating layer.

4. The portable communication device of claim 2, wherein the first PCB further includes a solder resist ink forming at least a part of an exterior surface of the first PCB portion or the third PCB portion.

5. The portable communication device of claim 1, wherein the first PCB further includes a fifth PCB portion in contact with the fourth PCB portion and being less flexible than the fourth PCB portion,

wherein the first PCB portion further includes a first connector for a connection with the antenna,

wherein the fifth PCB portion includes a second connector for a connection with the wireless communication circuitry, and

wherein a first ending portion of the first signal line and a second ending portion of the second signal line are in contact with the second connector.

6. The portable communication device of claim 1, further comprising:

a second PCB on which the wireless communication circuitry is located,

wherein the first ground line and the second ground line are electrically connected with a ground member located in the second PCB, or

wherein the first ground line and the second ground line are electrically connected with a ground member located in the first PCB portion.

7. A portable communication device comprising:

a first electronic component;

a second electronic component; and

a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:

a first rigid PCB portion;

a first flexible PCB portion in contact with the first rigid PCB portion;

a second rigid PCB portion in contact with the first flexible PCB portion; and

a second flexible PCB portion in contact with the second rigid PCB portion,

wherein each of the first rigid PCB portion and the second rigid PCB portion includes:

a first insulating layer including a first material and forming a first exterior surface of each of the first rigid PCB portion and the second rigid PCB portion;

a first conduction layer located on or above the first insulating layer;

a second insulating layer located on or above the first conduction layer and having a first thickness;

a second conduction layer located on or above the second insulating layer; and

a third insulating layer including the first material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first rigid PCB portion and the second rigid PCB portion, and

wherein each of the first flexible PCB portion and the second flexible PCB portion includes:

a first insulating layer including a second material different from the first material and forming a first exterior surface of each of the first flexible PCB portion and the second flexible PCB portion;

a first conduction layer located on or above the first insulating layer and being electrically connected with

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the first conduction layer of first rigid PCB portion and the first conduction layer of the second rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thinner than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first rigid PCB portion and the second conduction layer of the second rigid PCB portion; and

a third insulating layer including the second material and located on or above the second conduction layer, the third insulating layer forming a second exterior surface of each of the first flexible PCB portion and the second flexible PCB portion.

8. The portable communication device of claim 7, wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

9. The portable communication device of claim 7, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

10. The portable communication device of claim 9, wherein the fourth insulating layer includes the second material.

11. The portable communication device of claim 9, wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion,

wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and

wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via.

12. The portable communication device of claim 7, wherein the first material is a solder resist ink and the second material is a polyimide, and

wherein the second insulating layer of the first rigid PCB portion or the second insulating layer of the second rigid PCB portion includes a third material.

13. The portable communication device of claim 12, wherein the third material includes a resin or a glass fabric.

14. A portable communication device comprising:

a first electronic component;

a second electronic component; and

a printed circuit board (PCB) electrically connected with the first electronic component and the second electronic component, the PCB including:

a first flexible PCB portion including:

a first insulating layer forming a first exterior surface of the first flexible PCB portion and including a first material;

a first conduction layer located on or above the first insulating layer;

a second insulating layer located on or above the first conduction layer having a first thickness;

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a second conduction layer located on or above the second insulating layer; and

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the first flexible PCB portion, the third insulating layer including the first material;

a rigid PCB portion in contact with the first flexible PCB portion, the rigid PCB portion including:

a first insulating layer forming a first exterior surface of the rigid PCB portion and including a second material different from the first material;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the first flexible PCB portion;

a second insulating layer located on or above the first conduction layer and having a second thickness thicker than the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the first flexible PCB portion; and

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the rigid PCB portion, the third insulating layer including the second material;

a second flexible PCB portion in contact with the rigid PCB portion, the second flexible PCB portion including:

a first insulating layer forming a first exterior surface of the second flexible PCB portion and including the first material;

a first conduction layer located on or above the first insulating layer and being electrically connected with the first conduction layer of the rigid PCB portion;

a second insulating layer located on or above the first conduction layer and having the first thickness;

a second conduction layer located on or above the second insulating layer and being electrically connected with the second conduction layer of the rigid PCB portion;

a third insulating layer located on or above the second conduction layer and forming a second exterior surface of the second flexible PCB portion, the third insulating layer including the first material; and

a contact pad located on the third insulating layer.

15. The portable communication device of claim 14, wherein the second insulating layer of the first flexible PCB portion or the second insulating layer of the second flexible PCB portion includes the second material.

16. The portable communication device of claim 14, wherein each of the first flexible PCB portion and the second flexible PCB portion includes a fourth insulating layer located between the second conduction layer and the third insulating layer, and a third conduction layer located between the third insulating layer and the fourth insulating layer.

17. The portable communication device of claim 16, wherein the fourth insulating layer includes the second material.

18. The portable communication device of claim 16, wherein the first conduction layer of each of the first flexible PCB portion and the second flexible PCB portion includes a first ground portion,

wherein the second conduction layer of each of the first flexible PCB portion and the second flexible PCB

portion includes a second ground portion electrically connected with the first ground portion through a first conduction via, and

wherein the third conduction layer includes a third ground portion electrically connected with the second ground portion through a second conduction via. 5

19. The portable communication device of claim **14**, wherein the first material is a solder resist ink and the second material is a polyimide, and

wherein the second insulating layer of the rigid PCB portion includes a third material. 10

20. The portable communication device of claim **19**, wherein the third material includes a resin or a glass fabric.

* * * * *

Exhibit 2

REDACTED IN ITS ENTIRETY

Exhibit 3

Jas Vidwan

From: Trainor, D. Sean <dstrainor@omm.com>
Sent: Thursday, February 12, 2026 12:11 PM
To: Tiffany A. Miller; #OMM-SAMSUNG-OURA
Cc: WDC-CLIENT-OuraITCService-EXTERNAL-CONTACTS
Subject: RE: ITC Inv. 337-TA-1478: Potential Cross Use

Tiffany,

Yes, Samsung is interested in such a cross-use agreement.

Thanks,
Sean

D. Sean Trainor
O: +1-202-383-5114
M: +1-240-687-2031
dstrainor@omm.com

From: Miller, Tiffany A. <TMiller@mayerbrown.com>
Sent: Thursday, February 12, 2026 11:32 AM
To: #OMM-SAMSUNG-OURA <OMM-SAMSUNG-OURA@omm.com>
Cc: WDC-CLIENT-OuraITCService-EXTERNAL-CONTACTS <Oura-ITC@mayerbrown.com>
Subject: ITC Inv. 337-TA-1478: Potential Cross Use

Counsel,

Following up on ALJ Johnson Hines's suggestion at the preliminary conference, we wanted to explore whether Samsung is interested in a discovery cross-use agreement between the 1468 and the 1478 Investigations? At a minimum, we believe it would help streamline discovery and increase efficiency for the parties to agree to a cross-use of documents produced.

Please let us know if you agree, at least in principle, and we can move forward on the details. At this time, as we approach Monday's discovery deadline, Oura intends to operate as though an agreement on document productions can be reached.

Regards,
Tiffany

Tiffany A. Miller

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Exhibit 4

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January 26, 2026

BY ELECTRONIC DELIVERY

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Re: In re *Certain Smart Wearable Devices, Systems, and Components Thereof*, ITC Inv. No.
337-TA-1468 – Oura Document Production

Dear Counsel:

At the following link, please find Complainant Ouraring, Inc.'s ("Ouraring") document production consisting of documents bearing bates numbers OURA1468_00000001 - OURA1468_00097540 (OURA1468_V001):

https://ftpus.consilio.com/public/file/hplrecbo1u6zexqasbhbjja/OURA1468_V001.zip

A password will be sent separately for the link and production file. Please note that certain documents within the production have been designated CONFIDENTIAL BUSINESS INFORMATION – SUBJECT TO PROTECTIVE ORDER and should be treated accordingly under the agreed upon Protective Order.

January 26, 2026

Page 2

Please let us know if you have any issues accessing Ouraring's production.

Sincerely,

A handwritten signature in black ink, appearing to read 'Tiffany A. Miller', enclosed within a rectangular box.

Tiffany A. Miller

cc: Counsel of Record

Exhibit 5

REDACTED IN ITS ENTIRETY

Exhibit 6

REDACTED IN ITS ENTIRETY

Exhibit 7

REDACTED IN ITS ENTIRETY

Exhibit 8

Jas Vidwan

From: Tiffany A. Miller
Sent: Wednesday, February 11, 2026 5:09 PM
To: #OMM-SAMSUNG-OURA; z/Reebok_ITC_1468; ZeppHealthITC1468
Cc: WDC-CLIENT-OuraITCService-EXTERNAL-CONTACTS
Subject: ITC Inv. No. 337-TA-1468 - Source Code Review and Availability

Counsel,

Now that the Parties have moved to enter the supplement to the protective order with respect to source code, please let us know whether Respondents' source code and firmware is ready for inspection. Please also let us know where the source code laptops will be made available for inspection.

Oura's source code for Domestic Industry Product is ready for inspection in Mayer Brown's Washington D.C. office. Oura's source code computer is running the Windows 11 operating system. The following programs for analyzing source code are also installed on the source code computer: Notepad++, Powergrep, ExamDiff Pro, Cygwin, and Adobe Acrobat Reader. Please let us know if there are any additional software that Respondents want installed on the source code computer.

Regards,
Tiffany

Tiffany A. Miller

Partner

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February 25, 2026

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Re: In re *Certain Smart Wearable Devices, Systems, and Components Thereof*, ITC Inv. No.
337-TA-1468 – Oura Document Production

Dear Counsel:

At the following link, please find Complainant Ouraring Inc.'s ("Oura") document production consisting of documents bearing bates numbers OURA1468_00218120 - OURA1468_00220363 (OURA1468_V005):

https://ftpus.consilio.com/public/folder/y8ivihkswea06iy6dszetq/OURA1468_V003%20-%20OURA1468_V005

A password will be sent separately for the link and production file. Please note that certain documents within the production have been designated CONFIDENTIAL BUSINESS INFORMATION – SUBJECT TO PROTECTIVE ORDER and should be treated accordingly under the agreed upon Protective Order.

February 25, 2026

Page 2

Please let us know if you have any issues accessing Oura's productions.

Sincerely,

A handwritten signature in black ink, appearing to read 'Tiffany A. Miller', enclosed within a thin black rectangular border.

Tiffany A. Miller

cc: Counsel of Record

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March 17, 2026

BY ELECTRONIC DELIVERY

D. Sean Trainor
O'MELVENY & MYERS LLP
OMM-SAMSUNG-OURA@OMM.COM

Stephen R. Smith
COOLEY LLP
Reebok_ITC_1468@cooley.com

David M. Maiorana
JONES DAY
ZeppHealthITC1468@jonesday.com

Re: In re *Certain Smart Wearable Devices, Systems, and Components Thereof*, ITC Inv. No.
337-TA-1468 – Oura Document Production

Dear Counsel:

At the following links, please find Complainant Ouraring Inc.'s ("Oura") document productions consisting of documents bearing bates numbers OURA1468_00254513 - OURA1468_00255562 (OURA1468_V007) and OURA1468_00255563 - OURA1468_00255591 (OURA1468_V008):

https://ftpus.consilio.com/public/file/d8yj8f-4j0whx_tr7spwha/OURA1468_V007.zip

https://ftpus.consilio.com/public/file/13ar4p98le-8-xr6smsi2a/OURA1468_V008.zip

A password will be sent separately for the links and production files. Please note that certain documents within the production have been designated CONFIDENTIAL BUSINESS INFORMATION – SUBJECT TO PROTECTIVE ORDER and should be treated accordingly under the agreed upon Protective Order.

March 17, 2026

Page 2

Please let us know if you have any issues accessing Oura's productions.

Sincerely,

A handwritten signature in black ink, appearing to read 'Tiffany A. Miller', enclosed within a rectangular border.

Tiffany A. Miller

cc: Counsel of Record

Exhibit 11

Mayer Brown LLP
1999 K Street, N.W.
Washington, D.C. 20006-1101

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Main Fax +1 202 263 3300
www.mayerbrown.com

Tiffany A. Miller
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Direct Fax +1 202 263 3300
tmiller@mayerbrown.com

March 26, 2026

BY ELECTRONIC DELIVERY

D. Sean Trainor
O'MELVENY & MYERS LLP
OMM-SAMSUNG-OURA@OMM.COM

Stephen R. Smith
COOLEY LLP
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David M. Maiorana
JONES DAY
ZeppHealthITC1468@jonesday.com

Re: In re *Certain Smart Wearable Devices, Systems, and Components Thereof*, ITC Inv. No.
337-TA-1468 – Oura Document Production

Dear Counsel:

At the following link, please find Complainant Ouraring Inc.'s ("Oura") document production consisting of documents bearing bates numbers OURA1468_00256345 - OURA1468_00256376 (OURA1468_V010):

https://ftpus.consilio.com/public/file/osc_zfy97eykvxdy8jlexa/OURA1468_V010.zip

This production contains bates-stamped versions of the three files natively produced by Ms. Julia Haines on March 24, 2026. Additionally, please note that OURA1468_00256348 constitutes a reproduction of OURA1468_00255190, as requested by counsel for Samsung today.

A password will be sent separately for the link and production file. Please note that certain documents within the production have been designated CONFIDENTIAL BUSINESS INFORMATION – SUBJECT TO PROTECTIVE ORDER and should be treated accordingly under the agreed upon Protective Order.

March 26, 2026

Page 2

Please let us know if you have any issues accessing Oura's productions.

Sincerely,

A handwritten signature in black ink, appearing to read 'Tiffany A. Miller', written over a horizontal line.

Tiffany A. Miller

cc: Counsel of Record

Exhibit 12

REDACTED IN ITS ENTIRETY

Exhibit 13

REDACTED IN ITS ENTIRETY

Exhibit 14

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of:

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

**COMPLAINANTS SAMSUNG ELECTRONICS AMERICA, INC. AND SAMSUNG
ELECTRONICS CO., LTD.'S INITIAL INFRINGEMENT CONTENTIONS**

Complainants Samsung Electronics America, Inc. and Samsung Electronics Co., Ltd., (“Samsung”) hereby provide their Initial Infringement Contentions, which are incorporated by reference in Samsung’s Response to Respondent Ouraring Inc. and Oura Health Oy’s (“Oura”) Interrogatory No. 1 to Oura’s First Set of Interrogatories. Samsung incorporates herein by reference its responses and objections to Oura’s Interrogatory No. 1.

Samsung accuses the following wearable devices and corresponding applications of infringement: Oura Ring 4 and its corresponding software applications, including the Oura App (collectively, the “Oura Accused Product”). The Oura Accused Product directly infringes, literally or under the doctrine of equivalents, and/or indirectly infringes the following claims of U.S. Patent Nos. 10,642,359 (“the ’359 Patent”), 10,945,677 (“the ’677 Patent”), 10,231,675 (“the ’675 Patent”) and 10,978,789 (“the ’789 Patent”) (collectively, the “Asserted Patents”):

Patent Number	Asserted Claims
10,642,359	1-2, 4, 7-8, 10, 14-15, 20
10,945,677	1-7, 9-14, 16-20
10,231,675	13-19, 20
10,978,789	7-10

Samsung further incorporates herein by reference the March 16, 2026 Infringement Claim Charts being served herewith, including the documents and evidence cited in such charts (Exhibits

1-4).

Oura directly infringes each of the Asserted Patents by making, using, offering to sell, and/or selling the Oura Accused Product in the United States, and/or importing the Oura Accused Product in the United States.

Oura indirectly infringes each of the Asserted Patents by intentionally and actively inducing their customers, distributors, and/or end users of the Oura Accused Product to directly infringe the Asserted Patents at least by providing instructions explaining how to use the Accused Products after importation into the United States and encouraging their use in that manner. Such instructions include, for example, instructions on Oura's website <https://ouraring.com/>, User Guide <https://support.ouraring.com/hc/en-us/articles/4412081024147-User-Guides>, support pages <https://support.ouraring.com/>, blogs <https://ouraring.com/blog/>, press releases <https://ouraring.com/newsroom>, YouTube <https://www.youtube.com/channel/UCfxFf4xPcT9DVdOkcCaScg>, Facebook <https://www.facebook.com/ouraring>, Instagram <https://www.instagram.com/ouraring>, and other instructional and promotional media.

Oura also indirectly infringes each of the Asserted Patents by contributing to the infringement by their customers, distributors, and/or end users knowing that the Oura Accused Product is especially made or especially adapted for use in infringing the Asserted Patents. Additionally, the Oura Accused Product is not a staple article or commodity of commerce suitable for substantial non-infringing use.

Samsung further states that pursuant to Commission Rule 210.29(c), it has produced documents from which Respondents may ascertain relevant information responsive to this interrogatory, including without limitation those bearing Bates nos: SAMS382-0027560 to SAMS382-0028069.

PUBLIC VERSION

DATED: March 16, 2026

Respectfully submitted,

/s/ D. Sean Trainor

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2801 North Harwood Street
Suite 1600
Dallas, Texas 75201-2692
Telephone: (972) 360-1900

*Counsel for Complainants Samsung Electronics
Co., Ltd. and Samsung Electronics America, Inc.*

CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on March 16, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

<i>Counsel for Complainants Ouraring Inc. and Oura Health Oy</i> Jasjit S. Vidwan James A. Fussell, III Saqib J. Siddiqui Bryan C. Nese Michael L. Lindinger Tiffany A. Miller Clark S. Bakewell Alison T. Gelsleichter Tariq Javed Amanda E. Stephenson Seth W. Bruneel Courtney Krawice Séké Godo Angeline Premraj Paul Choi So Ra Ko Andrew Huntsinger Jaspreet Momi MAYER BROWN LLP 199 K. Street NW Washington, DC 20006 Tel: 202-263-5282 Robert G. Pluta Nicholas H. Ciulla MAYER BROWN LLP 71 S. Wacker Drive Chicago, IL 60606 Tel: 312-782-0600 Email: Oura-ITC@mayerbrown.com	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☐ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
--	---

/s/ Brian A. Treggs

Brian A. Treggs
Senior IPT Specialist
O'Melveny & Myers LLP
400 South Hope Street
Los Angeles, CA 90071

Exhibit 15

REDACTED IN ITS ENTIRETY

Exhibit 16

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Direct Fax +1 202 263 3300
tmiller@mayerbrown.com

May 5, 2026

BY ELECTRONIC DELIVERY

D. Sean Trainor
O'MELVENY & MYERS LLP
OMM-SAMSUNG-OURA@OMM.COM

Re: In re *Certain Wearable Devices*, ITC Inv. No. 337-TA-1478 – Oura Document
Production

Dear Counsel:

At the following link, please find Respondents Ouraring Inc. and Oura Health Oy's (collectively, "Oura") document production consisting of documents bearing bates numbers OURA1478_00004838 - OURA1478_00006316 (OURA1478_V004):

https://ftpus.consilio.com/public/file/luqxyn1fee2ygl73pqw1sq/OURA1478_V004.zip

A password will be sent separately for the production link and file.

Please let us know if you have any issues accessing Oura's production.

Sincerely,



Tiffany A. Miller

cc: Counsel of Record

Exhibit 17

Mayer Brown LLP
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tmiller@mayerbrown.com

May 29, 2026

BY ELECTRONIC DELIVERY

D. Sean Trainor
O'MELVENY & MYERS LLP
OMM-SAMSUNG-OURA@OMM.COM

Re: In re *Certain Wearable Devices*, ITC Inv. No. 337-TA-1478 – Oura Document
Production

Dear Counsel:

At the following link, please find Respondents Ouraring Inc. and Oura Health Oy's (collectively, "Oura") document production consisting of documents bearing bates numbers OURA1478_00013076 - OURA1478_00017180 (OURA1478_V010):

https://ftpus.consilio.com/public/file/yigeliujqk2c3lhgblrptg/OURA1478_V010.zip

A password will be sent separately for the production link and file.

Please let us know if you have any issues accessing Oura's production.

Sincerely,



Tiffany A. Miller

cc: Counsel of Record

Exhibit 18

REDACTED IN ITS ENTIRETY

Exhibit 19

REDACTED IN ITS ENTIRETY

Exhibit 20

REDACTED IN ITS ENTIRETY

Exhibit 21

REDACTED IN ITS ENTIRETY

Exhibit 22

Jas Vidwan

From: Jas Vidwan
Sent: Friday, May 1, 2026 5:07 PM
To: '#OMM-SAMSUNG-OURA'
Cc: WDC-CLIENT-OuraITCService-EXTERNAL-CONTACTS
Subject: RE: 337-TA-1478 - Oura's 30(b)(6) Deposition Designations

Counsel,

Below please find Oura's 30(b)(6) deposition witnesses, availability, and topic designations, subject to Oura's objections and responses.

Oura Witnesses	Location	Availability	30(b)(6) Topic Nos.
Lindsey Belknap	Boston, MA	May 6	27-28
Simon Ratner	San Francisco, CA	May 12	2-5 (As Topics Relate to App Source Code for A
Miguel Christophy	San Francisco, CA	May 15	2-3, 19, 26 (As Topics Relate to Hardware for Ac
Michael Larson	San Diego, CA	May 19	2-5, 26 (As Topics Relate to Firmware Source Co Ring)
Jack Murrin	New York City, NY	May 19	30, 33-35
CJ Vincent	Palo Alto, CA	May 19	1, 6-7, 18, 30, 36

Please note that we are still confirming the availability of one additional witness for Topics 20-23, 25, 30, 32. We'll get that information to you as soon as possible.

Best,

Jas

Jasjit S. Vidwan

Partner

Mayer Brown LLP

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Washington, DC 20006-1101

T +1 202 263 3065

mayerbrown.com

From: Jas Vidwan
Sent: Wednesday, April 29, 2026 9:39 PM
To: #OMM-SAMSUNG-OURA <OMM-SAMSUNG-OURA@omm.com>
Cc: WDC-CLIENT-OuraITCService-EXTERNAL-CONTACTS <Oura-ITC@mayerbrown.com>
Subject: 337-TA-1478 - Oura's 30(b)(6) Deposition Designations

Counsel,

Oura is in the process of getting availabilities of its Oura's 30(b)(6) deposition witness and topic designations. We'll provide a full list shortly.

In the meantime, one of Oura's witness, Lindsey Belknap, that will be designated on Topics 27 and 28, is available for deposition next week on May 6th in Boston. She has limited availability before close of fact discovery. Thus, please let us know whether Samsung accepts her offered date as soon as possible, but no later than Friday, May 1st.

Best,

Jas

Jasjit S. Vidwan

Partner

Mayer Brown LLP

1999 K Street, NW

Washington, DC 20006-1101

T +1 202 263 3065

mayerbrown.com

Exhibit 23

REDACTED IN ITS ENTIRETY

Exhibit 24

REDACTED IN ITS ENTIRETY

Exhibit 25

REDACTED IN ITS ENTIRETY

Exhibit 26

REDACTED IN ITS ENTIRETY

Exhibit 27

REDACTED IN ITS ENTIRETY

Exhibit 28

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

**COMPLAINANTS' FIRST SET OF
INTERROGATORIES TO RESPONDENTS (NOS. 1-10)**

Pursuant to Commission Rules 210.27 and 210.29, 19 C.F.R. §§ 210.27 and 210.29, Complainants Samsung Electronics Co., Ltd. ("SEC") and Samsung Electronics America, Inc. ("SEA") (collectively, "Samsung" or "Complainants") serve the following set of interrogatories on Respondents Ouraring, Inc. and Ōura Health Oy (collectively, "Oura" or "Respondents") in the above-captioned investigation. Samsung requests, pursuant to Commission Rule 210.29(b)(2) that Respondents serve a copy of their answers within 10 days after service hereof or at such other time as agreed to by the parties.

DEFINITIONS

The following definitions shall apply to these interrogatories.

1. The "'359 Patent" means U.S. Patent No. 10,642,359.
2. The "'677 Patent" means U.S. Patent No. 10,945,677.
3. The "'675 Patent" means U.S. Patent No. 10,231,675.
4. The "'789 Patent" means U.S. Patent No. 10,978,789.
5. The "Asserted Patents" means the '359, '677, '675, and '789 Patents. An "Asserted

Patent" refers to any one of the Asserted Patents.

6. The “Asserted Claims” means claims 1-2, 4, 7-8, 10, 14-15, and 20 of the ’359 Patent; claims 1-7, 9-14, and 16-20 of the ’677 Patent; claims 13-20 of the ’675 Patent; and claims 7-10 of the ’789 Patent. “Asserted Claim” refers to any of the preceding claims.

7. “Samsung” or “Complainants” means Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc.

8. “Oura” or “Respondents” means Respondents Ouraring, Inc. and Ōura Health Oy, along with their past and present predecessors, successors, subsidiaries, divisions, parents and affiliates, and all past and present officers, directors, agents, trustees, employees, consultants, accountants, attorneys, representatives, and any other person or entity acting in whole or in part on behalf of any of the foregoing.

9. “You” and “Your” means Respondents.

10. “Accused ’359 Products” means all of Respondents’ products that Samsung contends are covered by at least one Asserted Claim of the ’359 Patent, including at least Respondents’ Oura Ring products, its corresponding applications including the Oura App, the components within the Oura Ring, such as biosignal sensors, biometric sensors, and printed circuit boards, and any Oura wearable devices being imported by or on behalf of Oura, including yet to be released products with similar health tracking functionality as the ’359 accused features or functions.

11. “Accused ’677 Products” means all of Respondents’ products that Samsung contends are covered by at least one Asserted Claim of the ’677 Patent, including at least Respondents’ Oura Ring products, its corresponding applications including the Oura App, the components within the Oura Ring, such as biosignal sensors, biometric sensors, and printed circuit boards, and any Oura wearable devices being imported by or on behalf of Oura, including yet to

be released products with similar health tracking functionality as the '677 accused features or functions.

12. “Accused '675 Products” means all of Respondents’ products that Samsung contends are covered by at least one Asserted Claim of the '675 Patent, including at least Respondents’ Oura Ring products, its corresponding applications including the Oura App, the components within the Oura Ring, such as biosignal sensors, biometric sensors, and printed circuit boards, and any Oura wearable devices being imported by or on behalf of Oura, including yet to be released products with similar health tracking functionality as the '675 accused features or functions.

13. “Accused '789 Products” means all of Respondents’ products that Samsung contends are covered by at least one Asserted Claim of the '789 Patent, including at least Respondents’ Oura Ring products, the components within the Oura Ring, such as biosignal sensors, biometric sensors, and printed circuit boards, and any Oura wearable devices being imported by or on behalf of Oura, including yet to be released products with similar health tracking functionality as the '789 accused features or functions.

14. “Accused Product” or “Accused Products” means collectively the Accused '359 Products, Accused '677 Products, Accused '675 Products, and Accused '789 Products and any other of Respondents’ products identified in Your response to Interrogatory No. 1.

15. “'359 accused features or functions” includes features and functions in the Accused Products identified in the Infringement Chart for the '359 Patent (Complaint Exs. 13, 17) and any subsequent infringement contentions, including features and functions relating to the capability of the Accused Products to detect and process a user’s movement and biometric information.

16. “’677 accused features or functions” includes features and functions in the Accused Products identified in the Infringement Chart for the ’677 Patent (Complaint Exs. 14, 18) and any subsequent infringement contentions, including features and functions relating to the capability of the Accused Products to detect and process a user’s movement and biometric information.

17. “’675 accused features or functions” includes features and functions in the Accused Products identified in the Infringement Chart for the ’675 Patent (Complaint Exs. 15, 19) and any subsequent infringement contentions, including features and functions relating to the capability of the Accused Products to detect and process a user’s movement and biometric information.

18. “’789 accused features or functions” includes features and functions in the Accused Products identified in the Infringement Chart for the ’789 Patent (Complaint Exs. 16, 20) and any subsequent infringement contentions, including features and functions relating to printed circuit boards for the Accused Products.

19. “Accused Features or Functions” shall mean a combination of the ’359 accused features or functions, ’677 accused features or functions, ’675 accused features or functions, and ’789 accused features or functions.

20. “Implicated Components” shall mean biosignal sensors, biometric sensors, and printed circuit boards, and any other components configured to implement the Accused Features of Functions in the Accused Products and/or assemble the Accused Products.

21. “Investigation” means the investigation instituted by the United States International Trade Commission on January 14, 2026, captioned *Certain Wearable Devices*, and given Investigation No. 337-TA-1478.

22. “Complaint” refers to the Complaint filed by Samsung in this Investigation on December 12, 2025, and any corrections, supplements, or amendments thereto for which the ALJ or Commission has granted leave to file.

23. “Prior Art” means the activities and information encompassed by 35 U.S.C. § 102 and 35 U.S.C. § 103.

24. “Identify” or “Identification” when used with reference to a person means the full name, address, telephone number, and email address of that person.

25. “Communicate,” “Communication,” and “Communications” mean any oral or written utterance, notation, or statement of any nature, by and to whomsoever made including but not limited to conversations, dialogues, discussions, interviews, consultations, agreements, correspondence, letters, memoranda, notes, summaries, audio and video recordings, electronic communications, and other materials recording the interaction between one person and another.

26. “Document” shall have the broadest meaning allowable under Commission Rule 210.30, and includes any written, typed, printed, drawn or recorded material, however compiled or on whatever medium stored, including any photographic or other reproduction of any such material, including all non-identical copies of such Documents and including figures and drawings, either in sketch or printed form. Without limiting the generality of the foregoing, “Document” includes electronically stored information, including computer files and email communications.

27. “Thing” shall have the broadest meaning allowable under Commission Rule 210.30, and also includes tangible objects other than Documents including, without limitation, prototypes, models, or physical specimens thereof.

28. “Person(s),” includes any natural person, firm, proprietorship, partnership, joint venture, corporation, association, or other business entity and all present and former officers,

directors, agents, employees, and others acting or purporting to act on behalf of such natural person, firm, proprietorship, partnership, joint venture, corporation, association, or other business entity.

29. “Relate,” “Relating to,” and “Related to” shall include and be interpreted as meaning comprising, pertaining to, recording, evidencing, refuting, constituting, containing, setting forth, reflecting, showing, disclosing, describing, analyzing, explaining, summarizing, concerning or referring to, whether directly or indirectly, whether in whole or in part.

30. The word “or” refers to the conjunctive/inclusive meanings, so as to include any information within the scope of any of the requests in which it is contained.

31. The use of the singular form of any word includes the plural and vice versa.

INSTRUCTIONS

1. If, in answering these requests, You encounter any ambiguities construing either a question, instruction, or definition, please set forth the matter deemed ambiguous and the construction used in Your answer.

2. Any term not specifically defined herein shall be defined in accordance with the Commission Rules of Practice and Procedure (“Commission Rules”) and the Administrative Law Judge’s Ground Rules (“Ground Rules”), as well as the Federal Rules of Civil Procedure, to the extent not in conflict with the Commission Rules or Ground Rules.

3. These Interrogatories seek all responsive information in Respondents’ possession, custody, or control, including all responsive information known or available to Respondents.

4. In response to these Interrogatories, You shall provide written responses as specified by 19 C.F.R. § 210.29(b)(2) and/or records as specified by 19 C.F.R. § 210.29(c). In

either case, You shall answer each Interrogatory fully in writing and under oath in accordance with 19 C.F.R. §§ 210.27 and 210.29.

5. The answer to an Interrogatory shall not be supplied by referring to the answer to another Interrogatory unless the answer to the Interrogatory being referred to supplies a complete and accurate answer to the Interrogatory being answered.

6. If You withhold any information or decline to fully Identify a Person, Document, or Communication, or portion thereof, in response to any of the Interrogatories on grounds of privilege, work product protection, or any other claim of privilege or protection, then please provide the following:

- a. The specific ground(s) for not responding in full;
- b. A summary statement of the subject matter of such response, including any responsive Document, in sufficient detail to permit the ALJ to reach a determination in the event Complainant moves to compel pursuant to Commission Rule 210.33;
- c. The bases for any such claim of privilege and/or immunity; and
- d. The full identity of the information, Document, or material for which such privilege or immunity is asserted, including the name, date, and title of the Document; the name, address, and title of its author; each addressee; and each person to whom a copy of the Document or thing has been sent or received.

7. If the answer to any Interrogatory is “none,” then indicate so in Your response.

8. If Your response to any Interrogatory is qualified in any respect, then You shall set forth the details of, and reasons for, each such qualification.

9. You should assume that the time period for any Interrogatory is January 1, 2024, through the date of the evidentiary hearing pursuant to Respondents' duty to supplement, unless otherwise specified.

10. These Interrogatories are continuing in nature and shall be timely supplemented in accordance with 19 C.F.R. § 210.27(f). If You identify, acquire, or become aware of any additional information responsive to these Interrogatories at any time up to and including the time of the close of discovery in this Investigation, You shall promptly supplement your interrogatory Response. When supplementing Your interrogatory responses, You must also identify by Bates number any documentary information that prompted the supplementation and identify when You first became aware of identified Documents.

FIRST SET OF INTERROGATORIES

INTERROGATORY NO. 1:

Provide, in chart form, a complete identification of each Accused Product. In the context only of this interrogatory, “identification” means, for each Accused Product, all known product names, product numbers, part numbers, code names, and any other similar identification systems used for internal or external identification. For each Accused Product, also include the preinstalled version of firmware, all compatible versions of firmware, and all compatible versions of the corresponding application software including the Oura App software.

INTERROGATORY NO. 2:

Provide, in chart form, for each Accused Product, the supplier, internal model number, supplier model number, and code name used by You of the relevant components implicated by the Asserted Patents, such as the components implicated in performing the Accused Features or Functions, including the dates You utilized such components in each Accused Product.

INTERROGATORY NO. 3:

For each Accused Product, identify and describe the components of the printed circuit board (PCB), including any rigid and/or flexible PCB portions or parts, and for each PCB portion or part, its layer composition, including the number of layers, thickness of each layer, material composition of each layer, and whether the layer is conducting or insulating.

INTERROGATORY NO. 4:

For each Accused Product, identify each LED and sensor for detecting motion, acceleration, heart rate, heart rate variability, respiration rate, blood oxygen levels, and body temperature, and describe how each LED and sensor performs its detection function, including the conditions or factors that affect when each LED and sensor is active.

INTERROGATORY NO. 5:

For each Accused Product, identify and describe the sensors and algorithms used to detect whether a user is in a state of sleep or awake.

INTERROGATORY NO. 6:

For each Accused Product, identify the algorithms for determining whether to compute, store, transfer, discard, and display motion, acceleration, heart rate, heart rate variability, respiration rate, blood oxygen levels, and body temperature data, including based on whether a user is in a state of sleep or awake.

INTERROGATORY NO. 7:

For each Asserted Claim, state in detail all factual and legal bases for and identify all documents supporting your contention, if any, that You do not infringe and have not infringed that Asserted Claim (directly or indirectly, and literally or under the doctrine of equivalent), including identifying, for each Accused Product, each and every claim limitation you contend is not present in or practiced by that product and detailing all factual and legal bases for each such contention, including:

- a. a claim chart on a limitation-by-limitation basis identifying each claim limitation that you contend is not met by each Accused Product;
- b. an explanation of why each such limitation is not allegedly met (both literally and under the doctrine of equivalents); and
- c. an identification of any part of Samsung's infringement contentions (both those provided with the Complaint and any subsequent infringement contentions) with which you disagree and a detailed explanation of why you disagree.

INTERROGATORY NO. 8:

For each Asserted Claim that you contend is invalid, including without limitation under 35 U.S.C. §§ 112, 101, 102 or 103, state in detail all factual and legal bases for that contention on a claim-by-claim basis. To the extent that such a contention is based on 35 U.S.C. §§ 102 and/or 103, your response shall separately identify, on a limitation-by-limitation basis, all support for any contention that any alleged prior art references disclose such limitation. Your answer shall include:

- a. the statutory basis for your contention;
- b. all documents, witnesses, and things (such as prior art) on which you rely in support of your contention;
- c. if you contend the claim is anticipated or obvious, claim charts for each element of the prior art on which you rely in support of your contention;
- d. if you contend the claim is obvious, the specific combination or combinations of prior art that you contend renders the claim obvious to a person of ordinary skill in the art;
- e. if you contend the claim is obvious, what suggests the combination, combinations, and/or motivation to deviate from or combine the art;
- f. if you contend the claim is obvious, information relating to secondary considerations of non-obviousness, including without limitation commercial success, long-felt need, copying by others, prior failure by others, licensing, or unexpected results;
- g. the identification of all other documents relating to your contention, including without limitation those that rebut your contention; and
- h. if you contend the claim is invalid for failure to comply with the requirements of 35 U.S.C. § 112, a detailed description of your contention, including for any claim terms that you claim are indefinite an explanation regarding why.

INTERROGATORY NO. 9:

If You contend that Samsung has not satisfied the domestic industry requirement, please describe in detail all factual and legal bases for this contention and identify all relevant Documents.

INTERROGATORY NO. 10:

If You contend any affirmative defenses, please describe in detail all factual and legal bases for such contention and identify all relevant Documents.

Dated: January 20, 2026

Respectfully submitted,

/s/ D. Sean Trainor

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*Counsel for Complainants Samsung
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America, Inc.*

CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on January 20, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

<i>Counsel for Complainants Ouraring Inc. and Oura Health Oy</i> Jasjit S. Vidwan James A. Fussell, III Saqib J. Siddiqui Bryan C. Nese Michael L. Lindinger Tiffany A. Miller Clark S. Bakewell Alison T. Gelsleichter Tariq Javed Amanda E. Stephenson Seth W. Bruneel Courtney Krawice Séké Godo Angeline Premraj Paul Choi So Ra Ko Andrew Huntsinger Jaspreet Momi MAYER BROWN LLP 199 K. Street NW Washington, DC 20006 Tel: 202-263-5282 Robert G. Pluta Nicholas H. Ciulla MAYER BROWN LLP 71 S. Wacker Drive Chicago, IL 60606 Tel: 312-782-0600 Email: Oura-ITC@mayerbrown.com	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☐ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
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/s/ Brian A. Treggs

Brian A. Treggs
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Exhibit 29

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Exhibit 30

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Exhibit 31

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Exhibit 32

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Exhibit 57

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Exhibit 58

REDACTED IN ITS ENTIRETY

Certain Wearable Devices

Inv. No.: 337-TA-1478

CERTIFICATE OF SERVICE

I hereby certify that a copy of the foregoing **RESPONDENTS OURARING INC. AND OURA HEALTH OY'S MOTION FOR SUMMARY DETERMINATION OF NON-INFRINGEMENT BY OURA'S ALTERNATIVE PRODUCTS AND STATEMENT OF MATERIAL FACTS IN SUPPORT THEREOF (Public Version)** was served upon the following as indicated below on July 20, 2026:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, DC 20436	Via EDIS
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W., Room 317 Washington, DC 20436	Via email and Box: JohnsonHines1478@usitc.gov
Counsel for Complainants Samsung Electronics Co., Ltd and Samsung Electronics America, Inc. D. Sean Trainor O'MELVENY & MYERS LLP 1625 Eye Street, NW Washington, D.C. 20006-4061 Telephone: (202) 383-5300	Via email: OMM-SamsungOura@OMM.com

/s/ Jasjit S. Vidwan

Jasjit S. Vidwan

*Counsel for Respondents Ouraring Inc.
and Oura Health Oy*